On the Investment Network and Development *

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Abstract

Capital accumulation, productivity growth and sectoral reallocation are salient features of economic development. These features are interconnected through the means for production and heterogeneous uses of different capital types across sectors, i.e. the investment network. Our paper introduces the first harmonized measures of the investment network across the development spectrum and documents novel empirical regularities, including systematic changes in the network with development. We propose a simple theory linking these disparities, which are endogenous to the path of sectoral prices, to differences in income per capita across countries. We show that the elasticity of output to sectoral productivity depends on the standard Hulten effect, when capital is fixed; and on the investment network and the dynamics of the gross return on capital across sector along the transition. We show that sectoral influence varies systematically with income: it is highest in the Construction sector in developing countries, and in the ICT and Service sector in developed countries. The relative contribution of the static and dynamic effects varies across these sectors and income levels. Forces related to structural change, sectoral productivity and shifts in sectoral linkages drive this heterogeneity. For our sample of 58 countries, we show that 42% of cross-country differences in income per capita can be accounted for by disparities in the investment network. These differences are twice as large as the role of capital in income disparities obtained from standard development accounting.

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JEL classifications: E23; E21; O41.

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1 Introduction

Capital accumulation and the systematic reallocation of economic activity across sectors are two of the most salient features of economic development. Different sectors utilize various investment goods for production, which are either produced by other sectors of the economy or imported. As economic activity shifts across sectors, the economy's ability to produce new capital—or to export goods in exchange for this new capital—changes, facilitating further capital accumulation. The study of the nature of this continuous feedback is crucial for understanding the mechanics of economic development (Hirschman, 1958). Such a study requires measures of sectoral links in both the production and use of new capital, i.e., the investment network. This paper provides the first harmonized measures of investment networks across the development spectrum. We document novel facts about the characteristics of these networks and construct a theory to evaluate the impact of these differences on observed income disparities across countries.

Recent studies have documented systematic changes in the sectoral composition of inputs used for investment as countries develop, Garcia-Santana, Pijoan-Mas and Villacorta (2021); Herrendorf, Rogerson and Valentinyi (2021). Work has also examined how disparities in the bundles of capital goods used for sectoral production lead to structural change (Caunedo and Keller, 2023). We combine these two approaches and characterize the link between production and uses of capital, i.e. the investment network, and aggregate output. Through the lens of a neoclassical multisector open economy model, we show that the investment network is a key component of the elasticity of GDP to sectoral productivity. The extensive literature in networks labels this elasticity as "*sectoral influence*", Acemoglu, Carvalho, Ozdaglar and Tahbaz-Salehi (2012).

Sectional influence is a function of the input-output structure and the investment network, summarizing the direct and indirect impact of changes in sectoral productivity on aggregate economic activity. Our economy has several distinct features relative to other economies with sectoral linkages. First, welfare and GDP differ due to the presence of investment. Along a growth path with constant investment rates these two are proportional to each other. Second, Domar weights depend on equilibrium investment rates and therefore on the entire equilibrium path in an economy that displays non-trivial transition dynamics. Since welfare is a Domar-weighted sum of sectoral productivities, welfare also depends on the entire equilibrium path. This feature is a consequence of the durable nature of capital and a novel result in the production networks literature that, for the most part, focuses on nondurable inputs. Third, *sectoral influence* is independent of the investment rate and depends solely on the characteristics

of the IO and investment network at each point in time. The reason is that the effect on aggregate GDP includes both consumption and investment, while the effect on welfare includes the trade-off between current and future consumption.

To bring empirical content to the roundabout effects summarized in *sectoral influence*, one needs measures of intermediate and investment networks, as well as factor shares and investment rates. While estimates of the input–output structure have become increasingly available across countries, estimates of the investment network are only available for the US (vom Lehn and Winberry, 2022), and a handful of years across OECD economies.[1] We advance previous measurement efforts by providing cross-country and time-series harmonized estimates of the investment network for 58 countries with income per capita levels between $428 and $81,599 constant 2015 PPP dollars and time spans that date back to the 1960s for a subset of countries. In our analysis, capital is disaggregated into multiple equipment types, including ICT, electronics, machinery, transportation and other durables, as well as structures, measured through construction investment.[2]

To create our harmonized measures, we exploit a methodology similar to that of the Bureau of Economic Analysis (BEA) in the US. The BEA combines the occupational composition of each industry and an allocation rule for capital to workers to estimate investment by capital type and sector (Meade, Rzeznik and Robinson-Smith, 2003). Unfortunately, the apportioning of stocks to workers is not publicly available. Hence, to ensure replicability, we opt for an allocation of capital across sectors that follows Caunedo, Jaume and Keller (2023a) for equipment sectors and an allocation that follows intermediate inputs for construction and other sectors with positive investment in the national accounts. Our own estimates of the investment network for the US closely follow those published by the BEA, which yields confidence on our apportioning method. Still, the BEA and our apportioning depends on sectoral occupational compositions to impute capital uses. Ideally, one would like to have direct measures of investment by detailed categories in each sector. This data is broadly unavailable even within a country, let alone across countries and for long time series. There is a handful of countries with VAT data, that allow for measurement of firm-to-firm transactions, notably Chile between 2015-2024 Gillmore, Kohn, Heresi, Quevedo and Rivera (2025). We find that our estimates are fully consistent with VAT estimates which again is reassuring given the widely different sources of these estimates.

[1] These measures are self-reported by country offices to the OECD Statistics office, and it is unclear whether measurement is comparable across countries. Ding (2023) exploits these investment flows to estimate capital services in each sector from different sectors and countries. To do so, he uses bilateral import flows to input cross-country linkages and estimates user costs along a BGP. In other words, he treats the investment network as a primitive.

[2] Our benchmark estimates include 10 sectors, but estimates for as many as 19 sectors consistently defined across countries and time can be made readily available.

With our newly constructed measure, we document some novel facts. First, there are systematic changes in the composition of investment providers as countries develop, which we measure through *outdegrees*—the row sums of the network. Construction is the dominant provider at all income levels and follows a hump-shaped relationship with development; other services (including financial intermediation and real estate) are also major suppliers, but their importance declines with income and is overtaken by a strong increase in ICT, whose role nearly triples as countries get richer. Second, the bundle of capital goods that sectors use in production becomes more diversified as countries grow richer, with the exception of construction, where concentration rises. Finally, the investment network and the input–output tables differ substantially: sectors that are important providers of intermediate inputs are not necessarily those that matter for investment supply. This is the case of agriculture—a major provider of intermediate inputs (especially in low-income countries) that plays only a minor role in investment—and of construction, which is more than three times as important a provider of investment as of intermediate inputs. ICT is an important provider of both investment and intermediate goods. Moreover, homophily (the diagonal weight) is stronger in the input–output network, meaning sectors are more important providers of intermediate inputs to themselves than of investment goods. Hence, sectoral productivity shocks have different aggregate implications depending on whether a sector mainly supplies intermediates or investment.

A natural question to ask is whether sectors with high *outdegrees* in the investment network are also sectors where changes in productivity have the strongest impact on aggregate activity. The answer is no. Our theory predicts that their role is mediated by other features of the economy, including value added and capital shares in production, as well as the full input–output network. Hence, to assess the role of the investment network for income disparities, we exploit the characterization of *sectoral influence*.

Sectorial influence is an endogenous outcome of the model because it depends on endogenous value added shares, as well as the sectorial expenditure shares in investment which are functions of relative prices. We conduct two sets of exercises to assess the role of the technology for producing investment for income disparities. One exercise in partial equilibrium and another one in general equilibrium.

First, we conduct standard income accounting exercise in partial equilibrium, i.e. where the value added shares that enter into sectorial influence are taken exogenously. Given measured roundabout effects as summarized in an “augmented" Leontief inverse (augmented by investment linkages), we can calibrate sectoral productivity differences to match disparities in value added across sectors within countries and across countries for a given sector. Our main

exercise is then to compare our baseline economy, which by construction exactly matches income disparities across countries, to an economy where the investment network consists of a series of stacked columns equal to the sectorial value added shares. This counterfactual economy correspond to an economy with a single capital good (rather than sector-specific ones) which is produced with the same technology as final consumption. We show that 15% of the variation in observed income disparities is accounted for an investment technology that differs from that of the one-capital economy. This 15% is an additional contribution of the technology for producing capital beyond what measured capital disparities would dictate (in our sample, measured capital disparities account for 13% of income differences). Hence, the role of capital is 28% of the observed cross-country disparities in income. We conclude that accounting for roundabout effects in the technology for producing capital almost doubles its role for income disparities. In this partial equilibrium set up we also study the role of imported equipment for these results. The imported nature of capital is potentially relevant to our accounting because the magnitude of the roundabout effect is sensitive to how much capital is locally produced and because countries may buy productive capital by generating exports. Our theory accommodates these channels by modeling investment goods as a combination of locally and foreign-sourced investment and by explicitly modeling exports of final goods as a source to finance investment imports. We run counterfactual exercises where we construct the loadings of the investment network including only the domestically produced share of investment and shut down the productivity amplification through the production of tradable goods. We find that the counterfactual economy induces a 44% decline in steady-state income per capita on average across countries and that the variance of income falls by 12% relative to baseline. In other words, trade accounts for slightly more than one-third of the variance in income accounted for by the investment network.

Second, we reassess the role of the technology for producing capital in an economy where both the size of the sectors (VA shares) and the investment network are determined in equilibrium through differential rates of sectorial productivity growth, differential intensity in the use of sectorial investment for production and income effects. We solve the transition path in a simplified economy with five sectors, including agriculture, manufacturing, services, construction and ICT. We calibrate our economy to the path of development of Korea between 1965 and 2014 to target the path of sectorial relative prices and consumption shares, as well as the average level of value added shares in the economy. Despite not targeting trends in value added shares as well as the features of the cost shares of investment across sectors, the model does a remarkable good job at tracking structural change in value added, as well as the shifts in the investment network as Korea developed. We then run counterfactual exercises that provide

the general equilibrium counterpart to the partial equilibrium exercises. We run counterfactuals where we change sectorial productivities one-at-the time and evaluate sectorial influence in general equilibrium. We find relatively low role for productivity growth in ICT, except possibly for value added in the service sector, and a very important role for productivity growth in the manufacturing sector, as a key driver of improvements in aggregate output levels (shrinking to a quarter of its baseline growth) and capital output levels, shrinking by a factor of 3.

Contribution to the literature. There is a growing literature studying the relevance of sectoral linkages, mostly through intermediate input use, for differences in income per capita across countries; see Ciccone (2002); Jones (2011). This role is quantified in Fadinger, Ghiglino and Teteryatnikova (2022), who show that differences in the input–output structure amplify the role of sectoral productivity for differences in income per capita. We document that the input–output and investment networks are empirically different, leading to differential roles in driving income disparities across countries. We show that disparities in the technology for producing investment and consumption goods amplify or dampen the effect of sectorial productivity differences for income disparities. In other words, standard development accounting exercises Hall and Jones (1999) are misspecified when the technology for producing investment and consumption differ. Quantitatively, we find that disparities in measured capital are more important to income disparities than previously understood.

Garcia-Santana *et al.*, 2021; Herrendorf *et al.*, 2021 document systematic shifts in the inputs used for investment as economies develop. In the US time series, Gaggl, Gorry and vom Lehn (2023) document disparate shifts in the composition of inputs for investment and consumption, while Caunedo and Keller (2023) show how differences in the investment bundles used by different sectors drive sectoral reallocation. Our paper completes the puzzle by documenting disparities in the investment bundle and the composition of investment along the development spectrum. The study of the role of the investment network in determining GDP is relatively recent. In the US, vom Lehn and Winberry (2022) focus on short-run fluctuations, while Foerster, Hornstein, Sarte and Watson (2022) study implications for long term growth. [3] Baqaee and Malmberg (2025) characterize long-run comparative statics across balanced growth paths; we characterize the full GDP path, on impact, along the transition and in the long run, in an economy where investment goods become cheaper relative to final output and demand shifts across sectors, with an investment network.

Another novel finding of our theory is that sectoral investment rates mediate the impact of sectoral productivity on aggregate GDP and affect the rate of convergence of the economy to

[3] Differently from Foerster *et al.* (2022), we solve for the endogenous allocation of labor, our network changes endogenously with shifts in sectoral relative prices and we allow for equipment imports.

its asymptotic aggregate growth path. As in Liu (2019), output elasticities to sectoral productivity are different from Domar weights. In our model, this difference arises from the presence of investment, i.e., sectoral value-added shares and consumption shares do not equalize. Indeed, because our economy is efficient, welfare is a Domar-aggregated measure of sectoral productivities. These Domar weights are a function of equilibrium investment rates, a channel that is absent in economies where inputs are nondurable, as in Acemoglu *et al.* (2012) and the extensive literature that follows. Buera and Trachter (2024) is the closest paper to our study, characterizing differential GDP responses to sectoral productivity. Their work emphasizes the role of distortions and nonconvexities for endogenous technology choices. Our paper instead emphasizes the intensive margin of the adoption of capital-embodied technology, i.e., investment, and documents systematic disparities in its nature along the development spectrum.

The remainder of the paper is organized as follows: Section 2 discusses an illustrative example for the role of the investment network in accounting for income disparities. Section 3 discusses the methodology to construct estimates of the investment networks and novel empirical regularities of the investment network; Section 5 presents the model and Section 6 discusses the main result from income accounting exercises while Section 6.3 documents patterns of *sectoral influence* along the development spectrum. Section 8 concludes the paper.

2 An illustrative example

We begin with a simple economy in which the technology for producing new capital differs from that for producing consumption goods. Consider a two-sector economy with the following technologies for sectoral output, capital accumulation, and feasibility constraints:

$$v_{it} = z_{it} K_{it}^{\alpha} L_{it}^{1-\alpha}$$

$$K_{it+1} = (1-\delta) K_{it} + \prod_j x_{jit}^{\omega_j}$$

$$c_{it} + x_{it} = v_{it} \qquad x_{it} = \sum_j x_{ijt}$$

Sectoral output may be consumed or invested. Sectoral output itself is produced using sector-specific capital and labor via a Cobb-Douglas technology with heterogeneous productivity across sectors. Investment goods are assembled from sectoral inputs, with each sector j entering the investment aggregator with intensity ω_j: a common weight across sectors using input j. We relax both this symmetry and the Cobb-Douglas assumption later in the paper.

Because the investment aggregator is identical across sectors, the price of capital is equalized, which in turn equalizes capital-labor ratios: $k_i/l_i = K/L$. Log GDP is then an expenditure-share-weighted sum of log sectoral value added, $\ln(v_t) = \sum_{i=1,2} \tilde{\eta}_{it} \ln(v_{it})$.

The equalization of capital-labor ratios, combined with inelastic labor supply, implies that GDP takes the simple form $\ln(v_t) = \sum_{i=1,2} \tilde{\eta}_{it}(\ln z_{it} + \ln(\tilde{\eta}_i)) + \alpha \ln(K_t)$. If agents have no disutility from labor and derive utility from sectoral consumption in a log-separable manner, the economy admits a steady state in which output can be written as a function of the capital-output ratio:

$$\ln(v) = \frac{1}{1-\alpha} \sum_{i=1,2} \tilde{\eta}_i(\ln z_i + \ln(\tilde{\eta}_i)) + \frac{\alpha}{1-\alpha} \ln(K/v).^{4}$$

This expression mirrors that of the standard one-sector growth model, with aggregate TFP equal to a weighted average of sectoral productivities. For a given capital-output ratio, the pass-through from sectoral productivity to output is governed by expenditure shares, amplified through capital accumulation by the factor $1/(1-\alpha)$.

In the canonical one-sector model, the capital-output ratio is independent of TFP in steady state. Thus, one can hold the marginal product of capital fixed when varying TFP, cleanly separating the contributions of technology and capital to cross-country income differences (Klenow and Rodríguez-Clare, 1997; Hall and Jones, 1999). In our economy, this separation breaks down because the capital-output ratio, and hence the MPK measured in units of GDP, depends on sectoral productivities through the relative price of investment:

$$\text{MPK} \equiv \alpha \frac{v}{K} \prod_j z_j^{\omega_j - \tilde{\eta}_j} \prod_j \frac{\omega_j^{\omega_j}}{\tilde{\eta}_j^{\tilde{\eta}_j}}.$$

Using the steady-state capital-output ratio, GDP is

$$\ln(v) = \sum_j \left(\tilde{\eta}_j + \frac{\alpha\, \omega_j}{1-\alpha}\right) \ln z_j + \text{const.}$$

The marginal product of capital is independent of sectoral productivity only when the GDP and investment aggregators coincide. The dependence of the marginal product of capital on productivity has an important consequence: standard development accounting cannot isolate the direct effect of technology from its effect on capital accumulation. Crucially, the direction in which productivity improvements affect aggregate capital is ambiguous, depending on how intensively a sector's output is used in producing investment versus consumption. When productivity rises in a sector that contributes disproportionately to investment production, i.e.

[4] The term $\ln(\tilde{\eta}_i)$ captures the static allocation of labor.

$\omega_j > \tilde{\eta}_j$, the capital-output ratio increases. When it rises in a sector whose output is used predominantly for consumption, such as services, i.e. $\omega_j < \tilde{\eta}_j$, the capital-output ratio falls.

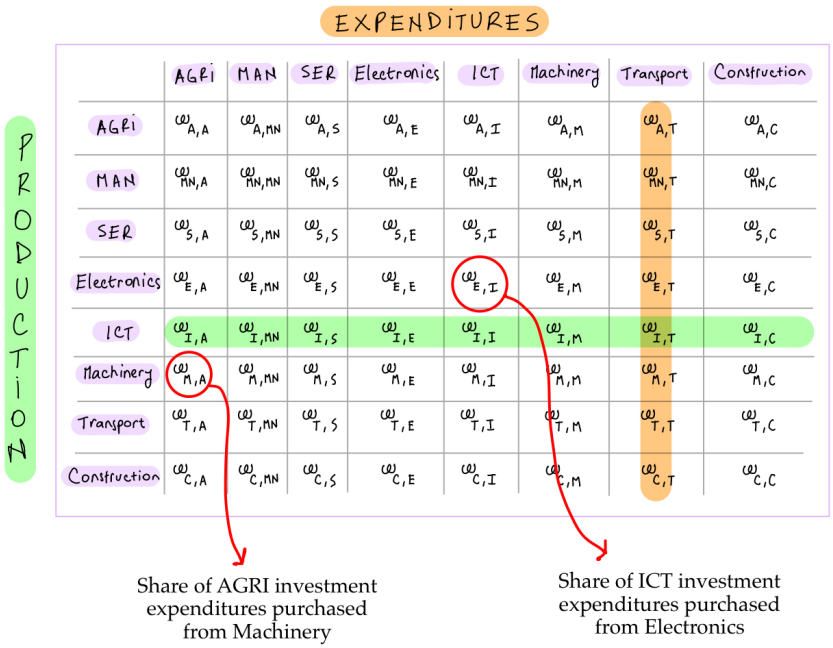
It follows that applying a misspecified one-sector framework, when investment and consumption technologies differ, will either overstate or understate the role of technology depending on the investment intensity of the sectors in which productivity grows. If productivity gains are concentrated in sectors that supply investment goods disproportionately, one-sector accounting will underestimate the contribution of technology and overestimate that of capital. If gains are concentrated in consumption-intensive sectors, the bias runs in the opposite direction. As we argue in Section 6, the cross-country evidence points to the latter case.

3 Building cross-country measures of investment networks

In this section we describe how we construct estimates of the investment technology along the income spectrum, where $\omega_{ij} \neq \omega_{jj}$, and can vary over time. This amounts to estimating expenditure shares in new capital, by capital type and purchasing sector, across countries and over time.

3.1 Methodology

Figure 1: Example: Allocation of ICT Investment Flows



Notes: Illustrative example of the investment network. Columns indicate consuming sectors, while rows indicate production sectors. Entries are expenditure shares by consuming sectors in different investment types, ω.

Figure 1 presents an example of an investment network. Each entry (i, j) records total investment expenditures by column sector j on goods purchased from row sector i. Summing across columns yields total investment production by each row sector; summing across rows yields total investment expenditures by each column sector. For instance, each element of the ICT row shows how much investment the ICT sector supplies to each purchasing sector j, while each element of the agriculture column shows how much investment agriculture purchases from each supplying sector i. Expenditure shares ω_{ij} are obtained by dividing each entry in column j by that column's total investment expenditures, so that column shares sum to one: $\sum_i \omega_{ij} = 1$.

Estimates of investment produced by each sector, that is, the row sums in Figure 1, are readily available from *use tables*, which record how sectoral output is allocated across intermediate and final uses, including consumption and investment. Importantly, our measure of investment flows encompasses both domestically produced and imported equipment; see Section 3.2. For ease of exposition, we refer to these flows as "produced" by a sector, though the category includes imports.

Our contribution is to develop an imputation strategy for determining how much of the investment produced (or imported and allocated to investment) by each sector is purchased by other sectors of the economy. Our approach follows the BEA's procedure for the US wherever it is replicable and wherever the required information is available across countries and over time. Meade *et al.* (2003) describes three stages of this imputation:

1. direct assignment;
2. allocation proportional to sectoral occupational composition;
3. allocation proportional to sectoral capital expenditures (used only for structures).

Direct assignment places all investment in a given category into its most likely use—for example, nuclear plant investment is assigned to the utilities sector. The second method assumes that a sector's occupational mix is informative about the types of capital it employs. For structures, the BEA allocates investment in proportion to sectoral capital expenditures.

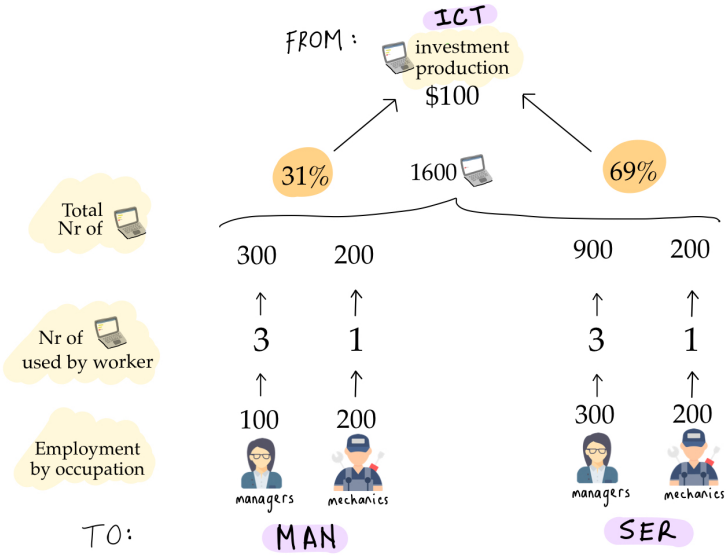
To maximize replicability, we forgo manual assignment of investment categories to user sectors (method 1), as this requires finely disaggregated data that are not consistently available internationally. We adopt method (2) as our benchmark and apply a variant of method (3) when (2) is infeasible; in particular, we allocate expenditures in proportion to each sector's demand for intermediate inputs.[5]

[5]The BEA typically uses micro-data on structures investment to allocate those flows, but such data are not available for our cross-country, cross-sector, and time-series sample.

3.1.1 Allocation of equipment investment flows

We allocate equipment investment flows primarily according to the occupational composition of each purchasing sector and the types of capital those occupations are likely to use.[6]

Figure 2: Example: Allocation of ICT Investment Flows



Notes: Illustrative example of the allocation of investment flows across purchasing sectors, obtained by exploiting the sectoral occupational composition and the equipment intensity across occupations from Caunedo *et al.* (2023a).

Figure 2 illustrates this assignment for ICT investment. Suppose there are three sectors—ICT, manufacturing, and services—and the ICT sector produces $100 of new capital goods (software). Our goal is to determine how much of this $100 is purchased by manufacturing and services respectively. To do so, we use the occupational mix of each purchasing sector together with the relative intensity with which those occupations use software.

In the example, manufacturing and services each employ 200 mechanics; manufacturing also employs 100 managers and services 300 managers. Managers use software three times as intensively as mechanics, so for every dollar assigned to mechanics, three are assigned to managers. These relative-use intensities come from the tool–occupation allocation in Caunedo *et al.* (2023a) (details below). It follows that for the manufacturing sector as a whole, out of every $500 of software, $300 is used by managers and $200 by mechanics. For the services sector, the corresponding allocation out of $1,100 is $900 to managers and $200 to mechanics. Rescaling these relative magnitudes to the $100 of software produced by ICT, we assign 31% to manufacturing and 69% to services.

The software example assumed a one-to-one mapping between equipment types and pro-

[6]The BEA's allocation follows its public documentation; however, the details of how capital is assigned to specific occupations and sectors are not publicly available. In Section 4.5, we validate our methodology by comparing our U.S. estimates with those of the BEA.

ducing sectors. While this holds for some equipment types, a given type of equipment is often produced by several sectors. This information is encoded in "bridge tables" that underlie the national accounts. For example, using the average U.S. bridge table from 2000 to 2018, 78% of computer investment is produced by the electronics sector and 22% by the ICT sector; conversely, 28% of electronics output consists of computers, 34% of communication equipment, and the remainder of other equipment categories.

Our methodology therefore proceeds in three mapping steps: (i) from investment-producing sectors to equipment types, using bridge tables; (ii) from investment-purchasing sectors to equipment types, using the occupational distribution and the tools used on the job; and (iii) combining (i) and (ii) to map investment-producing sectors to investment-purchasing sectors—that is, to construct the investment network.

From investment-producing sectors to equipment types. Bridge tables are not, to the best of our knowledge, available for countries other than the United States. We therefore use the average allocation of sectoral production to equipment types from U.S. bridge tables covering 2000–2018. [7] The dollar value of investment in equipment type e produced by sector i in country c at year t is obtained by multiplying the U.S. bridge-table share b_{ie}^{US} by the total investment produced (or imported and allocated to investment) by sector i in country c at year t, x_{it}^{c} (from use tables):

$$x_{iet}^{c} = x_{it}^{c}\, b_{ie}^{\text{US}}.^{8} \tag{1}$$

From investment-purchasing sectors to equipment types. We assign equipment investment to purchasing sectors using the tools employed within each occupation in the United States, as described by O*NET. Specifically, we implement the methodology of Caunedo *et al.* (2023a), which assigns equipment investment—and therefore equipment stocks—to workers in different occupations.[9] As illustrated in Figure 2, the key identifying assumption is that the relative intensity with which a manager uses a given piece of equipment compared to a mechanic is the same across countries and equal to the U.S. value.[10] The absolute investment allocated to each purchasing sector still varies across countries because ICT investment production differs

[7] As vom Lehn and Winberry (2022) point out, the bridge tables are available since 1997 for NAICS, and between 1987 and 1992 for SIC industry classifications. In any other year, they interpolate values.

[8] A sector's total production allocated to equipment types excludes replacement of used goods and trade margins. When we impute investment flows in other countries—which may include these margins—we effectively distribute them equally across equipment types.

[9] The methodology in Caunedo *et al.* (2023a) crosswalks equipment categories to the tools used within each SOC occupation. We use Dingel and Neiman (2020)'s crosswalk between SOC and ISCO to map these tools to harmonized cross-country occupational definitions.

[10] As a robustness check, we construct country-specific tool intensities within occupations by combining cross-country within-occupation variation in tasks with a projection of tool usage onto task intensity. We can only construct these projections for 19 mostly middle and high income countries. The Online Appendix compares baseline characteristics of the network across assignment rules.

internationally and because the occupational composition of industries varies with the level of development.

We first compute the share of total production of equipment type e purchased by industry j in country c at time t:

$$\gamma_{ejt}^{c} = \frac{\sum_{o} \tau_{e}^{o,\text{US}} \, n_{jot}^{c}}{\sum_{j} \sum_{o} \tau_{e}^{o,\text{US}} \, n_{jot}^{c}}, \tag{2}$$

where n_{jot}^{c} is the number of workers in occupation o and industry j in country c at time t, and $\tau_{e}^{o,\text{US}}$ is the intensity with which a worker in occupation o uses capital type e in the United States. Since γ_{ejt}^{c} distributes a given equipment type across industries, the shares sum to one: $\sum_{j} \gamma_{ejt}^{c} = 1$.

We then estimate the dollar value of investment in equipment type e purchased by sector j in country c at year t as the product of the value of investment in type e produced by sector i (equation 1) and the purchasing share in equation 2:

$$x_{ejt}^{c} = x_{iet}^{c} \, \gamma_{ejt}^{c}. \tag{3}$$

From investment-producing sectors to investment-purchasing sectors. For each purchasing sector j, we aggregate across all equipment types produced by sector i to obtain the total dollar value of investment that j purchases from i:

$$x_{ijt}^{c} = \sum_{e \in \mathcal{E}_i} x_{ejt}^{c}, \quad \text{for } i \in \text{equipment sectors}, \tag{4}$$

where $\mathcal{E}_i$ denotes the set of equipment types produced by sector i.

The investment network. Investment expenditure shares ω_{ijt}^{c}, which constitute the investment network, are defined as:

$$\omega_{ijt}^{c} = \frac{x_{ijt}^{c}}{\sum_{i} x_{ijt}^{c}}. \tag{5}$$

3.1.2 Allocation of non-equipment investment flows

For investment flows originating in the construction, agriculture, nondurables, transportation services, and other services sectors, no information on worker usage of capital is available. We therefore allocate these flows in proportion to each sector's role as a supplier of intermediate inputs to other sectors, drawing on information from input-output tables.

Let $\tilde{\mu}_{ijt}^{c}$ denote the share of total intermediate inputs produced by sector i that are purchased by sector j. For non-equipment sectors, the dollar value of each entry in the investment

table is:

$$x^c_{ijt} = \tilde{\mu}^c_{ijt}\, x^c_{it}, \quad \text{for } i \notin \mathcal{E}_i.$$

Investment expenditure shares are then computed as in equation 5.

3.2 Data Description

Based on the methodology described in the previous section, constructing the investment network requires five pieces of information: (i) sectoral production of investment goods, x^c_{it}; (ii) tools per worker by occupation, $\tau^{o,\text{US}}_e$; (iii) employment by occupation and sector, n^c_{iot}; (iv) a bridge table mapping sector i to equipment type e, b^{US}_{ie}; and (v) the input–output structure, $\tilde{\mu}^c_{ijt}$. Table 5 summarizes the sources, and Appendix Table 6 provides a detailed description.

Investment production by sector and input–output tables. We obtain gross fixed capital formation (GFCF) by sector from the *use tables* that underlie the input–output accounts. The *use tables* report investment production by sector, whether produced domestically or imported. For our baseline investment network in each country, we use total investment production by sector (domestic plus imported).[11] We also construct a domestic investment network for each country based on domestic investment production by sector. For the nine Sub-Saharan African countries in our sample, we use data from Mensah and de Vries (2023). For the remaining countries, we use the World Input–Output Database (WIOD) and the OECD input–output tables. We map equipment production into sectoral investment production using the U.S. bridge tables (as explained before).

Employment by occupation and sector. We use employment by occupation and sector from PIAAC, IPUMS International, and ILOSTAT. For countries with data from all three sources, we favor PIAAC over IPUMS International and ILOSTAT because PIAAC uses more detailed occupational categories.[12] In IPUMS International, the harmonized industry classification does not disaggregate equipment sectors within manufacturing; we instead used the raw datasets which include the entirety of the industry classification available in the labor survey. For IPUMS countries we manually construct crosswalks between the disaggregated (non-harmonized) industries and our 10 sectors. In PIAAC and ILOSTAT, sectors are classified according to ISIC Rev. 4 or ISIC Rev. 3, which we also crosswalk to our 10-sector scheme (see online appendix).

We use 2-digit occupational categories to assign relative intensity of tools by equipment

[11] GFCF flows are reported in nominal currency; we deflate them using output PPP prices from the Penn World Table. To abstract from business-cycle fluctuations, we apply a Hodrick–Prescott (HP) filter to sectoral GFCF flows.

[12] PIAAC measures aggregated at the 1-digit level correlate strongly with IPUMS data; see Caunedo, Keller and Shin (2023b).

type and then aggregate them up to 1-digit to have a common occupational classification across countries. We assign tools to workers in different occupations following the intensity measure in Caunedo *et al.* (2023a).

Country coverage. Our dataset covers 58 countries at different stages of development, with GDP per capita (PPP) ranging from $428 to $81599. For 20 of these countries, we construct time series of investment networks from 1965 to 2014; for the 9 countries in Sub-Saharan Africa, from 1990 to 2019. For the remaining 29 countries, the investment-network time series covers 2000–2014. See Table 6 in the Appendix for a full description.

4 The Investment Network along the Development Spectrum

We start by showing how the composition of sectors that provide investment changes with development. Then, we show how the investment network diversifies with income, and how it differs from the input-output network.

4.1 Investment Network Outdegrees

We begin by documenting each sector's *outdegree*—the row sum of the investment network. In Table 1, we group countries by the World Bank income classification and report the average sectoral outdegree for each group.[13] Several patterns emerge. Construction is the dominant supplier of investment goods across the distribution and follows a hump-shaped relationship with development. Other services—including financial intermediation and real estate—are also major suppliers, but their importance declines with income. The sector with the strongest increase is ICT: high-income economies exhibit outdegrees nearly three times those of low-income economies. Transportation equipment also rises with income, though more moderately. Electronics and transport services display hump-shaped profiles, whereas durables and nondurables fall from low to middle income and then tick up slightly at high income. Finally, Agriculture is a minor supplier in all groups, and its importance diminishes sharply with development.

[13] Country lists appear in Table 8. For cross-country comparisons we use 2005, the year with PPP deflators; results are similar with other reference years or when pooling all country–year observations.

Table 1: Investment Network Outdegrees

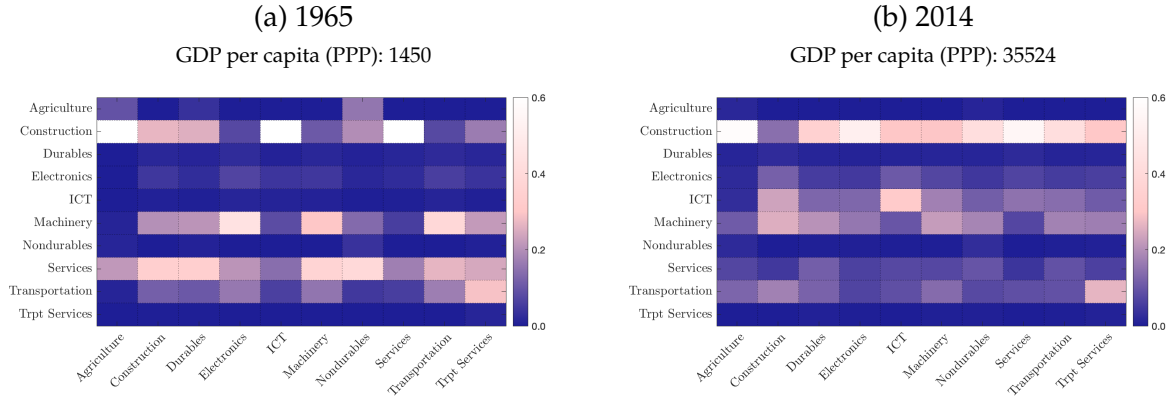
Sector	Low income	Middle income	High income
Agriculture	0.45	0.22	0.13
	(0.06 0.48)	(0.05 0.32)	(0.05 0.21)
Construction	3.18	3.39	2.98
	(2.69 3.89)	(3.07 3.63)	(2.55 3.35)
Durables	0.90	0.39	0.49
	(0.24 0.97)	(0.31 0.48)	(0.30 0.70)
Electronics	0.69	0.97	0.78
	(0.20 0.96)	(0.64 1.13)	(0.57 0.95)
ICT	0.41	0.46	1.21
	(0.01 0.90)	(0.28 0.58)	(0.84 1.38)
Machinery	1.42	1.31	1.29
	(0.36 1.80)	(1.08 1.68)	(0.99 1.64)
Nondurables	0.32	0.16	0.18
	(0.08 0.43)	(0.05 0.26)	(0.14 0.22)
Services	1.75	1.61	1.58
	(1.26 2.38)	(1.28 1.76)	(1.16 1.95)
Transportation	0.72	1.08	1.20
	(0.55 1.01)	(0.92 1.24)	(0.84 1.53)
TrptServices	0.14	0.41	0.16
	(0.00 0.12)	(0.15 0.44)	(0.08 0.18)

Notes: Data for 2005, for low-, middle-, and high-income countries (World Bank classification). Outdegrees are sectoral row sums in the investment network. Entries report means; values in parentheses are the 25th and 75th percentiles. Average GDP per capita (PPP): low 2587; middle 11110; high 35056.

Similar patterns emerge when we examine South Korea's evolution from the low-income group to the high-income group—i.e. from 1965, when GDP per capita (PPP) was 1450, to 2014, when income per capita increased 24-fold. Figure 3 depicts the investment network in both years. Comparing panels (a) and (b), the same patterns as in Table 1 hold: construction is a prominent supplier of investment goods at all stages of development; agriculture is a minor supplier; the role of ICT rises with income; and the importance of other services declines with development.

The investment network of Korea also becomes more diversified as the country grows richer, in the sense that the bundle of investment goods that a sector uses becomes more even across producing industries. This diversification is not specific to South Korea, it also characterizes the cross-section of countries along the development spectrum.

Figure 3: South Korea: Investment Network Over Time



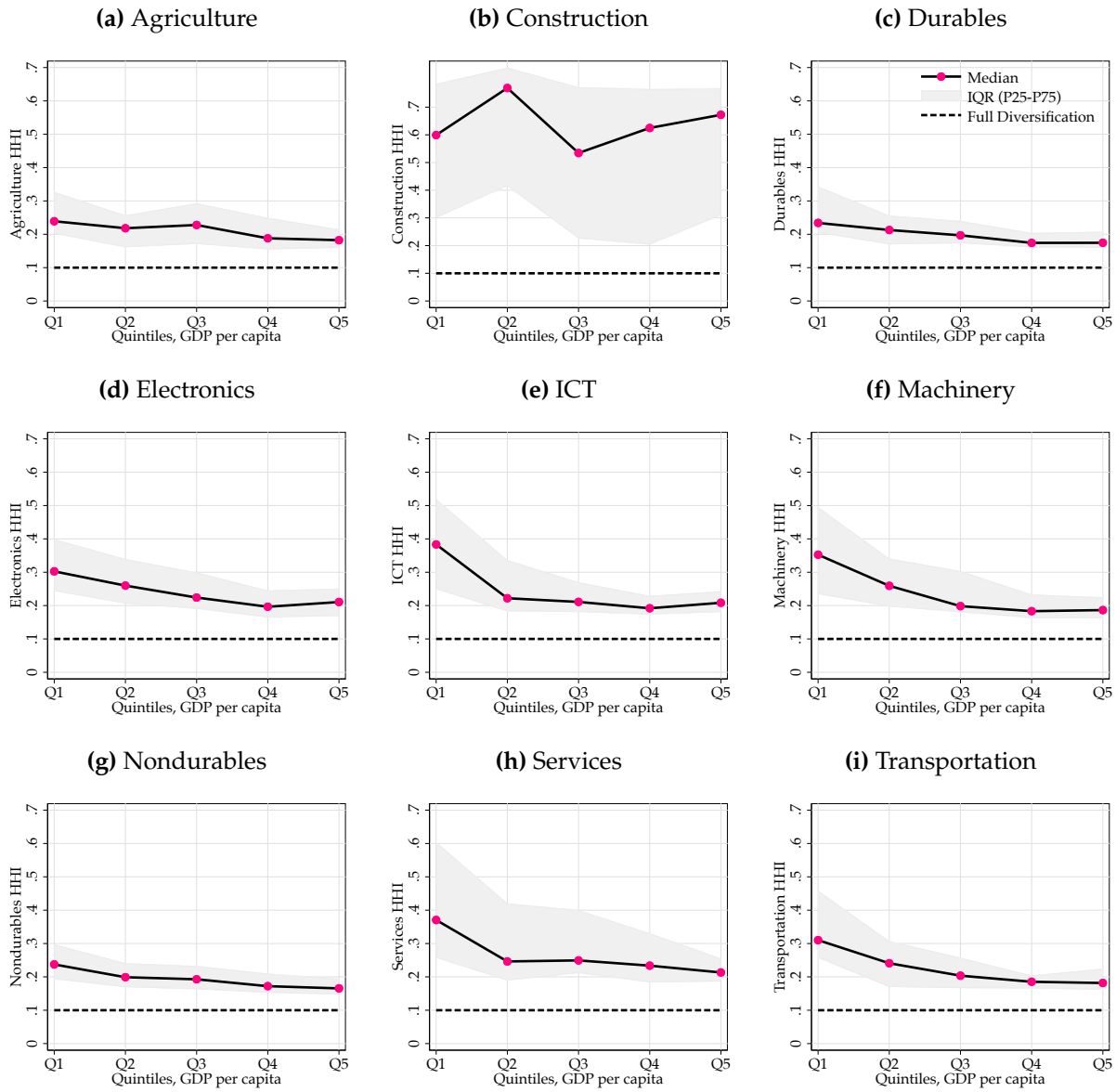
Notes: Estimates of South Korea's investment network over time. Each cell reports the investment-expenditure share of the *column* sector, as defined in Figure 1.

4.2 Investment Network Diversification

Figure 4 plots the Herfindahl–Hirschman Index (HHI) for each (column) sector that uses capital across quintiles of income per capita. Lower values indicate that a sector has a more diversified bundle. How does this structure change with development? For most sectors, the bundle of capital goods becomes more diversified as countries grow richer, moving closer to a fully diversified benchmark. The exception is construction, where concentration rises (about 56% from low to high income), largely because construction increasingly sources capital goods from its own sector. Diversification gains are most pronounced in capital-intensive activities—electronics, ICT, machinery, services, and transportation. In ICT, for instance, the HHI falls by roughly one-half (from about 0.45 in low-income to 0.19 in high-income countries), approaching the 0.10 value associated with full diversification.[14]

[14]Table 10 reports, for each sector, the mean and the 25th and 75th percentiles of the HHI for the low-, middle-, and high-income country groups.

Figure 4: Investment Network HHI



Notes: Data for 2005. The concentration index is the Herfindahl–Hirschman Index (HHI), computed as the sum of squared shares in each *column* of the investment network. Panels report medians and interquartile ranges of sectoral HHIs across quintiles of GDP per capita. The dashed line denotes the equal-split (full-diversification) benchmark, where investment is sourced equally from all capital types.

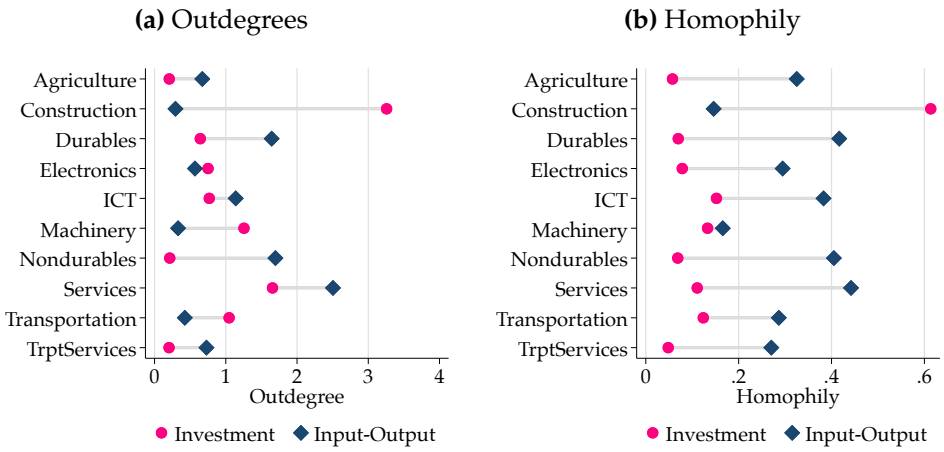
4.3 Investment Network vs. Input–Output Network

While the characteristics of the investment network were not known prior to this paper, there is considerably more evidence about the features of the input–output structure (see, for example, Fadinger *et al.* (2022)). Figure 5 documents differences between these two networks for the average country, as measured by (a) outdegrees, and (b) homophily. Moreover, we report these statistics in the Appendix for countries at different stages of development.[15]

Outdegrees. Several patterns stand out. Agriculture is a major supplier of intermediate

[15]Figure 14, reports the same statistics for low, middle, and high income country groups.

Figure 5: Investment Network vs. Input–Output Network



Notes: Data for 2005, averaged across countries. Circles denote the investment network; diamonds denote the input–output network. Panel (a) reports sectoral outdegrees (row sums); panel (b) reports sectoral homophily (diagonal elements of each network).

inputs—especially in low-income countries—but a minor provider of investment. Construction has the highest investment outdegree yet is among the least important intermediate input suppliers; its investment outdegree falls slightly with income, while its input–output outdegree rises. ICT outdegrees increase with development in both networks, with a large gap in low-income countries that narrows at high income. Electronics and transport services are hump-shaped, and transportation is predominantly a supplier of investment goods. Other services (including transport services) and nondurable manufacturing mainly supply intermediates rather than investment.s

Homophily. Comparing diagonal entries across the two matrices reveals a clear pattern: sectors heavily self-supply *intermediate* inputs (high input–output diagonals) but rely more on other sectors to source *investment* goods (lower investment-network diagonals). Two exceptions stand out: construction, which contributes little to its own intermediates but is a major supplier of investment goods to itself, and machinery, which shows similar homophily in both networks. These patterns hold across the income distribution and within countries over time.[16]

4.4 Sources of Cross-Country Variation in Investment Networks

The investment network for each country is constructed from two ingredients: sectoral investment flows and the occupational composition of the labor force. A natural question is how much of the cross-country variation in investment networks documented above is driven by differences in investment flows versus differences in employment composition. To answer this,

[16] Figure 15 compares Korea in 2014 and shows that diagonal loadings are systematically lower in the investment network than in the input–output network.

we construct two counterfactual investment networks for each country.

The first counterfactual replaces each country's occupational employment shares with those of the United States, keeping sectoral investment (GFCF) flows at their country-own levels. By equalizing the employment channel across countries, any remaining variation in the investment network is driven entirely by differences in investment flows. The second counterfactual does the reverse: it imposes U.S. GFCF levels on all countries while retaining each country's own occupational composition, so that residual variation reflects the employment channel alone. We assess the contribution of each channel both within countries—by computing the average distance between the baseline and counterfactual networks—and across countries, by comparing the cross-country variance of each element ω^c_{ijt} under each counterfactual to its baseline variance.

Differences in investment flows account for the bulk of cross-country heterogeneity in investment networks. When GFCF is equalized to U.S. levels, only 9% of the baseline cross-country variance in the typical element of $\mathbf{\Omega}$ remains—meaning that differences in the occupation composition explains little of the observed variation and investment flow differences explain roughly 91%. When occupational shares are equalized to U.S. levels instead, cross-country variance rises to about 130% of baseline, confirming that employment heterogeneity does not drive network dispersion—if anything, it slightly dampens it. Within countries, the GFCF channel plays a more important role for less developed economies relative to richer ones, consistent with their investment structures being furthest from the U.S. benchmark; the employment channel is broadly similar across all stages of development. Appendix Figure 16 and Table 15 report the full results.

4.5 Validation with other sources

Comparison with US Investment Networks. We compare our U.S. investment-network estimates to those constructed by vom Lehn and Winberry (2022) ("*VLW*"), which are based on the BEA capital-flows tables. We use 2012 as the primary reference year, as this is when data on employment by sector and occupation were collected by the PIAAC survey.[17] To control for potential disparities between BEA and WIOD in sectoral GFCF flows, we apply our methodology using the same sectoral investment-production estimates as in vom Lehn and Winberry (2022) (estimates are also robust to using WIOD investment flows). We then aggregate their 41-sector investment network to our 10 sectors, consistent with the ISIC Rev. 4 crosswalk in

[17] The Online Appendix shows that the patterns observed in 2012 are largely consistent with those from 1972 and 1992. Since occupational composition has changed over time, it is surprising that the assignment works relatively well 20 and 40 years earlier.

Table 7.

Appendix Table 16 compares estimates of (a) the sectoral outdegrees of the investment network—a measure of each sector's relevance as an investment provider to other sectors—and (b) the homophily of the investment network—a measure of each sector's relevance as a provider of investment to itself. The *outdegree* is the row sum of the investment network, and *homophily* summarizes the diagonal elements of the matrix. Comparing the estimates in the second and third columns of Table 16, panels (a) and (b), we find that our methodology aligns closely with vom Lehn and Winberry (2022), especially considering that the only common inputs are sectoral investment flows.

For further comparison, we regress the elements of our investment-network estimates on those from VLW and report the mean squared error (MSE) as a measure of fit. The MSEs are 0.005 for 2012, 0.008 for 1992, and 0.017 for 1972, indicating relatively small differences between our estimates and theirs.

Comparison with Chile's Investment Networks. A potential concern with ours and the BEA estimates of the investment network is that the allocation of investment to its uses depends on the occupational composition of the sector demanding equipment, rather than on information on actual investment in each sector. This data is rarely available. Chile provides a rare opportunity because of the availability of data on the universe of firm-to-firm transactions through the VAT system. We use investment networks estimates from Gillmore *et al.* (2025) based (on domestic) firm-to-firm transactions for comparison.

We compare our estimates with theirs through summary statistics of the network, including outdeegres and homophily, see Table 17. We find that for all sectors, except for ICT and Services our estimate of domestic investment networks align very well with their estimates from firm-to-firm transactions. The difference for services is that their methodology overestimates the effects of wholesale trade which are intermediaries of imported good and therefore show up as "traded" component in our series but domestic in theirs. The differences for ICT arises from disparities in the capitalization of certain investments, in particular software.

Having shown the key characteristics of the investment network along the development spectrum, we are now ready to analyze its importance for the development process. To do so, we build a model that rationalizes the distinctive features of the network, and its change with development, through differences in investment elasticities and sectoral productivities.

5 A Model of the Investment Network and Economic Development

The model is an open-economy version of Long and Plosser (1983), augmented to include a sectoral investment aggregator. Production, intermediate inputs and investment are aggregated with general constant-returns-to-scale technologies; the Cobb–Douglas and CES specifications used in the quantitative analysis are special cases. Differences in aggregate GDP per capita across countries are associated with disparities in investment and input–output networks. The investment network adjusts endogenously to sectoral productivity trends.

5.1 Environment

The economy consists of N sectors that combine capital, labor, and intermediate inputs to produce output. Gross output in sector n combines value added, v_{nt}, and a bundle of intermediate inputs, m_{nt}, with a constant-returns-to-scale (CRS) technology,

$$y_{nt} = G_n(v_{nt}, m_{nt}).$$

Sectoral value added is

$$v_{nt} = z_{nt}\, V_n(k_{nt}, l_{nt}),$$

where V_n is CRS, and depends on sectoral TFP z_{nt} and on capital and labor inputs k_{nt} and l_{nt}. Intermediate inputs are aggregated with a CRS technology, $m_{nt} = M_n(m_{1nt}, \ldots, m_{Nnt})$, where m_{int} denotes inputs from sector i used in sector n. All technologies are increasing, concave and differentiable. Let the share of value added in gross output be $\gamma_{nt} \equiv P^v_{nt} v_{nt} / (p_{nt} y_{nt})$, the capital share in value added be $\alpha_{nt} \equiv R_{nt} k_{nt} / (P^v_{nt} v_{nt})$, and the share of inputs from sector i in intermediate spending be $\mu_{int} \equiv p_{it} m_{int} / (\mathcal{M}_{nt} m_{nt})$. These expenditure shares are endogenous given prices: p_{nt} is the price of good n, P^v_{nt} and $\mathcal{M}_{nt}$ are the prices of value added and of the intermediate bundle, and R_{nt} is the rental rate of capital. The input shares μ_{int} define the input–output matrix M_t. Each column of M_t describes the composition of intermediate inputs used by a sector, with $\sum_i \mu_{int} = 1$, while each row captures the importance of a sector as a supplier of intermediates to the rest of the economy.

The capital stock used in each sector follows the law of motion

$$k_{n,t+1} = x_{nt} + (1 - \delta_n) k_{nt},$$

where x_{nt} is a composite investment good for sector n. Investment is produced using inputs sourced from multiple sectors. Let χ_{int} denote investment goods from sector i used in the

production of capital for sector n at time t. The investment aggregator is a CRS technology,

$$x_{nt} = X_n(\chi_{1nt}, \ldots, \chi_{Nnt}). \tag{6}$$

The allocation of investment expenditures across sectors is summarized by the investment network Ω_t, whose typical element is $\omega_{int} \equiv \frac{p_{it}^{\chi}\chi_{int}}{p_{nt}^{x}x_{nt}}$, where p_{it}^{χ} is the price of investment goods from sector i, defined below, and p_{nt}^{x} is the price of the investment composite. Investment shares vary with relative prices. Rows of Ω_t capture the production of investment goods by each sector, while columns describe the composition of investment used by each sector. Since the investment technology exhibits constant returns to scale, column shares sum to one, $\sum_{i=1}^{N} \frac{p_{it}^{\chi}\chi_{int}}{p_{nt}^{x}x_{nt}} = 1$.

Moreover, investment goods from sector i used by other sectors can be sourced domestically or imported: $\chi_{int} = \left(\frac{\chi_{int}^{d}}{1-\phi_i}\right)^{1-\phi_i}\left(\frac{\chi_{int}^{f}}{\phi_i}\right)^{\phi_i}$, where χ_{int}^{d} and χ_{int}^{f} denote, respectively, domestic and imported investment goods from sector i used by sector n, and ϕ_i is the import share in investment goods from sector i.[18] Domestic goods from sector i trade at price p_{it}, and imported goods at the exogenous world price p_{it}^{f}, so that the price of investment goods from sector i is $p_{it}^{\chi} = p_{it}^{1-\phi_i}(p_{it}^{f})^{\phi_i}$.

Output in each sector can be allocated to final goods c_{nt}, intermediate inputs, or domestic investment:

$$y_{nt} = c_{nt} + \sum_i m_{nit} + \sum_i \chi_{nit}^{d}.$$

Aggregate output Y_t is defined implicitly by the non-homothetic CES aggregator of Comin, Lashkari and Mestieri (2021),

$$\sum_{n=1}^{N} \theta_n^{\frac{1}{\sigma}} (c_{nt})^{\frac{\sigma-1}{\sigma}} Y_t^{\frac{\epsilon_n-1}{\sigma}} = Y_t^{\frac{\sigma-1}{\sigma}}, \qquad \theta_n > 0,$$

where θ_n governs the loading of final goods from sector n, σ is the elasticity of substitution across goods, and ϵ_n is the sector-specific income elasticity.[19] Final goods can be allocated to exports, e_t, or to consumption, C_t, by the representative household.[20] The household has logarithmic utility, $U(C_t) = \ln C_t$, and discounts the future at rate $\beta \in (0,1)$. Final goods

[18] Most results follow with a general CRS aggregator for foreign and domestic investment but the definition of the terms of trade is more convoluted. We keep this unitary elasticity aggregator for simplicity.

[19] The induced expenditure shares take the form $s_{nt} \equiv \frac{p_{nt}c_{nt}}{p_{Yt}Y_t} = \frac{\theta_n p_{nt}^{1-\sigma} Y_t^{\epsilon_n-1}}{\sum_{j=1}^{N}\theta_j p_{jt}^{1-\sigma} Y_t^{\epsilon_j-1}}$. When $\epsilon_n = 1$ for all n, the aggregator collapses to standard CES and shares depend only on relative prices.

[20] Our findings are robust to allowing for separate aggregators for exports and consumption. When the price of consumption is defined in units of exports, the results are unchanged. Alternatively, exports can be specified as a constant fraction of final output Y_t.

market clearing therefore implies

$$Y_t = C_t + e_t.$$

We consider a small open economy that exports final goods in exchange for investment goods of different types, as in Jones (2011). Net exports are defined as the difference between the value of exports and the value of imported investment goods:

$$NX_t = p_{Yt} e_t - \sum_{i=1}^{N} p_{it}^f \chi_{it}^f, \qquad \chi_{it}^f \equiv \sum_{n=1}^{N} \chi_{int}^f,$$

and trade is balanced, $NX_t = 0$. The terms of trade for investment goods of type i are defined as the ratio of the price of value added (the GDP deflator p_t, defined in Section 5.3) to the price of imported goods from sector i, $\mathcal{T}_{it} \equiv \frac{p_t}{p_{it}^f}$.

5.2 Competitive Equilibrium

Definition 1 (Competitive Equilibrium). *Given initial capital stocks $\{k_{j0}\}_{j=1}^N$, sequences of sectoral productivities $\{z_{jt}\}_{j,t}$ and world prices of imported investment goods $\{p_{jt}^f\}_{j,t}$, and model parameters, a* competitive equilibrium *is a collection of sequences of allocations*

$$\{C_t, Y_t, e_t, c_{jt}, x_{jt}, \chi_{ijt}, \chi_{ijt}^d, \chi_{ijt}^f, m_{jt}, m_{ijt}, l_{jt}, y_{jt}, k_{j,t+1}\}_{t \geq 0,\, i,j}$$

and prices $\{p_{jt}, p_{jt}^\chi, p_{jt}^x, p_{Yt}, W_t, R_{jt}, \mathcal{M}_{jt}, r_t\}_{t \geq 0,\, j}$ such that, taking prices as given, (i) households choose consumption and sectoral investment to maximize lifetime utility subject to their budget constraint; (ii) firms in each sector choose labor, capital, and materials to maximize profits subject to the production technology; (iii) capital evolves according to the law of motion; and (iv) all markets clear and trade is balanced. Here W_t is the wage, R_{jt} the rental rate of capital in sector j, $\mathcal{M}_{jt}$ the price index of intermediate inputs of sector j, and r_t the interest rate. We take the final good as the numeraire, $p_{Yt} = 1$.

5.3 Aggregate Growth Path

Definition *An aggregate growth path (AGP) is an allocation such that aggregate output, consumption, and wages grow at constant rates, and where the economy's interest rate is constant.*

Such an allocation imposes no restrictions on the sectoral allocation of resources, including capital and investment. When capital-labor ratios are not equalized, as it is the case in our economy, sectoral employment reallocation that is consistent with structural change prevents a constant interest rate. In other words, an aggregate growth path may only exist in the limit, as in Acemoglu and Guerrieri (2008). Since productivity from different sectors affects the rate

of capital accumulation in the economy, at this level of generality in the value added aggregator, an AGP may not exist: we impose a unitary elasticity aggregator of capital and labor to overcome Uzawa (1961). In addition, we restrict the remaining aggregators to be CES rather than general CRS because the AGP relies on how cost shares behave when relative input prices grow at different constant rates.[21] All these restrictions apply only from some date $\bar{t}$ onwards, so technologies along the transition are general CRS. With an elasticity above one, the share of the input whose price grows slowest converges to one; with an elasticity below one, the share of the input whose price grows fastest does.

Assumption 1.

(i) Sectoral productivity grows at constant (possibly heterogeneous) rates, g_{z_j} for all j.

(ii) World prices of imported equipment are constant in units of the exported final good, $q_{it} \equiv p_{it}^f / p_{Yt} = q_i$ for all i and t.[22]

(iii) From date $\bar{t}$ onwards,

(a) value added is Cobb–Douglas in capital and labor $v_{nt} = z_{nt}(k_{nt}/\alpha_n)^{\alpha_n}(l_{nt}/(1-\alpha_n))^{1-\alpha_n}$, with capital shares α_n;

(b) the gross-output and intermediate aggregators of each sector n are CES, with elasticities of substitution e_n^G and e_n^M, and a limiting network $(\Gamma^\infty, M^\infty)$ exists, as defined in Appendix A.1;

(c) the investment aggregators are CES, with weights $\tilde{\omega}_{in} > 0$ and elasticities of substitution ι_n either above one in every sector, $\iota_n > 1$ for all n (gross substitutes), or below one in every sector, $\iota_n < 1$ for all n (gross complements).

Assumption 1(iii)(c) guarantees that a single sector dominates the investment composite of all destination sectors. This limiting sector need not be the same limiting sector that overtake consumption (which will be determined by income effects too).

Units and notation. Along the AGP, the investment composite of every sector converges to a single dominant supplier, which we denote by n, and final expenditure concentrates on a single sector, which we denote by n^*. The final good is the numeraire, $p_{Yt} = 1$. Let $\hat{g}_w \equiv g_W - g_{p_n^x}$

[21] For a general CRS aggregator, shares need not converge: if the elasticity of substitution oscillates around one as relative prices diverge, the input that dominates the aggregator can switch infinitely often, and the price of the aggregate does not grow at a constant rate. A CES aggregator rules this out.

[22] World prices are exogenous to the small open economy, and under this assumption they are common across countries and have no trend relative to the exported good. Hsieh and Klenow (2007) assume that investment goods are fully tradable at a common world price, normalized to one, and Lee (1995) takes the price of imported capital goods as given to a developing economy. Appendix A.3 discusses two alternatives: world prices that decline at a constant rate relative to the exported good; and world prices that move with domestic prices, which reproduce the growth rates of the closed economy.

denote the growth rate of the wage relative to the price of the investment composite of the dominant supplier. It does not depend on the choice of numeraire, and it summarizes how capital accumulation raises wages: it is the growth rate of the quantity of the dominant investment good that a unit of labor can buy. Let $\Gamma^\infty = \mathrm{diag}\{\gamma_n^\infty\}$ and M^∞ denote the limiting value-added shares and input–output matrix of Assumption 1(iii)(b). Let $\Psi \equiv (I - (I - \Gamma^\infty)M^{\infty\prime})^{-1}$ be the Leontief inverse of the limiting network, and define the parameters

$$\mathcal{A} \equiv \Psi\Gamma^\infty, \qquad \mathcal{B} \equiv \mathcal{A}(I - \alpha)\mathbf{1}, \qquad \kappa_{n^*} \equiv \frac{\sigma - 1}{\sigma - \epsilon_{n^*}},$$

where $\mathcal{A}$ is an $N \times N$ matrix, $\mathcal{B}$ is an $N \times 1$ vector, and κ_{n^*} is a scalar, with ϵ_{n^*} the income elasticity of the dominant final good n^*. We write $\mathcal{A}g_z$ for the $N \times 1$ vector with typical element $(\mathcal{A}g_z)_j = \sum_h \Psi_{jh}\gamma_h^\infty g_{z_h}$. Finally, let $p_t\nu_t$ denote nominal aggregate value added, $\zeta_{jt} \equiv p_{jt}(c_{jt} + \chi_{jt}^d)/(p_t\nu_t)$ the share of good j in final uses and domestic investment, where $\chi_{jt}^d \equiv \sum_i \chi_{jit}^d$. The price of value added, p_t, is the Divisia GDP deflator with weights ζ_{jt}.

Proposition 1. *Under Assumption 1, there exists an asymptotic AGP where aggregate GDP and wages grow at constant rates and sectoral employment converges. Along the AGP, in units of the final good:*

(a) Wages. *The wage grows at the rate of output, $g_W = g_Y$, and relative to the price of the dominant investment good at*

$$\hat{g}_w = \frac{(1 - \phi_n)\,(\mathcal{A}g_z)_n + \phi_n\,\kappa_{n^*}(\mathcal{A}g_z)_{n^*}}{(1 - \phi_n)\,\mathcal{B}_n + \phi_n\big[1 - \kappa_{n^*}(1 - \mathcal{B}_{n^*})\big]}, \tag{7}$$

where $(\mathcal{A}g_z)_j$ and $\mathcal{B}_j$ denote the j-th elements of the $N \times 1$ vectors $\mathcal{A}g_z$ and $\mathcal{B}$, so every term in (7) is a scalar, provided the denominator is positive, which holds whenever $\epsilon_{n^*} = 1$.

(b) Prices. *The prices of the investment composites and rental rates of all sectors grow at $g_{p_j^x} = g_{R_j} = g_Y - \hat{g}_w \;\forall\, j$, so investment becomes cheaper relative to final output whenever $\hat{g}_w > g_Y$. Output prices grow at $g_{p_j} = g_Y - (1 - \mathcal{B}_j)\,\hat{g}_w - (\mathcal{A}g_z)_j \;\forall\, j$. World prices of equipment are constant, $g_{p_i^f} = 0$, and the price of investment goods supplied by sector i grows at*

$$g_{p_i^x} = (1 - \phi_i)\,\big[g_Y - (1 - \mathcal{B}_i)\hat{g}_w - (\mathcal{A}g_z)_i\big], \tag{8}$$

which equals $g_Y - \hat{g}_w$ for the dominant supplier $i = n$.

(c) Capital, value added and the interest rate. *Sectoral capital grows at*

$$g_{k_j} = g_W - g_{p_j^x} + g_{l_j} = \hat{g}_w + g_{l_j} \qquad \forall\, j, \tag{9}$$

where $g_{l_j} = 0$ for every sector with a positive limiting employment share. The interest rate is

constant, $1 + r = (1 + g_Y)/\beta$*, whenever the growth rate of nominal aggregate output is constant.*

(d) Output. *Aggregate output grows at*

$$g_Y = \kappa_{n^*}\left(g_W - g_{p_{n^*}}\right) = \kappa_{n^*}\left[(1 - \mathcal{B}_{n^*})\hat{g}_w + (\mathcal{A}g_z)_{n^*}\right], \tag{10}$$

where the dominant final sector is $n^* \equiv \arg\max_j \Lambda_j$*, with* $\Lambda_j \equiv (1-\sigma)\, g_{p_j} + (\epsilon_j - 1)\, g_Y$*, and satisfies* $(1-\sigma)\left[(\mathcal{A}g_z)_j - (\mathcal{A}g_z)_{n^*}\right] + (1-\sigma)(\mathcal{B}_{n^*} - \mathcal{B}_j)\hat{g}_w + (\epsilon_{n^*} - \epsilon_j)\, g_Y \;>\; 0 \qquad \forall\, j \neq n^*$ *with* $\hat{g}_w$ *and* g_Y *given by* (7) *and* (10)*.*

(e) Real value added. *Nominal value added deflated by the price of value added* p_t *grows at*

$$g_v = \left(g_W - g_{p_{n^*}}\right) - \zeta_n^{\infty}\left(g_{p_n} - g_{p_{n^*}}\right), \tag{11}$$

where $g_W - g_{p_{n^*}} = g_{c_{n^*}}$ *is the growth rate of the dominant final good. Thus,* $g_v = g_{c_{n^*}}$ *when* $n = n^*$ *and* $g_{c_{n^*}} = g_Y$ *when* $\epsilon_{n^*} = 1$*.*

Proofs to all propositions can be found in Appendix A.4.

The typical element of $\mathcal{A}$, $\mathcal{A}_{jh} = \Psi_{jh}\gamma_h^{\infty}$, is the share of the value added of sector h in the cost of producing good j, directly and through intermediates. Because Ψ is the Leontief inverse of the cost (price) system, and the columns of M^{∞} sum to one, each row of $\mathcal{A}$ sums to one, $\mathcal{A}\mathbf{1} = \mathbf{1}$: all costs are ultimately payments to value added. The typical element of $\mathcal{B}$, $\mathcal{B}_j = \sum_h \Psi_{jh}\gamma_h^{\infty}(1 - \alpha_h) \in (0, 1]$, is the labor share embodied in good j, directly and through intermediates, and $(\mathcal{A}g_z)_j$ is the TFP growth embodied in good j, which we refer to as the network-adjusted TFP growth of sector j. In the closed economy, $\phi_j = 0$ for all j, (7) reduces to

$$\hat{g}_w = \varpi_n, \qquad \varpi_j \equiv \frac{(\mathcal{A}g_z)_j}{\mathcal{B}_j} = \frac{\sum_h \Psi_{jh}\gamma_h^{\infty}\, g_{z_h}}{\sum_h \Psi_{jh}\gamma_h^{\infty}(1 - \alpha_h)},$$

so the wage, relative to the dominant investment good, grows at the network-adjusted TFP growth of the dominant investment supplier per unit of its embodied labor share.[23]

Because the investment composite of every sector converges to the same supplier, investment prices grow at a common rate, and capital grows at the wage deflated by the investment price, $\hat{g}_w$, in all sectors that retain employment in the limit. Step 5 of the proof of Proposition 1 in Appendix A.1 formally shows this result.

Real value added and the weight of investment. Let $s_{jt}^I \equiv \frac{p_{jt}^x x_{jt}}{p_t v_t}$ be the investment expenditure of sector j relative to aggregate value added, and s^I the limit along the AGP of

[23] In a one-sector economy without intermediate inputs, $\Psi = 1$ and $\gamma = 1$, so $\mathcal{A} = 1$, $\mathcal{B} = 1 - \alpha$ and $\varpi = g_z/(1 - \alpha)$, the growth rate of wages along a balanced growth path with capital accumulation.

the aggregate investment-to-value-added ratio, $\sum_j s^I_{jt}$. Since every sector buys its investment from n, a share $1-\phi_n$ of it domestically, the limiting weight of the dominant investment sector in the price of value added is $\zeta^\infty_n = (1-\phi_n)\, s^I$ if $n \neq n^*$ and 1 otherwise. Along the AGP, the investment expenditure of each sector is a constant share of its capital income, $p^x_j x_j = \rho_j R_j k_j$: with a constant interest rate and capital growing at g_{k_j}, the user cost of capital gives $\rho_j = \beta(g_{k_j} + \delta_j)/(1 + \hat{g}_w - \beta(1-\delta_j))$. With common capital shares and depreciation rates, $\alpha_j = \alpha$ and $\delta_j = \delta$, every sector that retains employment accumulates capital at $\hat{g}_w$ and has the same investment rate, $\rho = \beta(\hat{g}_w + \delta)/(1 + \hat{g}_w - \beta(1-\delta))$.

Whether $n = n^*$ is governed by the relative strength of two forces: the *price effect*, $(1-\sigma)(g_{p_n} - g_{p_j})$, which favors sectors with fast-falling prices when $\sigma > 1$ and sectors with fast-rising prices when $\sigma < 1$; and the *income effect*, $(\epsilon_n - \epsilon_j) g_Y$, which favors sectors with high income elasticity when $g_Y > 0$. Corollary 1 in Appendix A.1 gives conditions under which the two forces align, so that $n = n^*$.

We next characterize equilibrium allocations at each date, both along the transition and on the AGP.

5.4 Equilibrium Characterization

Domar weights. Let $\tilde{\eta}_n \equiv \frac{p_n y_n}{pv}$ be the Domar weight of sector n, where pv is nominal aggregate value added, $\tilde{\zeta}_n \equiv \frac{p_n c_n}{pv}$ the share of value added allocated to final uses of good n (consumption and exports), and $\zeta_n \equiv \tilde{\zeta}_n + \frac{p_n \chi^d_n}{pv}$ the share of value added allocated to final uses and domestic investment of good n, which we refer to as the final-demand share, where $\chi^d_n \equiv \sum_i \chi^d_{ni}$. By goods market clearing, nominal value added equals the value of final goods plus domestic investment, $pv = \sum_n p_n y_n - \sum_n \sum_i p_n m_{ni} = \sum_n p_n (c_n + \chi^d_n)$, so $\sum_n \zeta_n = 1$.As in Section 5.3, the price of value added p_t is the Divisia GDP deflator with weights ζ_{nt}, and real value added v_t is nominal value added deflated by p_t, so that $p_t v_t = \sum_n \left(p_{nt} y_{nt} - \mathcal{M}_{nt} m_{nt}\right)$ and $d \ln v_t = \sum_n \zeta_{nt}\, d \ln(c_{nt} + \chi^d_{nt})$.

Let the (diagonal) matrices of value-added shares $\mathbf{\Gamma}_t = \text{diag}\{\gamma_{nt}\}$, capital shares in value added $\boldsymbol{\alpha}_t = \text{diag}\{\alpha_{nt}\}$, import shares of investment goods $\boldsymbol{\phi} = \text{diag}\{\phi_n\}$, and investment rates $\boldsymbol{\rho}_t = \text{diag}\{\rho_{nt}\}$, where

$$\rho_{nt} \equiv \frac{\xi_{nt}}{r^x_{nt} + \delta_n}, \qquad \xi_{nt} \equiv \frac{x_{nt}}{k_{nt}}, \qquad r^x_{nt} \equiv \frac{R_{nt}}{p^x_{nt}} - \delta_n,$$

so that $\xi_{nt} = g_{k_n, t+1} + \delta_n$ is the ratio of investment to capital and r^x_{nt} the net return on capital in units of the investment good. Capital installed at t is rented at $t+1$, and under no arbitrage

the return equals the own interest rate, $1 + r^x_{n,t+1} = (1 + r_{t+1})\, p^x_{nt} / p^x_{n,t+1}$, where $1 + r_{t+1}$ is the gross interest rate in units of the numeraire, so $r^x_{nt} + \delta_n$ is the user cost of capital per unit of equipment value. As before, the term ρ_{nt} scales investment expenditure relative to capital income, $p^x_{nt} x_{nt} = \rho_{nt}\, \alpha_{nt} \gamma_{nt}\, p_{nt} y_{nt}$. In what follows, I corresponds to the identity matrix, and $\boldsymbol{M}_t$ is the input–output matrix with typical element μ_{int}.

Proposition 2. *The equilibrium Domar weights satisfy*

$$\tilde{\boldsymbol{\eta}}_t := \left[I - (I - \boldsymbol{\phi}) \boldsymbol{\Omega}_t \boldsymbol{\Gamma}_t \boldsymbol{\alpha}_t \boldsymbol{\rho}_t - \boldsymbol{M}_t (I - \boldsymbol{\Gamma}_t) \right]^{-1} \tilde{\boldsymbol{\zeta}}_t \tag{12}$$

In vector form,

$$\tilde{\eta}_{nt} = \tilde{\zeta}_{nt} + \sum_{i=1}^{N} \rho_{it}\, \alpha_{it} \gamma_{it} (1 - \phi_n) \omega_{ni,t} \tilde{\eta}_{it} + \sum_{i=1}^{N} (1 - \gamma_{it}) \mu_{ni,t} \tilde{\eta}_{it}.$$

Note that the role of the investment network for the Domar weight scales with the importance of domestic investment across sectors, $(1 - \phi_n) \in (0, 1)$.[24] In our economy, equilibrium Domar weights are nontrivial functions of the investment rates along the transition to the AGP. Thus, Domar weights are functions of the entire path of future Domar weights.

Proposition 3 (Welfare). *The welfare elasticity to sectoral TFP satisfies*

$$\frac{d \ln C_t}{d \ln z_{nt}} = \frac{p_t \nu_t}{C_t} \tilde{\eta}_{nt} \gamma_{nt},$$

where $d \ln C_t$ is the consumption-equivalent change in welfare at date t, $p_t \nu_t / C_t$ is the ratio of nominal value added to consumption, both in units of the final good, and $\tilde{\eta}_{nt}$ is the equilibrium Domar weight of sector n.[25]

The proof of this result follows from the envelope theorem. Value-added shares scale productivity levels because productivity enters sectoral production with value-added elasticity γ_{nt}. Proposition 3 is consistent with results in vom Lehn and Winberry (2022) for short-run fluctuations: Domar weights are scaled by the ratio between GDP and final consumption. Along the AGP, this ratio is a constant. In the transition, the welfare effect of sectoral productivity shocks can be amplified or dampened depending on the relative allocation of value added between consumption and investment uses.

[24] This is driven by our modeling of exports from a composite of final uses. If all equipment is tradable, then roundabout effects from the investment network will appear directly in GDP through trade amplification; see Φ in Proposition 4. If instead we model each sector's allocation of gross output to exports, these flows appear in the equilibrium representation of the Domar weight.

[25] The term γ_{nt} rescales productivity because value added enters gross output with elasticity γ_{nt}, the equilibrium value-added share.

For the level representation in Proposition 4, we measure real value added and the terms of trade with the geometric index $\bar{p}_t \equiv \prod_n p_{nt}^{\zeta_{nt}}$.[26]

Proposition 4 (Aggregate value added). *The equilibrium level of GDP in the economy satisfies*

$$\ln(\nu_t) = \Phi_t \boldsymbol{\eta}_t \Gamma_t \Big[\ln \boldsymbol{z}_t + \alpha_t \Omega_t' \boldsymbol{\phi} \ln(\boldsymbol{\mathcal{T}}_t) + \alpha_t \big(\ln \boldsymbol{\rho}_t - \ln \boldsymbol{\xi}_t \big) \Big] + \Phi_t \varrho_t,$$

or, equivalently,

$$\ln(\nu_t) = \underbrace{\tilde{\boldsymbol{\eta}}_t' \Gamma_t \ln \boldsymbol{z}_t}_{\text{on impact}} + \underbrace{\Big[\overbrace{\tilde{\boldsymbol{\eta}}_t' \Gamma_t \alpha_t \Omega_t' (I - \boldsymbol{\phi}) \Xi_t \Gamma_t}^{\text{domestic } \Delta k} + \overbrace{(\Phi_t - 1) \boldsymbol{\eta}_t \Gamma_t}^{\text{imported } \Delta k} \Big] \ln \boldsymbol{z}_t + \overbrace{\Phi_t \boldsymbol{\eta}_t \Gamma_t \alpha_t \Omega_t' \boldsymbol{\phi} \ln(\boldsymbol{\mathcal{T}}_t)}^{\text{TOT}}}_{\text{capital deepening}}$$
$$+ \underbrace{\Phi_t \boldsymbol{\eta}_t \Gamma_t \alpha_t \big(\ln \boldsymbol{\rho}_t - \ln \boldsymbol{\xi}_t \big)}_{\text{capital deepening: gross rate of return}} + \underbrace{\Phi_t \varrho_t}_{\text{curvature and labor share}},$$

where $\Phi_t \boldsymbol{\eta}_t \Gamma_t$ is the row vector of sectoral influence; $\Phi_t \equiv \big(1 - \boldsymbol{\eta}_t \Gamma_t \alpha_t \boldsymbol{\Omega}_t' \boldsymbol{\phi} \mathbf{1}\big)^{-1}$ is a scalar amplification factor from tradable investment; $\ln(\boldsymbol{\mathcal{T}}_t)$ is the vector of log terms of trade with typical element $\ln \mathcal{T}_{it}$; $\ln \boldsymbol{\rho}_t - \ln \boldsymbol{\xi}_t$ is the vector with typical element $-\ln(R_{nt}/p_{nt}^x)$, minus the log gross rate of return on capital; and ϱ_t is a scalar term that collects the curvature of the production, intermediate and investment aggregators and the labor share. [27] *The vector $\boldsymbol{\eta}_t$ is the product of final-demand shares, $\boldsymbol{\zeta}_t'$, and an adjusted Leontief inverse $\Xi_t \equiv (I - \Gamma_t \alpha_t \Omega_t' (I - \boldsymbol{\phi}) - (I - \Gamma_t) M_t')^{-1}$, i.e., $\boldsymbol{\eta}_t \equiv \boldsymbol{\zeta}_t' \Xi_t$.*

Influence and Domar weights. The two expressions in Proposition 4 coincide because of the following identity between the influence of a sector and its Domar weight:

$$\boldsymbol{\eta}_t = \tilde{\boldsymbol{\eta}}_t' + \tilde{\boldsymbol{\eta}}_t' \Gamma_t \alpha_t \Omega_t' (I - \boldsymbol{\phi}) \Xi_t. \tag{13}$$

In words, the cost-based influence of a sector, $\boldsymbol{\eta}_t$, equals its Domar weight, $\tilde{\boldsymbol{\eta}}_t$, plus the productivity it builds into the capital of other sectors through the investment network. The Domar weight governs the response of real value added on impact, as in Hulten (1978); the second term, which exists only because capital is produced, is realized as capital accumulates.

The second expression splits the influence of sectoral productivity, $\Phi_t \boldsymbol{\eta}_t \Gamma_t$ (as in Acemoglu *et al.*, 2012), into two parts. The first is the on-impact influence, the Domar weight times the value-added share, $\tilde{\boldsymbol{\eta}}_t' \Gamma_t$, which governs the response of real value added while capital stocks

[26] The Divisia deflator defines growth rates, and the growth rates of the geometric index coincide with those of the Divisia deflator whenever shares are constant.

[27] When the gross-output, value-added and intermediate aggregators are Cobb–Douglas, so that Γ, α and M are constant, $\varrho_t = \boldsymbol{\eta}_t \Gamma \alpha \boldsymbol{\kappa}_t^x - \boldsymbol{\eta}_t \Gamma (I - \alpha) \mathbf{1} \ln s_t^L$, where $s_t^L = \mathbf{1}' \Gamma (I - \alpha) \tilde{\boldsymbol{\eta}}_t$ is the labor share of value added, and $\boldsymbol{\kappa}_t^x$ captures the curvature of the CES investment aggregator, with typical element $\kappa_{nt}^x = \frac{1}{1-\iota_n} \sum_i \omega_{int} \ln(\omega_{int}/\tilde{\omega}_{in})$, which is zero when $\iota_n = 1$. The general expression, with curvature terms for every aggregator, is given in the proof in Appendix A.4.

are fixed (Proposition **??**). The second is the capital-deepening effect: productivity lowers the cost of the goods each sector invests in, which raises capital stocks. It has a domestic component, $\tilde{\boldsymbol{\eta}}_t'\Gamma_t\alpha_t\Omega_t'(I-\boldsymbol{\phi})\Xi_t\Gamma_t$, which traces productivity through the cost of domestically produced investment goods, weighted by each sector's capital income share, and a trade component, $(\Phi_t-1)\boldsymbol{\eta}_t\Gamma_t$, through which higher productivity raises exports and finances imported equipment. The terms of trade operate through capital deepening, a channel that Kehoe and Ruhl (2008) identify.[28] As $\phi \to 0$, the economy's dependence on tradable investment vanishes, and Propositions 2 and 4 collapse to their closed-economy counterparts, which we derive in the Online Appendix.

The two capital-deepening terms play different roles. The first, the long-run component, is the capital deepening that the current level of productivity would support once capital has fully adjusted to it, evaluated at date-t shares. The second, the transition component, records how far actual capital is from that level. By the definition of ρ_{nt}, it depends only on the gross rate of return on capital, $\ln\rho_{nt} - \ln\xi_{nt} = -\ln\left(R_{nt}/p^x_{nt}\right)$: a high gross rate of return signals that capital is scarce relative to what productivity supports. After a permanent increase in productivity at date t, capital is predetermined, so the gross rate of return rises to exactly compensate for the higher marginal product of capital (Proposition **??**). As capital accumulates, the gross rate of return falls back and the capital-deepening effect builds up gradually, at a pace set by investment and depreciation in each sector (Proposition **??**(a)). Along the AGP, the gross rate of return, $R_n/p^x_n = r^x + \delta_n$, and the investment to capital ratios, $\xi_{nt} = g_{k_n} + \delta_n$, depend only on growth rates, depreciation and discounting, which a permanent change in the level of productivity does not alter.

The remaining term, ϱ_t, collects the curvature of the aggregators, which is zero under Cobb–Douglas, and a labor-share term that depends on the distribution of sectors' shares of the aggregate wage bill.

5.5 On-impact vs. dynamic responses.

Long-run responses: capital as a marked-up input. By the definition of ρ_{nt}, capital income and investment expenditure in sector n satisfy

$$R_{nt}k_{nt} = \mu^k_{nt}\, p^x_{nt}x_{nt}, \qquad \mu^k_{nt} \equiv \frac{1}{\rho_{nt}}. \tag{14}$$

[28] When all imported equipment faces a common price, $\mathcal{T}_{it} = \mathcal{T}_t$ for all i, the terms-of-trade term equals $\boldsymbol{\eta}_t\Gamma_t\alpha_t\Omega_t'\boldsymbol{\phi}\mathbf{1}\ln\mathcal{T}_t = (1-\Phi_t^{-1})\ln\mathcal{T}_t$, and $\ln(\nu_t) = \Phi_t\left[\boldsymbol{\eta}_t\Gamma_t\left(\ln z_t + \alpha_t(\ln\boldsymbol{\rho}_t - \ln\boldsymbol{\xi}_t)\right) + \varrho_t\right] + (\Phi_t-1)\ln\mathcal{T}_t$. The elasticity of GDP to the terms of trade is then exactly the amplification from tradable investment, $\Phi_t - 1$.

So the capital of sector n is an input produced from the investment composite and sold to sector n at a "wedge/markup" μ^k_{nt} over its cost, $\frac{p^x_{nt}}{R_{nt}}$. This representation in the long-run, when ρ_{nt} is a constant, is what Baqaee and Malmberg (2025) exploit to show that balanced growth paths of multisector economies are equilibria of static economies in which capital services are produced from investment goods and sold at an as-if markup equal to the ratio of capital income to investment expenditure. Thus, the tools for economies with markups of Baqaee and Farhi (2020) apply.

Let $\boldsymbol{s}_t$ be the vector of final expenditure shares s_{nt} and define

$$\boldsymbol{\lambda}_t \equiv \left[I-(I-\boldsymbol{\phi})\boldsymbol{\Omega}_t\Gamma_t\alpha_t\boldsymbol{\rho}_t-\boldsymbol{M}_t(I-\Gamma_t)\right]^{-1}\boldsymbol{s}_t, \qquad \tilde{\boldsymbol{\lambda}}_t \equiv \left[I-(I-\boldsymbol{\phi})\boldsymbol{\Omega}_t\Gamma_t\alpha_t-\boldsymbol{M}_t(I-\Gamma_t)\right]^{-1}\boldsymbol{s}_t = \Xi'_t\boldsymbol{s}_t.$$

By Proposition 2, $\boldsymbol{\lambda}_t = \frac{p_t v_t}{p_{Y_t}Y_t}\tilde{\boldsymbol{\eta}}_t$ are the Domar weights in units of final output. They are revenue-based: they count capital at the income it earns.[29] The weights $\tilde{\boldsymbol{\lambda}}_t$ are cost-based: they count capital at the cost of the investment that produces it. The two sets of weights are related by

$$\tilde{\boldsymbol{\lambda}}_t - \boldsymbol{\lambda}_t = \Xi'_t(I-\boldsymbol{\phi})\boldsymbol{\Omega}_t\Gamma_t\alpha_t(I-\boldsymbol{\rho}_t)\,\boldsymbol{\lambda}_t.$$

The right-hand side traces a wedge from the sectors that use capital back to the sectors that produce it. For each sector i, $\lambda_{it}\gamma_{it}\alpha_{it}$ is its capital income, $R_{it}k_{it}$, relative to consumption, and by (14) a fraction $1-\rho_{it}$ of that income is not spent on investment. Along the AGP, $1-\rho_i = (r^x - g_{k_i})/(r^x+\delta_i)$, which is zero when $r^x = g_{k_i}$: capital income is then exactly reinvested, as at the golden rule of Phelps (1961). The investment network assigns this unspent income to the goods that make up each sector's capital, with $\omega_{ji,t}$ the share of good j in the investment composite of sector i. The factor $I-\boldsymbol{\phi}$ keeps the domestically produced part, since the imported part accrues to foreign suppliers, and Ξ'_t passes it further upstream, to the suppliers of intermediate inputs and of capital to the producers of these goods. When $\rho_{it}=1$ for all i, revenue and cost measures of capital coincide and so do the two sets of weights.

Element by element,

$$\tilde{\lambda}_{jt} - \lambda_{jt} = \sum_l \Xi_{lj,t}\,(1-\phi_l)\sum_i \omega_{li,t}\,\gamma_{it}\alpha_{it}\lambda_{it}\,(1-\rho_{it}).$$

Hence $\tilde{\boldsymbol{\lambda}}_t \geq \boldsymbol{\lambda}_t$ whenever $\rho_{it}\leq 1$ for all i, and the gap of sector j is large when three conditions hold. First, j is in the supply chain of domestically produced investment goods: $\Xi_{lj,t}$ measures how much the cost of good l depends on productivity in j, whether j produces l itself, supplies inputs to its producers, or supplies their capital. Second, these goods are bought as investment

[29] In the closed economy $\lambda_{nt}\gamma_{nt}$ is the welfare elasticity of Proposition 3.

by large, capital-intensive sectors, with high capital income $\gamma_{it}\alpha_{it}\lambda_{it}$. Third, those sectors are far from the golden rule, with a large $1-\rho_{it}$, because their capital grows slowly relative to the interest rate. A sector that sells only for consumption or exports, or whose output is invested in sectors close to the golden rule, has a gap close to zero.

Proposition 5 (Long-run responses). *Let Assumption 1 hold, and compare two economies on the AGP that differ only in the levels of sectoral productivity and of world equipment prices and share the same limiting network. As $t \to \infty$:*

(a) capital markups are constant, $\mu_n^k = 1/\rho_n$, with

$$\rho_n = \frac{\beta\,(g_{k_n} + \delta_n)}{1 + \hat{g}_w - \beta(1-\delta_n)} < 1,$$

and $\rho_n \to 1$ as $\beta \to 1$ in every sector with a positive limiting employment share;

(b) the difference in real value added is

$$\Delta \ln v = \Phi\Big[\boldsymbol{\eta}\Gamma\,\Delta \ln \boldsymbol{z} + \boldsymbol{\eta}\Gamma\alpha\Omega'\boldsymbol{\phi}\,\Delta \ln \boldsymbol{\mathcal{T}}\Big],$$

where Φ, $\boldsymbol{\eta}$, Γ and Ω take their limiting values.

(c) in the closed economy, $\boldsymbol{\phi} = 0$, the difference in consumption is

$$\Delta \ln C = \kappa_{n^*}\,\tilde{\boldsymbol{\lambda}}'\Gamma\,\Delta \ln \boldsymbol{z} = \kappa_{n^*}\Big[(\mathcal{A}\Delta \ln \boldsymbol{z})_{n^*} + (1-\mathcal{B}_{n^*})\frac{(\mathcal{A}\Delta \ln \boldsymbol{z})_n}{\mathcal{B}_n}\Big],$$

where $\tilde{\boldsymbol{\lambda}} = \Xi'\mathbf{e}_{n^*}$ are the limiting cost-based weights, and $\tilde{\boldsymbol{\lambda}} \geq \boldsymbol{\lambda}$, with equality when $\rho_n = 1$ for all n.

Along the AGP. Part (a) shows that the markup on capital is constant along the AGP and pinned by the growth of capital, depreciation and discounting. Comparing AGPs is therefore a long-run comparative static in their sense, with constant markups: productivity is aggregated with cost-based weights, which treat capital as a produced input and trace the productivity of each sector through the investment network into the cost of capital of every other sector. Part (c) is the technological effect of Baqaee and Malmberg (2025): because the two economies share the same limiting network, labor shares coincide and their reallocation effect, which works through changes in labor shares, is zero. The non-homothetic demand system adds the factor κ_{n^*}, which converts real consumption expenditure into the index C.

Along the transition. Away from the AGP, ρ_{nt} depends on the growth of capital between t and $t+1$ and on the return on capital, which includes capital gains on equipment, so markups

vary over time and the revenue-based weights, i.e. the Domar weights, depend on the entire future path.

Since $\ln \rho_{nt} - \ln \xi_{nt} = -\ln(r^x_{nt} + \delta_n)$, real value added at each date loads on the log user costs weighted by cost-based influence $-\boldsymbol{\eta}_t \Gamma_t \alpha_t \ln(\boldsymbol{r}^x_t + \boldsymbol{\delta})$. Writing $\ln(r^x_{nt} + \delta_n) = \ln \mu^k_{nt} + \ln \xi_{nt}$, this is the misallocation term of Baqaee and Farhi (2020): log markups weighted by cost-based influence, together with investment rates. After a permanent shock at date t, the path between the two AGP is governed by the dynamics of the markups, which depend on how fast each sector accumulates capital and therefore on the investment network.

6 Income Accounting

The goal of this exercise is to quantify how much of the observed differences in income per capita between poor and rich economies can be explained by the differences in the investment network. Such an exercise requires inferring sectorial and cross-country productivity differences to fully rationalize the observed disparities in aggregate value added across countries, and sectorial value added across sectors.

6.1 Inferring productivity-like shifters

For each country, we consider a vector of sectoral "productivity-like" shifters,

$$\boldsymbol{a} \equiv \ln \boldsymbol{z} + \boldsymbol{\alpha}\, \boldsymbol{\Omega}' \boldsymbol{\phi} \ln \boldsymbol{\mathcal{T}},$$

which depends on sectoral TFP and the terms of trade. We infer this object residually using the structural restrictions of Proposition 4,

$$\boldsymbol{a} = \left(\boldsymbol{\Phi} \boldsymbol{\Xi} \boldsymbol{\Gamma}\right)^{-1} \ln(\boldsymbol{\nu}),$$

where $\ln(\boldsymbol{\nu})$ is the vector of logs of sectoral real value added.[30] This inversion is exact whenever the data are generated by the model of Section 5.

Since sectors combine capital, labor, and materials with heterogeneous elasticities, the productivity residuals a_j and $a_{j'}$ for $j \neq j'$ stem from two different composite inputs, and are not directly comparable. In other words, a_j is output per unit of composite input in sector j, but

[30] Recall that in the model, the terms of trade operate like a sectoral TFP shock, scaled by the capital share and the sector's reliance on imported equipment, $\alpha_n \sum_i \omega_{in} \phi_i$. Separating TFP from the terms of trade would require constructing equipment-specific terms of trade.

these composite inputs are different in different sectors. [31] In what follows, we never compare disparities in a_j across sectors within a country directly.

However, our income-accounting exercise (Section 6) requires comparing at minimum a_j for some baseline sector j *across* countries. In other words, to be able to say what is the contribution of productivity to income disparities, one needs that a_j^c and a_j^{US} are expressed in the same units. This in turn requires that curvature in sector j's technology be *common* across countries. It is sufficient that sector j's log value-added technology takes the separable, Hicks-neutral form

$$\log v_j^c = \log z_j^c + h_j\big(\log k_j^c, \log l_j^c, \log m_j^c\big), \tag{15}$$

where $h_j(\cdot)$ is a function common to every country c and z_j^c is a country-specific Hicks-neutral shifter entering additively (in logs), separably from h_j. Under competitive factor markets, each country's realized cost shares α_j^c equal h_j's own elasticities evaluated at that country's factor ratios; these need not coincide across countries, and generically will not, because countries face different relative factor prices r^c/w^c. If one further assumes that h_j is quadratic in logs (translog), or well approximated by a translog function, then the difference in Tornqvist output and input indexes identifies these residuals, Diewert (1976). Under this condition, $\log z_j^{c_1} - \log z_j^{c_2}$ is comparable across countries and productivity residuals satisfy transitivity.[32]

6.2 The accounting exercise

We are now ready to conduct income accounting. First, sectoral expenditure shares capture the direct impact of a sector's productivity on aggregate value added. Second, sectoral productivity also has an indirect effect through investment and intermediate-input linkages. Both the direct and indirect effects are captured by the vector η, and are further amplified through trade of investment goods by Φ; together they form the sectoral influence vector $\Phi\eta\Gamma$. A useful feature of Proposition 4 is that it expresses the direct and indirect effects of sectoral productivity in terms of sufficient statistics that can be measured directly using available sectoral data together with our new measures of the investment network across countries. We then conduct a series of counterfactual exercises to assess the role of the investment network in income disparities,

[31] Appendix Figure 17 illustrates this point graphically: isoquants from a common technology are radial rescaling of one another, so the Hicks-neutral gap between them is the same along any ray; isoquants from technologies with different curvature cross, so no single rescaling reconciles them everywhere. Thus effects of differences in curvature and differences in levels are confounded.

[32] Common curvature across countries, with only the Hicks-neutral level z_j^c varying, is the standard maintained assumption throughout the development-accounting literature (e.g. Hall and Jones 1999, Klenow and Rodríguez-Clare 1997); Boppart, Kiernan, Krusell and Malmberg (2023) operationalize this assumption by projecting sectoral variables onto a continuous development index and allowing for $\sigma \neq 1$ and assuming a common technology across countries, separable from the Hicks neutral term.

Table 2: Development Accounting

Exercise	Income Variance
Baseline	1.00
Investment Links Only	0.42
Investment Links Only + US TFP	0.41
Investment Links Only + α avg.	0.62
Domestic (no trade)	1.04
Ω = Identity	1.26
Ω = VA Shares	1.11
Ω = VA Shares, TFP (VA model)	4.23
Baseline Ω, TFP (VA model)	1.16

and its interaction with sectoral productivity, the share of capital expenditures in value added, and international trade.

To ensure cross-country comparability, we conduct the accounting for 2005, a year in which we have PPP sectoral prices, which we use to convert nominal sectoral value added into real units in the WIOD sample. For countries not in WIOD, we use GDP PPP price deflators from the Penn World Table (PWT). We provide a full description of the data sources used in the accounting in Appendix B.2.

We normalize the observed income variance across countries to one as a benchmark, and report the results of the accounting exercise in Table 2. We begin by quantifying the contribution of observed differences in the investment network to cross-country income differences through a counterfactual economy in which the indirect impact of sectoral productivity operates solely through investment linkages (*Investment Links Only* in Table 2), abstracting from intermediate input linkages (i.e., we set $\ln(\boldsymbol{\nu}) \equiv \Phi\tilde{\zeta}' \left(I - \Gamma\alpha(I-\phi)\Omega\right)^{-1}\Gamma\boldsymbol{a}$ and recompute income variances). We find that investment links account for 39% of the observed income variance.

Since the role of the network in aggregate output is mediated by sectoral productivity, we further isolate its contribution by setting sectoral productivities equal to those of the US for all countries (*Investment Links Only + US TFP* in Table 2). The residuals from the structural expression for value added are comparable across countries if one assumes a common technology per sector across countries, as in 15. Countries may still differ in their allocations through disparities in the relative prices of inputs. After imposing the same sectoral productivity in all countries, investment linkages account for 32% of the observed income disparities, implying that sectoral influence is responsible for 82% of the investment network's overall contribution. The remaining contribution reflects the fact that poor countries tend to have low productivity precisely in the sectors whose output is used intensively as investment.

The capital expenditure share in value added, α, also mediates the role of the investment

network and varies with the stage of development (see Appendix Figure 18). When we set α equal to the cross-country average (*Investment Links Only + avg. α* in Table 2), the contribution of investment linkages to income disparities rises to 54%. This implies that observed differences in α dampen income disparities stemming from differences in the investment network. Imposing the average α on all countries, which is higher than that in poor economies, magnifies the role of investment composition for these economies. [33]

In the remaining counterfactuals, we incorporate the interaction with international trade and intermediate input linkages. First, we quantify the role of capital goods trade (*Domestic (no trade)* in Table 2). We shut down the trade amplification term, Φ, and recompute the investment network using only domestic sectoral investment; $\ln(\boldsymbol{\nu}) \equiv \tilde{\zeta}' \left(I - \Gamma\alpha\Omega^{DOM} - (I-\Gamma)M\right)^{-1}\Gamma\boldsymbol{a}$.

Eliminating trade in capital goods increases income variance by 31%, suggesting that trade in capital goods generates income convergence. Average GDP per capita also declines by 60% in our sample (Appendix Figure 19), reflecting the substantial cost of closing these economies to capital goods imports.

Next, we quantify the amplification of investment linkages and their interaction with intermediate linkages by shutting down investment linkages entirely (Ω = *Identity* in Table 2). Moreover, guided by the insights from our simple economy in Section 2, we replace the investment network with a matrix whose columns are identical and equal to sectoral aggregate value added shares (Ω = *VA shares* in Table 2). In both scenarios, income disparities increase by 39%, implying that investment network heterogeneity actually reduces income disparities.

6.3 Sectoral influence

Before moving to general equilibrium, it is useful to assess the differential role of sectors in driving income differences as summarized by sectoral influence, Proposition 4.

$$\Phi\boldsymbol{\eta}\Gamma \equiv \underbrace{\Phi}_{\substack{\text{trade}\\ \text{amplification}}} \quad \underbrace{\boldsymbol{\zeta}'}_{\substack{\text{final-demand}\\ \text{shares}}} \quad \underbrace{\left(I - \Gamma\alpha\Omega'(I-\boldsymbol{\phi}) - (I-\Gamma)M'\right)^{-1}}_{\substack{\text{augmented}\\ \text{Leontief inverse } \Xi}}\Gamma, \tag{16}$$

The influence vector characterizes how changes in productivity in a given sector contribute to aggregate value added. It is disciplined by the sector's direct contribution to value added and its indirect contribution through investment and intermediate-input linkages, and it is amplified by trade via the imported fraction of investment goods. In the model, $\boldsymbol{\zeta}$ collects final-demand shares, the share of value added allocated to final uses and domestic investment of

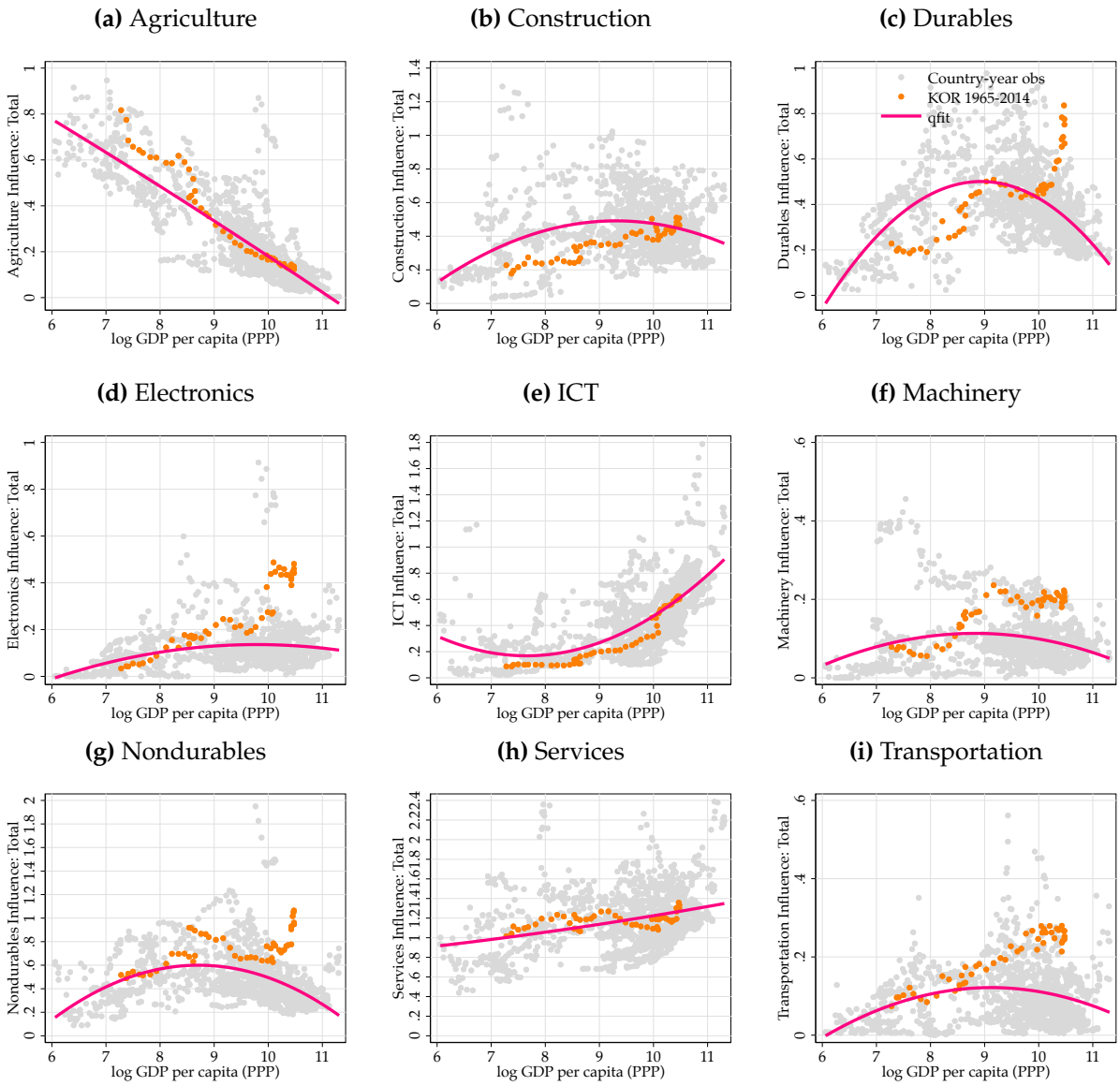
[33] As Appendix Figure 18 shows, α exhibits a hump-shaped relationship with development.

each good. In the data, we proxy ζ with sectoral value-added shares (Appendix B.2). Figure 6 shows sectoral influence for country-year observations using our full sample. We highlight the development path of South Korea from 1965 to 2014, which aligns surprisingly well with the fitted average across the sample.

Some empirical patterns emerge. A 1% increase in the productivity of the construction sector translates into a 0.13% increase in GDP among low income countries. This influence almost doubles among middle income countries. Sectoral influence for durable goods follows a similar trajectory and magnitude as that of the construction sector despite its lesser role as a producer of investment to other sectors. Sectoral influence for electronics, machinery and transportation is smaller in magnitude along the development spectrum.

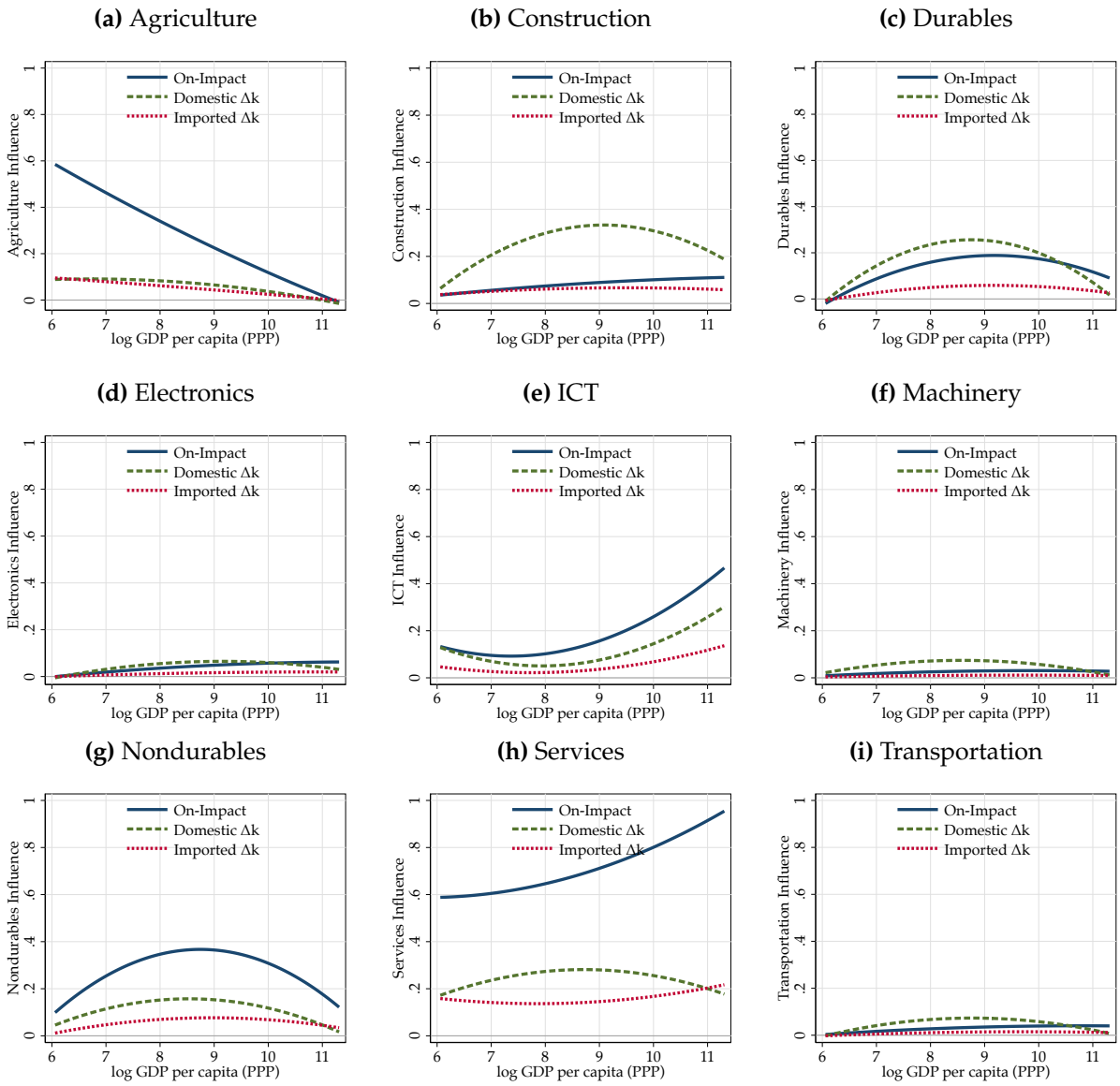
In some sectors, the importance of productivity gains changes substantially as countries develop. The influence of agriculture declines sharply with development, from 0.3% on average in low-income countries to 0.04% in high-income countries. Moreover, a 1% productivity improvement in ICT is linked to a 0.12% increase in GDP per capita in low-income countries, but a much larger 0.34% increase in high-income countries. This likely reflects complementarities and linkages with the rest of the economy, which we explore further in out equilibrium counterfactuals. The influence of other (non-ICT) services deserves special mention, as they exhibit the highest level of influence across countries at all stages of development. Their contribution to aggregate GDP rises with development—from 0.6% to 0.7%—despite a reduction in their role as providers of investment inputs as countries become richer, documented in Table 1.

Figure 6: Sectoral Influence Along Development



Notes: Data for all country-years in the sample. sectoral influence is defined as $\Phi\eta\Gamma$ in Proposition 4, and measures the change in aggregate value added following a change in sectoral productivity. Dots represent country–year observations; South Korea (1965–2014) is highlighted.

Figure 7: Decomposition of Sectoral Influence Along Development



Notes: Data for all country-years in the sample. The figure decomposes sectoral influence, $\Phi\boldsymbol{\eta}\Gamma$ in Proposition 4, into the on-impact component, $\tilde{\boldsymbol{\eta}}'\Gamma$, the domestic capital-deepening component, $\tilde{\boldsymbol{\eta}}'\Gamma\alpha\Omega'(I-\boldsymbol{\phi})\Xi\Gamma$, and the imported capital-deepening component, $(\Phi-1)\boldsymbol{\eta}\Gamma$.

One concern with pooling country–year observations across income levels is that the technologies available fifty years ago differ from those available to countries catching up more recently. Our new data also allow further exploration of how investment paths vary across development trajectories. As a first step, in Figure 20 we highlight the paths of India and China, two countries that have experienced relative rapid growth since 2000s, when the technologies available for production were arguably different than those available in the 1960s. Even for these cases, the path of sectoral influence—especially for newer technologies such as ICT—tracks the pattern predicted by their income levels.[34]

[34]One exception is other services, where sectoral influence lies below the average path for China and above it for

A second concern is that these patterns may reflect standard structural-transformation forces—namely, the reallocation of sectoral value-added shares (ζ) as countries develop. To isolate the role of the investment network from this reallocation channel, we recompute sectoral influence from equation (16) after imposing U.S. value-added shares on every country in the sample. The results are reported in Appendix Figure 21. For the main investment-providing sectors—Construction, Durables, Machinery, Electronics, and Transportation—the influence pattern along development is essentially unchanged, indicating that their role is driven by the structure of the investment network rather than by reallocation in ζ. By contrast, the slope flattens substantially for agriculture (a sector used mainly in consumption), and more mildly for non-ICT services. For ICT, influence rises at low income levels, but the positive slope with development is preserved.

Mapping back to the illustrative example in Section 2, when the value-added and investment aggregators coincide, the marginal product of capital does not depend on productivity and we are back in the canonical one-sector model. Appendix Figure 22 shows how the difference between the investment and value-added bundles matters for the shape and magnitude of sectoral influence: we recompute equation (16) after setting every column of Ω equal to the value-added share vector of that country-year. Consumption-intensive sectors such as agriculture and nondurables keep roughly the same influence, and services gains a steeper slope with development. The equipment-heavy investment sectors, by contrast, lose influence in this counterfactual—most starkly construction and durable goods. ICT keeps its slope with development, since value-added shares are still ICT-intensive, but the counterfactual understates ICT influence at both low and high income levels.

Because value-added shares and the investment network are themselves equilibrium objects that respond to sectoral productivity, a complete decomposition requires a general equilibrium analysis, to which we turn next.

7 The role of the investment network in general equilibrium

We study the transition dynamics of the model economy, calibrated to South Korea over the period 1965–2014. South Korea is an ideal setting to study transition dynamics: over 1965–2014 its GDP per capita grew roughly twentyfold, while its investment and value-added composition underwent a structural transformation consistent with the cross-country patterns documented in Section 4 and in Figure 6. With the calibrated model, we run a series of counterfactual exercises. We first study the role of relative price changes in shaping how the investment network

India.

and sectoral influence evolve along the development path. We then measure the contribution of the investment network, and of sectoral productivity growth, to aggregate output and (measured) aggregate capital, comparing the baseline to three benchmark economies: (i) one in which the investment network composition equals the sectoral value-added composition; (ii) one without investment network linkages; and (iii) one in which the productivity growth of individual sectors is shut down, one sector at a time. To ease computation, we solve an economy with five sectors: Agriculture, Manufacturing, Construction, ICT, and Services.

Table 3 summarizes the calibrated parameter values. We discipline the elasticity of substitution in the investment aggregator, ι_s, by matching the key coefficients of a panel regression of log relative investment expenditure shares on log relative prices, using South Korean data from 1965 to 2014. Given ι_s, we choose the CES loadings $\tilde{\omega}_{in}$ so that the model matches two features of the observed investment network: the sectoral composition of aggregate investment and the odds ratios of the bilateral investment shares.[35] The input–output network $\mu_{in,t}$, a Cobb-Douglas aggregator in the model, is interpolated linearly between its observed 1965 and 2014 values. Sectoral TFP growth rates and levels are chosen to match the path of relative prices in this time horizon,[36] while its growth rate in the manufacturing sector (the base sector) disciplines the levels of aggregate GDP per capita change observed in the data. We set the discount factor to $\beta = 0.96$, consistent with a 4% annual interest rate, and the depreciation rate to $\delta = 0.10$. We also discipline the degree of income effects in preferences by targeting consumption shares, for a given elasticity of substitution in consumption σ, which we set to 0.5, treating consumption goods as complements, as is standard in the literature.

The economy starts at the balanced growth path consistent with initial productivity levels and the 1965 input–output structure, and transitions as sectoral productivity grows over the sample period 1965–2014. After 2014, sectoral productivities are held constant at their terminal level, and the economy converges to the corresponding balanced growth path.

Model Fit

Figure 8 compares the model with the data along the targeted dimensions: sectoral consumption shares (panel a), sectoral investment shares (panel b), GDP per capita (panel c), and relative prices (panel d). Overall, the model matches the targeted moments well. It closely reproduces

[35] In the baseline calibration we let the CES weights $\tilde{\omega}_{in,t}$ vary over time, capturing sector-specific changes in investment technology over Korea's rapid growth between 1965 and 2014. In one of our counterfactual exercises we instead hold them constant, at their historical average, to see how far relative price changes alone can go in explaining the evolution of the investment network along the development path.

[36] The baseline calibration features sector-specific capital shares in value added, α_i, and value-added shares in gross output, γ_i, which govern the equilibrium allocation of capital and value added. When recovering sectoral TFP from relative prices, we use common values of α and γ.

Table 3: Calibration: Korea Transition Dynamics

Parameter	Description	Value	Moment/Calibration
Sector order for vectors: Agriculture, Construction, ICT, Manufacturing, Services			
Technology			
γ_i	VA share in gross output	$\{0.58, 0.35, 0.46, 0.25, 0.57\}$	Sectoral VA/gross-output ratios
α_i	Capital share in VA	$\{0.09, 0.15, 0.42, 0.55, 0.31\}$	Sectoral capital income shares
ι	Elast. of subst., inv.	2.41	Regression investment shares on prices
$\tilde{\omega}_{in,t}$	Investment weights	see Table 18	Sectoral investment shares (aggregate) Investment-network odds ratios
$\mu_{in,t}$	IO network	see Table 19	Input–output network
$z_{j,t}$	Sectoral productivities	see Figure 24	Relative prices GDP per capita, 2014/1965
Preferences			
ϵ_j	Income elasticities	$\{0.20, 1.00, 1.20, 1.00, 0.80\}$	Income variation of consumption shares
θ_j	Consumption weights	$\{0.07, 0.00, 0.07, 0.29, 0.57\}$	Average consumption shares, 1965–2014

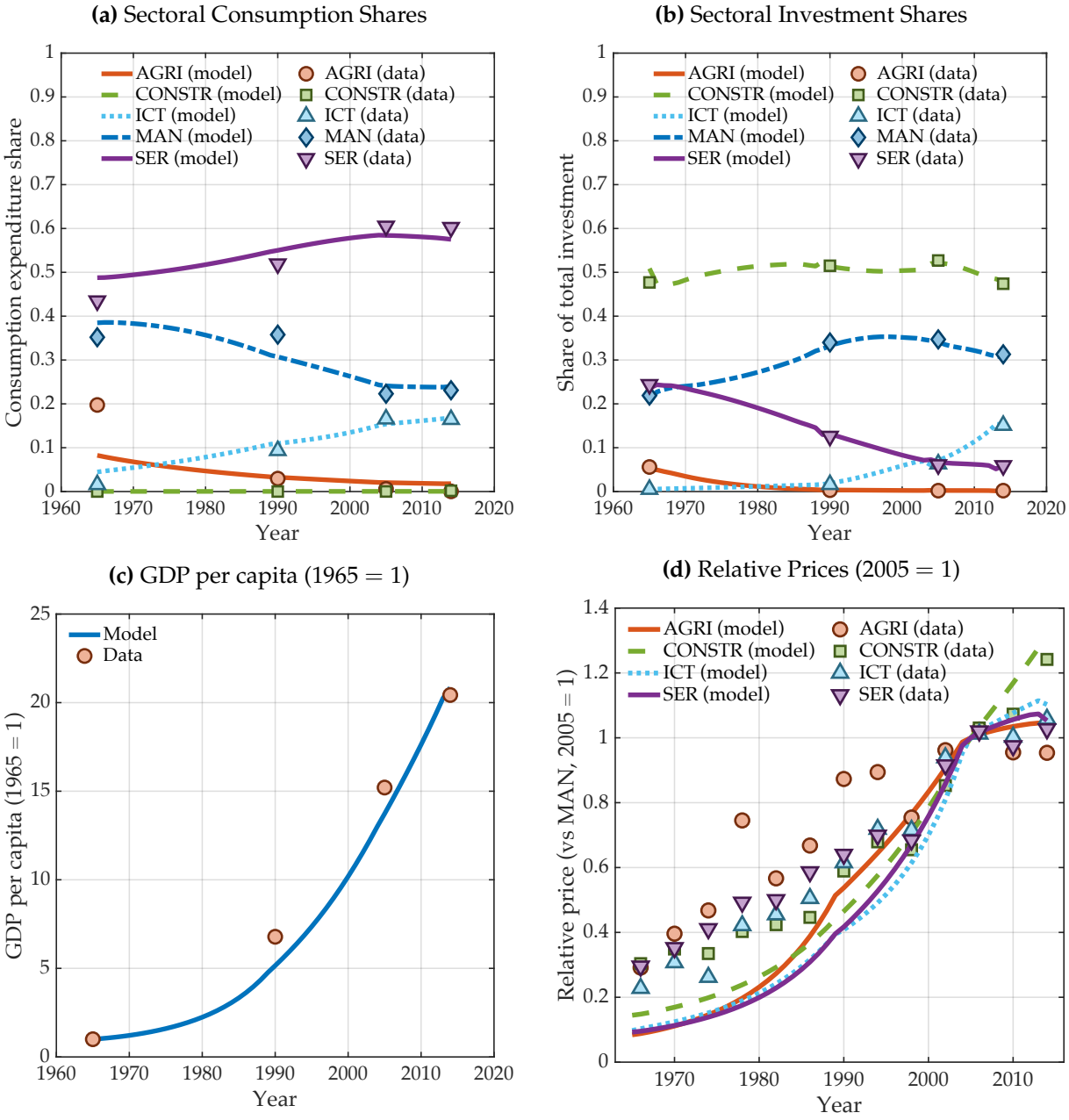
Notes: See the text for a full description of the calibration. The calibration also includes a wedge in the Euler equation, τ_t, recovered so that the model's aggregate investment share $I/(C+I)$ matches the data, and sector-specific labor wedges $\tau_i^\ell = \{0.06, -0.07, -0.04, 0.00, 0.06\}$ that capture sectoral wage gaps. *Data sources:* value-added and consumption shares, the input–output network, sectoral gross-output prices, and capital income shares are from the World Input–Output Database (WIOD); aggregate GDP per capita is from the Penn World Table (PWT); the investment network is constructed as described in Section 3.

the sectoral consumption and investment shares, and the rise in GDP per capita. The model places relative prices somewhat below the data early in the sample; it also puts the consumption share of agriculture a little below the data in the 1960s, likely because preferences lack a subsistence (Stone-Geary) term.

Korea's structural transformation looks different in consumption than in investment, with manufacturing at a different point of the hump in each. In consumption, its share falls from about 36% to 21% while services rise from about 41% to 66%, so manufacturing is already past its peak and on the way down. In investment, manufacturing is only reaching the top of the hump: its share rises from about 22% to a peak near 35% around 2000 and then levels off. Services run the other way: their share of aggregate investment falls from about 24% to 6%, even though they grow as a consumption good.

One salient feature of calibrating our economy to the path of Korea rather than other industrialized countries is that the relative price of agriculture relative to manufacturing is rising over the sample period, rather than declining as it is commonly featured for the process of structural change, Herrendorf, Rogerson and Ákos Valentinyi (2014). The model rationalizes the rise in the relative price with faster productivity growth in manufacturing.

Figure 8: Targeted Moments: Model vs Data (South Korea, 1965–2014)

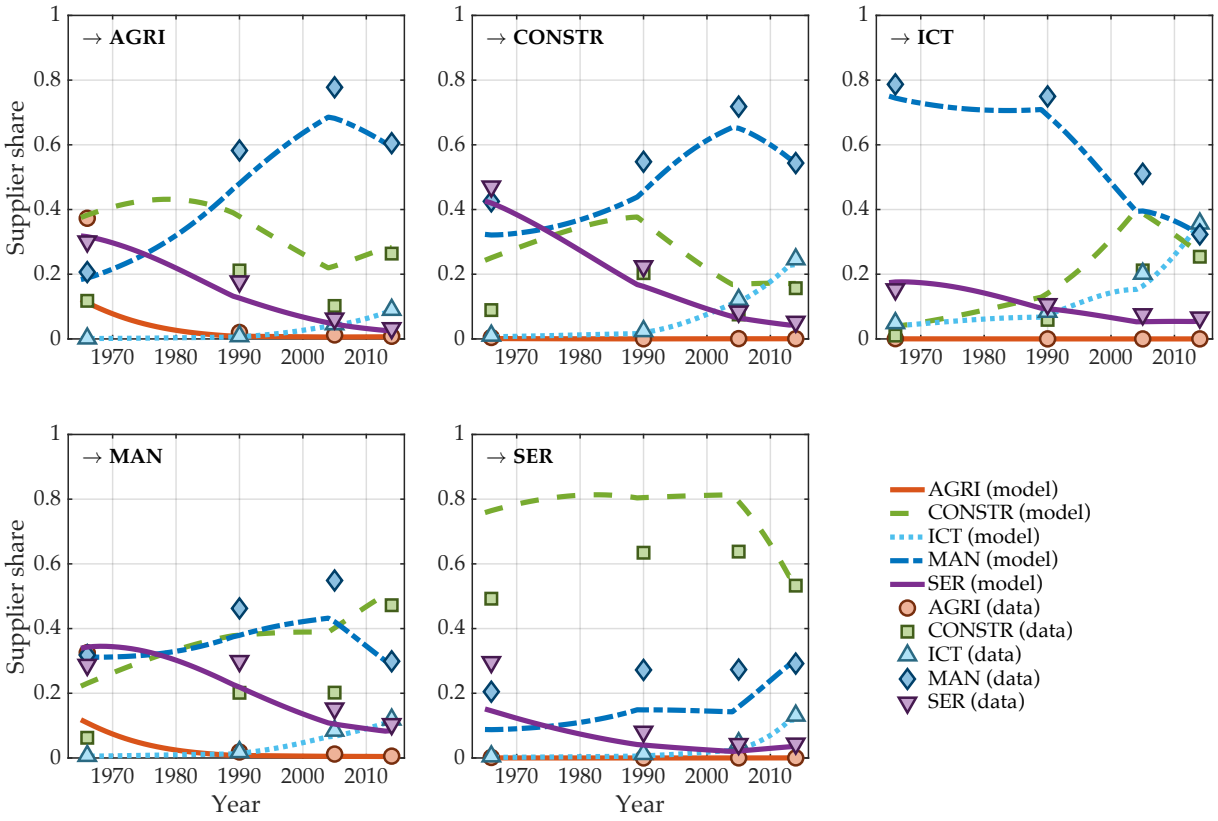


Notes: Model (lines) versus data (markers) for South Korea, 1965–2014. Panel (a): sectoral consumption expenditure shares. Panel (b): sectoral shares of aggregate investment. Panel (c): GDP per capita (1965 = 1). Panel (d): relative prices (2005 = 1). Sectors are agriculture, construction, ICT, manufacturing, and services.

We target the bilateral investment shares only indirectly. What we match exactly are the network's odds ratios—how the supplier mix differs across destination sectors, independent of how much each sector invests or supplies in total—and the sectoral composition of aggregate investment. The bilateral shares then follow from these, together with each destination sector's total investment, which the model determines endogenously. The model captures the shape of the network well (Figure 9). The share of manufacturing investment goods rises throughout the period in all sectors except ICT, the share of services falls in all sectors, and ICT equipment

increases substantially as an investment input in all sectors.

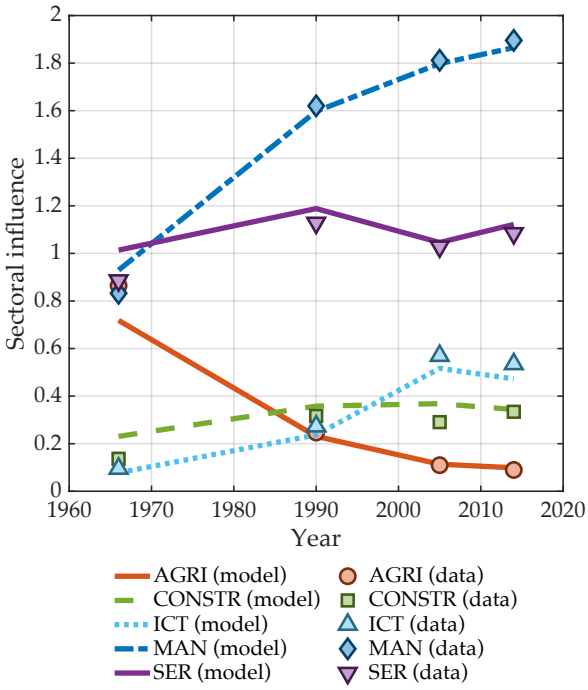
Figure 9: Investment Input Shares by Destination Sector: Model vs Data



Notes: Model (lines) versus data (markers) for South Korea, 1965–2014. Each panel shows the share of one destination sector's investment goods supplied by each source sector (the arrow indicates the destination, e.g. →MAN), so the shares in each panel-year sum to one.

The model reproduces the broad reallocation of value added reasonably well—agriculture falls while manufacturing, ICT, and services rise—as Appendix Figure 25 shows. Figure 10 turns to sectoral influence, which we compute using Proposition 4. The model captures the patterns well. Manufacturing influence roughly doubles over the period, from about 0.9 to 1.8, and ICT influence climbs steeply, from below 0.2 to nearly 0.6. Agricultural influence falls just as sharply, from about 0.7 to 0.1, while services and construction influence stay roughly stable.

Figure 10: Sectoral Influence: Model vs Data (South Korea, 1965–2014)

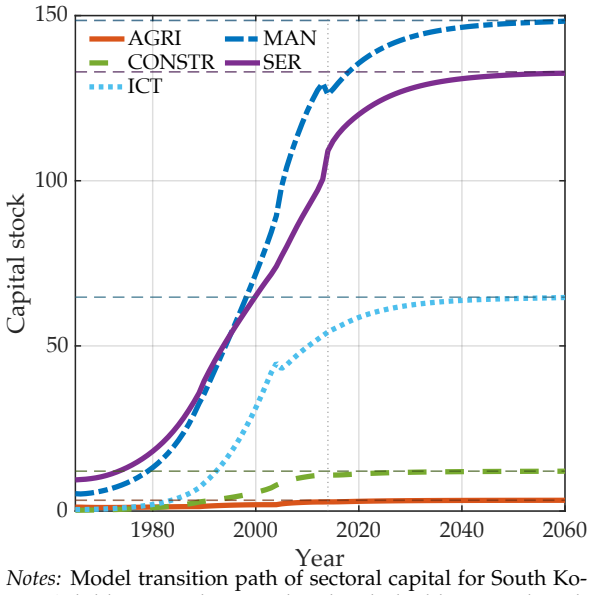


Notes: Model (lines) versus data (markers) for South Korea, 1965–2014. Sectoral influence is computed from Proposition 4.

Transition Paths

Figure 11 shows the path of sectoral capital. Manufacturing and services build up the most capital: manufacturing grows about 24-fold and services about 12-fold from 1965 to 2014. ICT grows the fastest, ending more than a hundred times its 1965 level, while construction and agriculture stay small—agriculture barely doubles. Appendix Figure 26 shows the aggregate paths. Aggregate capital rises from about 4% of its steady-state level in 1965 to about 84% by 2014, and reaches the steady state by the 2040s. Aggregate consumption rises from about 16% to about 90% of the steady state by 2014, close to a six-fold gain. These paths reflect standard neoclassical forces behind Korea's transition.

Figure 11: Sectoral Capital along the Transition (South Korea)



Notes: Model transition path of sectoral capital for South Korea. Solid lines are the capital paths; dashed lines mark each sector's steady-state level. The vertical dotted line marks 2014, the end of the data sample.

7.1 Counterfactual Exercises

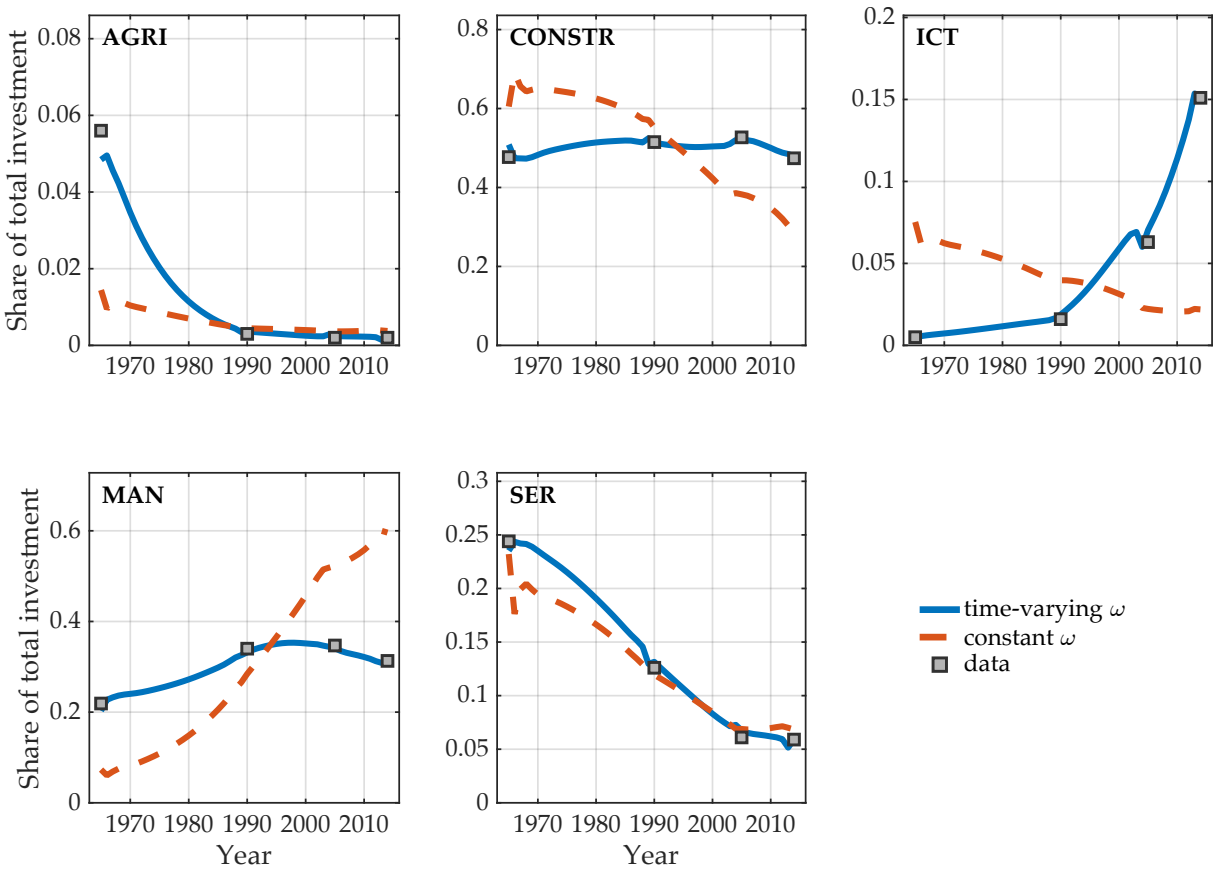
7.1.1 The role of relative prices

In the baseline we allow for time-varying investment loadings $\tilde{\omega}_{in,t}$—investment-specific wedges that match, in each period, both the odds ratios of the investment network and each sector's share of aggregate investment. These wedges stand in for forces we do not model directly: changes in production technology that draw on different capital types over time, sector-specific investment elasticities, and other frictions in how sectors combine their capital. In this counterfactual we remove these wedges and hold the loadings constant, so the investment network moves only through relative prices, and we ask how far prices alone can explain the sectoral composition of aggregate investment and the evolution of the network along the development path.

Figure 12 shows the sectoral shares of aggregate investment in the baseline and in this counterfactual. With constant loadings the model misses the rise in the ICT investment share: in the data the price of ICT rises relative to manufacturing, and because investment goods are substitutes, a higher relative price pushes ICT's share *down*—the opposite of what we see. The same logic applies to construction, whose relative price also rises: the model predicts a sharp decline in construction as a provider of investment, while in the data its share stays high. Matching these two patterns needs something the constant loadings shut down—production structures that lean more on ICT and construction goods over time, or sector-specific invest-

ment elasticities. Prices do, however, capture services: the model reproduces the fall of services as an investment provider, from about 25% to 5% as in the data, since the rising relative price of services pushes its share down in the right direction. It then follows that the manufacturing share is too low early (7% in 1965 versus 22% in the data) and too high late (60% in 2014 versus 31%). Manufacturing's investment share grows eight-fold in the model, whereas in the data it traces a much milder hump.[37]

Figure 12: Sectoral Investment Composition: Time-Varying vs. Constant Investment Loadings



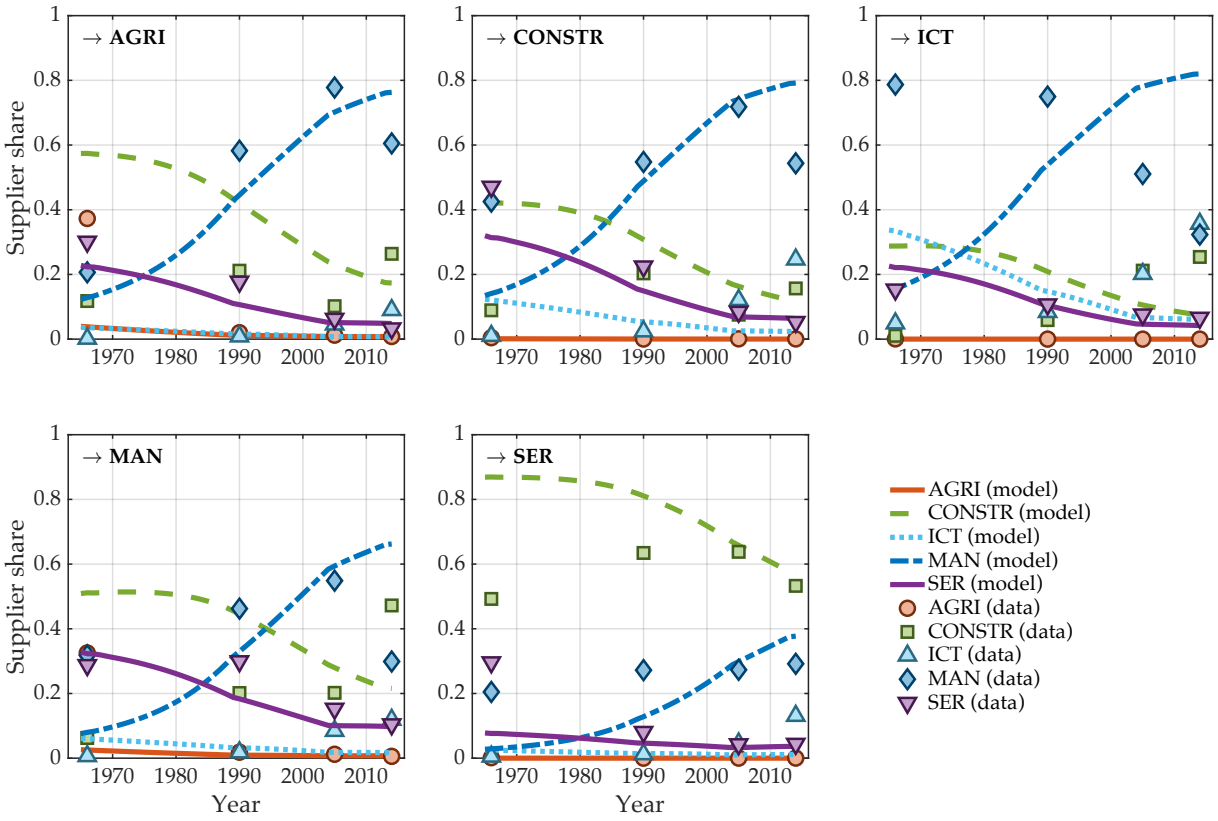
Notes: Each panel plots a sector's share of aggregate investment for South Korea, 1965–2014, under time-varying investment loadings $\tilde{\omega}_{in,t}$ (solid), constant loadings (dashed), and in the data (squares). Holding $\tilde{\omega}$ constant lets the investment network move only through relative prices.

These missed compositional shifts also keep the model from matching how the investment network changes along the development path. Figure 13 looks past each sector's weight in aggregate investment to the sector-specific investment bundles, which our network estimates let us measure. Relative prices fail most clearly on the ICT bundle: in the data ICT investment becomes far more intensive in ICT and construction goods and much less reliant on manufacturing, while with constant loadings the model keeps it manufacturing-heavy. It also misses

[37] This over-prediction reflects not only the constant loadings but also that the investment bundle likely combines complements and substitutes across source sectors, which a single elasticity cannot capture. We estimated sector-specific investment elasticities of substitution, but the five-sector data do not offer enough variation to pin them down: the estimates are noisy and add little explanatory power, so we keep a common elasticity throughout.

the growing role of construction goods, most of all in manufacturing's bundle, and the rising share of ICT goods across the manufacturing, construction, and service bundles.

Figure 13: Investment Network by Destination under Constant Investment Loadings



Notes: Model (lines) versus data (markers) for South Korea, 1965–2014, with the investment loadings $\tilde{\omega}$ held constant so the network moves only through relative prices. Each panel shows the share of one destination sector's investment goods supplied by each source sector (the arrow indicates the destination), so the shares in each panel-year sum to one.

7.1.2 The role of sectoral productivity

With a fully calibrated economy we can now revisit the role of capital accumulation for the transition dynamics. Relative to the income accounting exercises, the equilibrium counterfactuals reflect the endogenous response of value added shares and the investment network to relative prices.

We freeze one sector's productivity growth relative to manufacturing at a time, holding its detrended, manufacturing-relative TFP $z_{j,t}$ at its 1966 level and keeping all other primitives at their baseline-calibrated values. Figure 27 reports the aggregate capital–output ratio, Figure 28 the sectoral capital paths against the baseline, and Table 20 the change in output, the change in the capital–output ratio, and their ratio.

7.1.3 The role of the investment network

Table 4: Output and the Capital–Output Ratio across Investment-Network Scenarios

Scenario	$\Delta \ln Y$	$\Delta \ln(K/Y)$	$\frac{\Delta \ln Y}{\Delta \ln(K/Y)}$
Baseline	2.04	0.88	2.31
Constant $\tilde{\omega}$	1.97	0.69	2.84
$\Omega =$ KOR 1965	1.93	1.10	1.75
$\Omega =$ value-added shares	1.92	0.87	2.20
$\Omega =$ identity	2.04	0.69	2.98

Notes: South Korea, 1965–2014. Log changes from anchor-chained Törnqvist output (Y) and capital (K), detrended (excluding the common 2% balanced-growth trend). The last column is the elasticity of output to the capital–output ratio, $\Delta \ln Y / \Delta \ln(K/Y)$. The $\tilde{\omega}$ = value-added-shares case uses a Cobb–Douglas investment aggregator ($\iota = 1$); the others use $\iota = 2.41$.

8 Final Remarks

We have constructed novel measures of the investment network for 58 countries across the development spectrum and time-series estimates that cover the period 1965 to 2014. Our analysis reveals systematic disparities in the sectors' roles as providers of investment goods as economies progress. We also document significant empirical disparities between the investment network and the input–output network for countries at different income levels.

Leveraging our estimates of the investment network across countries at different development stages, we conduct an income accounting exercise, finding that disparities in the investment network can account for 28% of observed differences in income per capita across countries, almost double the effect of capital in the standard income accounting exercise.

We argue that this role relates to systematic shifts in the investment network as countries develop. The extent to which comparative advantage, distortions, or variations in human capital endowments explain shifts in the network is an exciting avenue for future research. Is there a systematic ladder in the type of investment required to transition an economy from low to high income levels? We hope that this work can serve as a foundation for studying this and other critical questions.

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A Proofs & Derivations

[Under construction]

B Data Appendix

B.1 Data Sources and Coverage

Table 5: Investment Network: Data Sources

Data description	Source
Sector–equipment bridge	US Bridge Tables (BEA)
Number of tools per worker by occupation	Caunedo *et al.* (2023a)
Investment production by sector	WIOD; OECD; Mensah and de Vries (2023)
Input–output structure	WIOD; OECD; Mensah and de Vries (2023)
Employment by occupation and sector	IPUMS; ILOSTAT; PIAAC

Notes: Data sources to construct investment networks. WIOD = World Input–Output Database; OECD = Organisation for Economic Co-operation and Development; BEA = U.S. Bureau of Economic Analysis; IPUMS = Integrated Public Use Microdata Series; ILOSTAT = ILO Statistics; PIAAC = Programme for the International Assessment of Adult Competencies. "Investment production by sector" and "Input–output structure" come from WIOD, OECD and national Supply–Use / IO tables (Mensah and de Vries (2023)). The sector–commodity bridge uses BEA bridge tables to map commodities to industries. "Tools per worker" by occupation is from Caunedo *et al.* (2023a). Employment by occupation and sector combines household microdata (IPUMS), labor statistics (ILOSTAT), and skills microdata (PIAAC).

Table 6: Country Sample and Data Sources

	Country	GDP per capita (2005 PPP)	Use/IO Source	Employment Source	Investment Network Available Years
1	Ethiopia	679	MDV	ILOSTAT	1990–2019
2	Rwanda	1.246	MDV	ILOSTAT	1990–2019
3	Tanzania	1.507	MDV	ILOSTAT	1990–2019
4	Zambia	1.710	MDV	IPUMS	1990–2019
5	Kenya	1.972	MDV	ILOSTAT	1990–2019
6	Cambodia	2.048	OECD	ILOSTAT	2005–2015
7	Senegal	2.728	MDV	IPUMS	1990–2019
8	India	2.872	WIOD	IPUMS	1965–2000; 2000–2014
9	Vietnam	3.128	OECD	IPUMS	2005–2015
10	Ghana	3.219	MDV	ILOSTAT	1990–2019
11	Nigeria	3.481	MDV	IPUMS	1990–2019
12	Philippines	4.366	OECD	IPUMS	2005–2015
13	Indonesia	4.602	WIOD	IPUMS	2000–2014
14	Morocco	4.672	OECD	IPUMS	2005–2015
15	China	6.681	WIOD	IPUMS	1965–2000; 2000–2014
16	Peru	6.832	OECD	PIAAC	2005–2015
17	Colombia	8.367	OECD	ILOSTAT	2005–2015
18	Tunisia	9.353	OECD	ILOSTAT	2005–2015
19	Brazil	9.610	WIOD	IPUMS	1965–2000; 2000–2014
20	Thailand	10.293	OECD	IPUMS	2005–2015
21	South Africa	11.311	OECD	IPUMS	2005–2015
22	Costa Rica	11.580	OECD	IPUMS	2005–2015
23	Turkey	13.941	WIOD	PIAAC	2000–2014
24	Argentina	14.247	OECD	ILOSTAT	2005–2015
25	Mauritius	14.325	MDV	IPUMS	1990–2019
26	Chile	14.534	OECD	PIAAC	2005–2015
27	Mexico	15.230	WIOD	PIAAC	1965–2000; 2000–2014
28	Russia	15.450	WIOD	PIAAC	2000–2014
29	Poland	16.838	WIOD	PIAAC	2000–2014
30	Malaysia	17.412	OECD	IPUMS	2005–2015
31	Lithuania	17.646	WIOD	ILOSTAT	2000–2014
32	Slovakia	20.168	OECD	PIAAC	2000–2014
33	Hungary	20.819	WIOD	PIAAC	2000–2014
34	Czechia	26.624	WIOD	PIAAC	2000–2014
35	Portugal	27.149	WIOD	IPUMS	1965–2000; 2000–2014
36	Slovenia	28.821	OECD	PIAAC	2000–2014
37	Greece	30.138	WIOD	PIAAC	1965–2000; 2000–2014
38	South Korea	30.784	WIOD	PIAAC	1965–2000; 2000–2014
39	New Zealand	31.485	OECD	PIAAC	2005–2015
40	Israel	32.358	OECD	PIAAC	2005–2015
41	Spain	32.769	WIOD	PIAAC	1965–2000; 2000–2014
42	Cyprus	33.025	OECD	ILOSTAT	2005–2015
43	Italy	36.167	WIOD	PIAAC	1965–2000; 2000–2014
44	France	36.651	WIOD	PIAAC	1965–2000; 2000–2014
45	Japan	38.466	WIOD	PIAAC	1965–2000; 2000–2014
46	Germany	38.475	WIOD	PIAAC	1965–2000; 2000–2014
47	Belgium	39.220	WIOD	PIAAC	1965–2000; 2000–2014
48	United Kingdom	39.308	WIOD	PIAAC	1965–2000; 2000–2014
49	Denmark	40.344	WIOD	PIAAC	1965–2000; 2000–2014
50	Sweden	40.381	WIOD	PIAAC	1965–2000; 2000–2014
51	Austria	41.678	WIOD	ILOSTAT	1965–2000; 2000–2014
52	Australia	43.333	WIOD	ILOSTAT	1965–2000; 2000–2014
53	Netherlands	44.662	WIOD	PIAAC	1965–2000; 2000–2014
54	Ireland	47.211	WIOD	PIAAC	1965–2000; 2000–2014
55	Switzerland	49.859	WIOD	IPUMS	2000–2014
56	Norway	54.200	WIOD	PIAAC	2000–2014
57	United States	54.210	WIOD	PIAAC	1965–2000; 2000–2014
58	Singapore	63.949	OECD	PIAAC	2005–2015

Notes: Country-year coverage and main sources of data. "Use/IO Source" refers to VA shares and GFCF imported shares. GDP per capita in chained PPPs (2017 international dollars), 2005.

Table 7: Aggregate Sectors Definition

ISIC Rev.4 Industry Code	Industry Description	Aggregate Sector
A01	Crop and animal production, hunting and related service activities	Agriculture
A02	Forestry and logging	Agriculture
A03	Fishing and aquaculture	Agriculture
B	Mining and quarrying	Mining
C10–C12	Manufacture of food products, beverages and tobacco products	Non-Durables
C13–C15	Manufacture of textiles, wearing apparel and leather products	Non-Durables
C16	Manufacture of wood and of products of wood and cork, except furniture; manufacture of articles of straw and plaiting materials	Durables
C17	Manufacture of paper and paper products	Non-Durables
C18	Printing and reproduction of recorded media	Non-Durables
C19	Manufacture of coke and refined petroleum products	Non-Durables
C20	Manufacture of chemicals and chemical products	Non-Durables
C21	Manufacture of basic pharmaceutical products and pharmaceutical preparations	Non-Durables
C22	Manufacture of rubber and plastic products	Non-Durables
C23	Manufacture of other non-metallic mineral products	Durables
C24	Manufacture of basic metals	Durables
C25	Manufacture of fabricated metal products, except machinery and equipment	Durables
C26	Manufacture of computer, electronic and optical products	Electronics
C27	Manufacture of electrical equipment	Electronics
C28	Manufacture of machinery and equipment n.e.c.	Machinery
C29	Manufacture of motor vehicles, trailers and semi-trailers	Transportation
C30	Manufacture of other transport equipment	Transportation
C31, C32	Manufacture of furniture; other manufacturing	Durables
C33	Repair and installation of machinery and equipment	Durables
D35	Electricity, gas, steam and air conditioning supply	Services
E36	Water collection, treatment and supply	Services
E37–E39	Sewerage; waste collection, treatment and disposal activities; materials recovery; remediation activities and other waste management services	Services
F	Construction	Construction
G45	Wholesale and retail trade and repair of motor vehicles and motorcycles	Services
G46	Wholesale trade, except of motor vehicles and motorcycles	Services
G47	Retail trade, except of motor vehicles and motorcycles	Services
H49	Land transport and transport via pipelines	Trpt Services
H50	Water transport	Trpt Services
H51	Air transport	Trpt Services
H52	Warehousing and support activities for transportation	Trpt Services
H53	Postal and courier activities	Trpt Services
I	Accommodation and food service activities	Services
J58	Publishing activities	ICT
J59, J60	Motion picture, video and television programme production, sound recording and music publishing activities; programming and broadcasting activities	ICT
J61	Telecommunications	ICT
J62, J63	Computer programming, consultancy and related activities; information service activities	ICT
K64	Financial service activities, except insurance and pension funding	Services
K65	Insurance, reinsurance and pension funding, except compulsory social security	Services
K66	Activities auxiliary to financial services and insurance activities	Services
L68	Real estate activities	Services
M69, M70	Legal and accounting activities; activities of head offices; management consultancy activities	ICT
M71	Architectural and engineering activities; technical testing and analysis	ICT
M72	Scientific research and development	ICT
M73	Advertising and market research	ICT
M74, M75	Other professional, scientific and technical activities; veterinary activities	ICT
N	Administrative and support service activities	ICT
O84	Public administration and defence; compulsory social security	Services
P85	Education	Services
Q	Human health and social work activities	Services
R, S	Other service activities	Services
T	Activities of households as employers; undifferentiated goods- and services-producing activities of households for own use	Services
U	Activities of extraterritorial organizations and bodies	Services

Notes: Crosswalk between ISIC Rev.4 classification and our 10-sector disaggregation. "Trpt Services" refers to Transportation Services.

Table 8: Country Groups

Low income	Middle income	High income
Ethiopia	Argentina	Australia
Ghana	Brazil	Austria
India	China	Belgium
Kenya	Colombia	Switzerland
Morocco	Costa Rica	Chile
Nigeria	Indonesia	Cyprus
Philippines	Mexico	Czech Republic
Rwanda	Mauritius	Germany
Senegal	Malaysia	Denmark
Tanzania	Peru	Spain
Zambia	Thailand	France
	Turkey	United Kingdom
	South Africa	Greece
		Hungary
		Ireland
		Israel
		Italy
		Japan
		South Korea
		Lithuania
		Netherlands
		Norway
		New Zealand
		Poland
		Portugal
		Russia
		Singapore
		Slovakia
		Slovenia
		Sweden
		United States

Notes: Countries are grouped into income categories per the World Bank classification. *Low income* comprises "Low income" and "Lower middle income"; *Middle income* corresponds to "Upper middle income"; *High income* corresponds to "High income." Average real GDP per capita (chained PPPs, 2017 international dollars) for 2005, from the Penn World Table (PWT): low 2587; middle 11110; high 35056.

B.2 Data Sources for Income Accounting

To estimate sectoral influence in equation 16 we need data on sectoral value-added shares in gross output ($\mathbf{\Gamma}$), sectoral value-added shares, which proxy for the final-demand shares $\boldsymbol{\zeta}$, capital shares in value added ($\boldsymbol{\alpha}$), and the sectoral imported share of investment ($\boldsymbol{\phi}$), together with estimates of input-output linkages ($\boldsymbol{M}$) and our newly estimated investment networks ($\mathbf{\Omega}$), both already described in Section 3.2.

Value-added shares in production (Γ) and sectoral value-added shares (ζ). We compute

sectoral value-added shares in gross output and sectoral value-added shares using the same data sources as the input-output tables for each country: Mensah and de Vries (2023), WIOD, and OECD. In the model, ζ_n is the share of good n in final uses and domestic investment; we use sectoral value-added shares as a proxy for these final-demand shares.

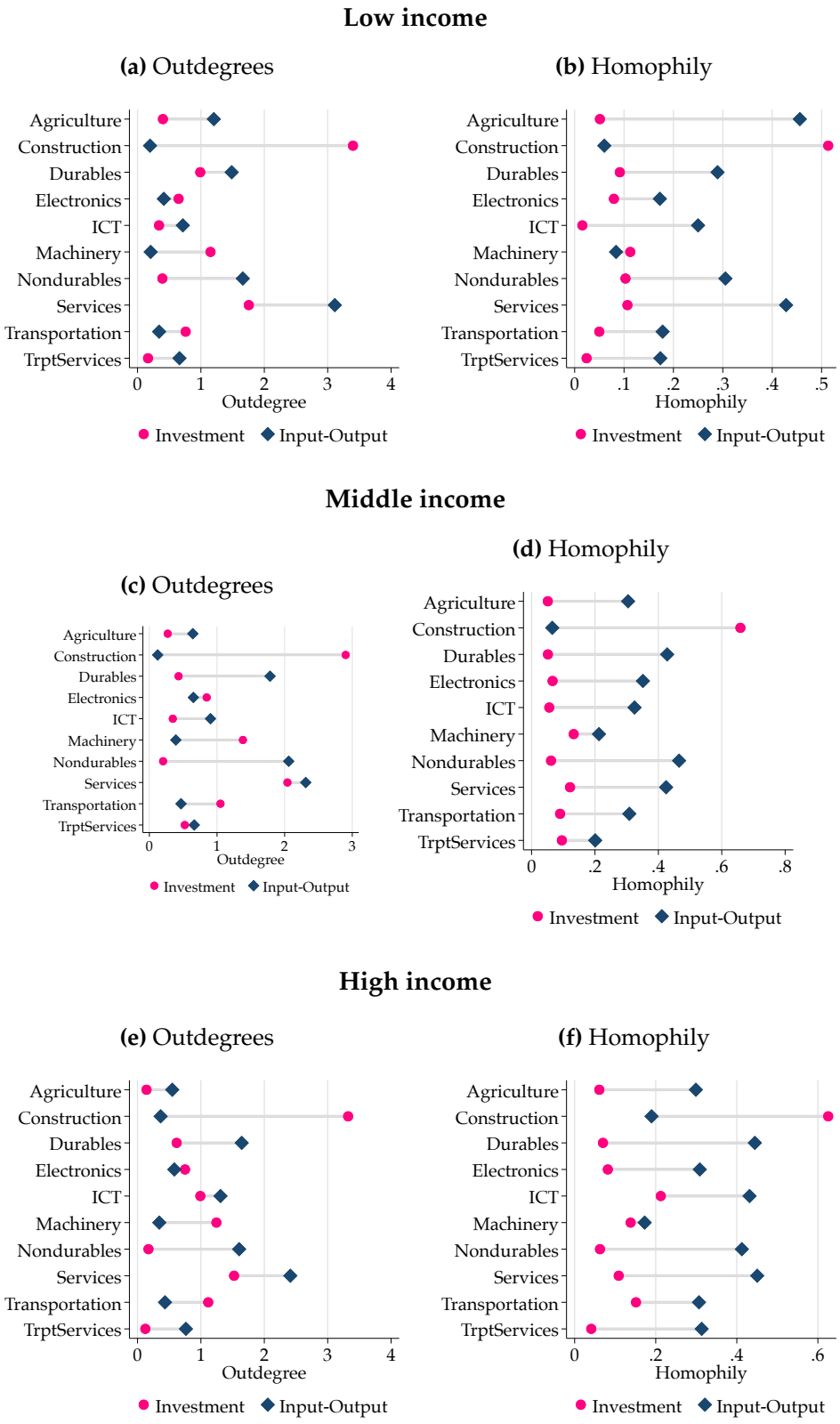
Capital share in value added (α). We exploit data from PWT version 10.01 to compute the labor expenditure share. We estimate capital shares as residuals from labor expenditure shares, under the assumption of CRS value-added production technologies. The capital expenditure share is computed at the aggregate level and therefore country specific but common across sectors.[38]

sectoral imported investment shares (ϕ). For each sector and country, we compute the share of sectoral investment that is sourced from abroad using the *use tables* described in Section 3.2, which contain information on imported and domestic sectoral investment.

[38]Some high- and medium-income countries have data on sectoral capital shares in value added from WIOD. However, we lack this information for more than half of the countries in our sample. For consistency, we use country-specific aggregate capital shares.

B.3 Investmenet Network vs. Input-Output Network

Figure 14: Investment Network vs. Input-Output Network



Notes: Data for 2005, for Low, Middle and High income countries per the World Bank classification. Circles denote the investment network; diamonds denote the input–output network. Panel (a) reports sectoral outdegrees (row sums); panel (b) reports sectoral homophily (diagonal elements of each network).

Table 9: Input Output Outdegrees

Sector	Low income	Middle income	High income
Agriculture	0.97	0.53	0.42
	(0.76 1.25)	(0.44 0.69)	(0.32 0.53)
Construction	0.20	0.10	0.38
	(0.05 0.28)	(0.04 0.14)	(0.19 0.54)
Durables	1.87	1.79	1.50
	(1.16 2.47)	(1.58 1.94)	(1.37 1.63)
Electronics	0.47	0.67	0.62
	(0.04 0.69)	(0.51 1.01)	(0.43 0.73)
ICT	0.77	0.96	1.46
	(0.44 0.88)	(0.69 1.13)	(1.17 1.66)
Machinery	0.17	0.34	0.32
	(0.03 0.25)	(0.28 0.45)	(0.24 0.43)
Nondurables	1.72	2.08	1.48
	(1.39 2.17)	(1.95 2.22)	(1.30 1.65)
Services	2.91	2.38	2.56
	(2.35 3.57)	(2.20 2.59)	(2.28 2.81)
Transportation	0.25	0.43	0.45
	(0.09 0.47)	(0.32 0.53)	(0.32 0.55)
TrptServices	0.68	0.71	0.81
	(0.42 0.78)	(0.62 0.81)	(0.62 0.87)

Notes: Data for 2005, for Low, Middle and High income countries per the World Bank classification. Outdegrees are sectoral row sums in the investment network. Entries report means; values in parentheses are the 25th and 75th percentiles. Average GDP per capita (PPP): low 2587; middle 11110; high 35056.

Table 10: Investment Network Concentration Index

Sector	Low income	Middle income	High income
Agriculture	0.26	0.22	0.20
	(0.19 0.35)	(0.17 0.25)	(0.15 0.22)
Construction	0.54	0.49	0.59
	(0.35 0.80)	(0.28 0.70)	(0.40 0.78)
Durables	0.31	0.19	0.19
	(0.18 0.44)	(0.17 0.19)	(0.16 0.20)
Electronics	0.36	0.24	0.20
	(0.22 0.50)	(0.19 0.29)	(0.17 0.21)
ICT	0.43	0.36	0.21
	(0.30 0.50)	(0.20 0.39)	(0.18 0.25)
Machinery	0.39	0.27	0.20
	(0.21 0.60)	(0.20 0.29)	(0.17 0.21)
Nondurables	0.28	0.17	0.17
	(0.17 0.42)	(0.16 0.18)	(0.14 0.18)
Services	0.36	0.36	0.26
	(0.26 0.43)	(0.21 0.48)	(0.18 0.29)
Transportation	0.34	0.22	0.22
	(0.23 0.48)	(0.18 0.25)	(0.17 0.23)
TrptServices	0.33	0.27	0.22
	(0.26 0.38)	(0.18 0.30)	(0.16 0.24)

Notes: Data for 2005. Concentration index refers to the Herfindahl–Hirschman index computed as the squared sum of the elements of each column of the investment network. The lower the value of the index, the more diversified, with full-diversification achieved at 0.1. Entries report means; values in parentheses are the 25th and 75th percentiles. Average GDP per capita (PPP): low 2587; middle 11110; high 35056.

Table 11: Input-Output Concentration Index

Sector	Low income	Middle income	High income
Agriculture	0.34 (0.25 0.39)	0.30 (0.27 0.32)	0.27 (0.24 0.28)
Construction	0.38 (0.22 0.47)	0.28 (0.25 0.31)	0.23 (0.20 0.25)
Durables	0.35 (0.27 0.36)	0.28 (0.25 0.33)	0.28 (0.24 0.32)
Electronics	0.32 (0.23 0.35)	0.26 (0.21 0.32)	0.25 (0.20 0.28)
ICT	0.31 (0.25 0.36)	0.29 (0.26 0.31)	0.35 (0.31 0.38)
Machinery	0.30 (0.23 0.35)	0.23 (0.21 0.25)	0.21 (0.18 0.24)
Nondurables	0.31 (0.29 0.33)	0.32 (0.28 0.36)	0.28 (0.24 0.29)
Services	0.27 (0.22 0.32)	0.29 (0.27 0.33)	0.29 (0.26 0.31)
Transportation	0.28 (0.19 0.33)	0.23 (0.20 0.24)	0.23 (0.18 0.25)
TrptServices	0.31 (0.25 0.39)	0.26 (0.23 0.29)	0.26 (0.23 0.27)

Notes: Data for 2005. Concentration index refers to the Herfindahl–Hirschman index computed as the squared sum of the elements of each column of the input-output network. The lower the value of the index, the more diversified, with full-diversification achieved at 0.1. Entries report means; values in parentheses are the 25th and 75th percentiles. Average GDP per capita (PPP): low 2587; middle 11110; high 35056.

Table 12: Investment Network Homophily

Sector	Low income	Middle income	High income
Agriculture	0.09 (0.00 0.16)	0.07 (0.01 0.07)	0.07 (0.02 0.12)
Construction	0.45 (0.00 0.90)	0.60 (0.48 0.83)	0.70 (0.61 0.88)
Durables	0.09 (0.02 0.12)	0.04 (0.04 0.05)	0.06 (0.03 0.08)
Electronics	0.08 (0.01 0.14)	0.09 (0.06 0.11)	0.10 (0.06 0.12)
ICT	0.02 (0.00 0.03)	0.06 (0.04 0.10)	0.23 (0.15 0.30)
Machinery	0.16 (0.01 0.25)	0.16 (0.09 0.24)	0.18 (0.14 0.22)
Nondurables	0.08 (0.01 0.14)	0.05 (0.01 0.09)	0.06 (0.04 0.08)
Services	0.10 (0.03 0.15)	0.09 (0.07 0.10)	0.13 (0.09 0.16)
Transportation	0.05 (0.02 0.07)	0.11 (0.06 0.13)	0.16 (0.10 0.19)
TrptServices	0.03 (0.00 0.04)	0.11 (0.03 0.12)	0.06 (0.03 0.08)

Notes: Data for 2005, for Low, Middle and High income countries per the World Bank classification. Homophily consists on the diagonal elements of the network, i.e., how important are sectors are provider of investment inputs for themselves. Entries report means; values in parentheses are the 25th and 75th percentiles. Average GDP per capita (PPP): low 2587; middle 11110; high 35056.

Table 13: Input-Output Homophily

Sector	Low income	Middle income	High income
Agriculture	0.41 (0.27 0.54)	0.26 (0.18 0.32)	0.24 (0.16 0.33)
Construction	0.07 (0.00 0.13)	0.04 (0.01 0.05)	0.22 (0.06 0.35)
Durables	0.33 (0.13 0.49)	0.44 (0.41 0.50)	0.43 (0.37 0.47)
Electronics	0.20 (0.01 0.30)	0.36 (0.20 0.51)	0.33 (0.22 0.46)
ICT	0.27 (0.14 0.52)	0.38 (0.32 0.45)	0.50 (0.44 0.55)
Machinery	0.06 (0.01 0.10)	0.18 (0.12 0.22)	0.16 (0.09 0.21)
Nondurables	0.32 (0.23 0.44)	0.48 (0.41 0.54)	0.39 (0.35 0.44)
Services	0.39 (0.27 0.46)	0.46 (0.44 0.50)	0.47 (0.42 0.50)
Transportation	0.15 (0.01 0.25)	0.31 (0.25 0.40)	0.32 (0.21 0.40)
TrptServices	0.18 (0.05 0.35)	0.25 (0.18 0.31)	0.35 (0.25 0.42)

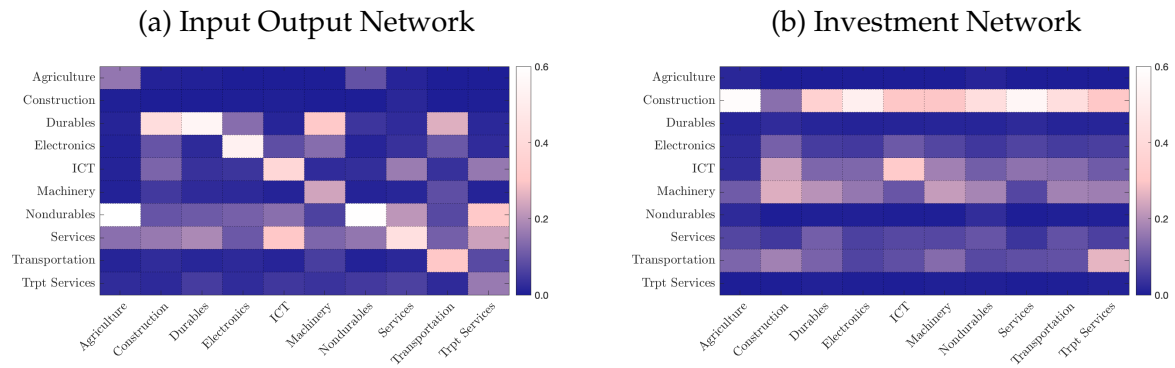
Notes: Data for 2005, for Low, Middle and High income countries per the World Bank classification. Homophily consists on the diagonal elements of the network, i.e., how important are sectors are provider of intermediate inputs for themselves. Entries report means; values in parentheses are the 25th and 75th percentiles. Average GDP per capita (PPP): low 2587; middle 11110; high 35056.

Table 14: sectoral Influence

Sector	Low income	Middle income	High income
Agriculture	0.38 (0.28 0.51)	0.15 (0.08 0.20)	0.04 (0.02 0.06)
Construction	0.17 (0.07 0.26)	0.25 (0.21 0.33)	0.19 (0.13 0.21)
Durables	0.13 (0.05 0.18)	0.20 (0.13 0.21)	0.13 (0.10 0.16)
Electronics	0.05 (0.01 0.05)	0.10 (0.05 0.10)	0.06 (0.04 0.07)
ICT	0.14 (0.10 0.15)	0.23 (0.16 0.29)	0.31 (0.25 0.37)
Machinery	0.03 (0.01 0.04)	0.07 (0.05 0.08)	0.04 (0.03 0.06)
Nondurables	0.20 (0.10 0.31)	0.34 (0.23 0.44)	0.13 (0.09 0.15)
Services	0.67 (0.59 0.80)	0.69 (0.51 0.84)	0.75 (0.69 0.81)
Transportation	0.03 (0.01 0.05)	0.07 (0.05 0.08)	0.04 (0.03 0.06)
TrptServices	0.10 (0.07 0.13)	0.14 (0.09 0.15)	0.11 (0.08 0.12)

Notes: Data for 2005, for Low, Middle and High income countries per the World Bank classification. sectoral influence is defined as $\Phi\eta\Gamma$ in Proposition 4, and measures the change in aggregate value added following a change in sectoral productivity. Entries report means; values in parentheses are the 25th and 75th percentiles. Average GDP per capita (PPP): low 2587; middle 11110; high 35056.

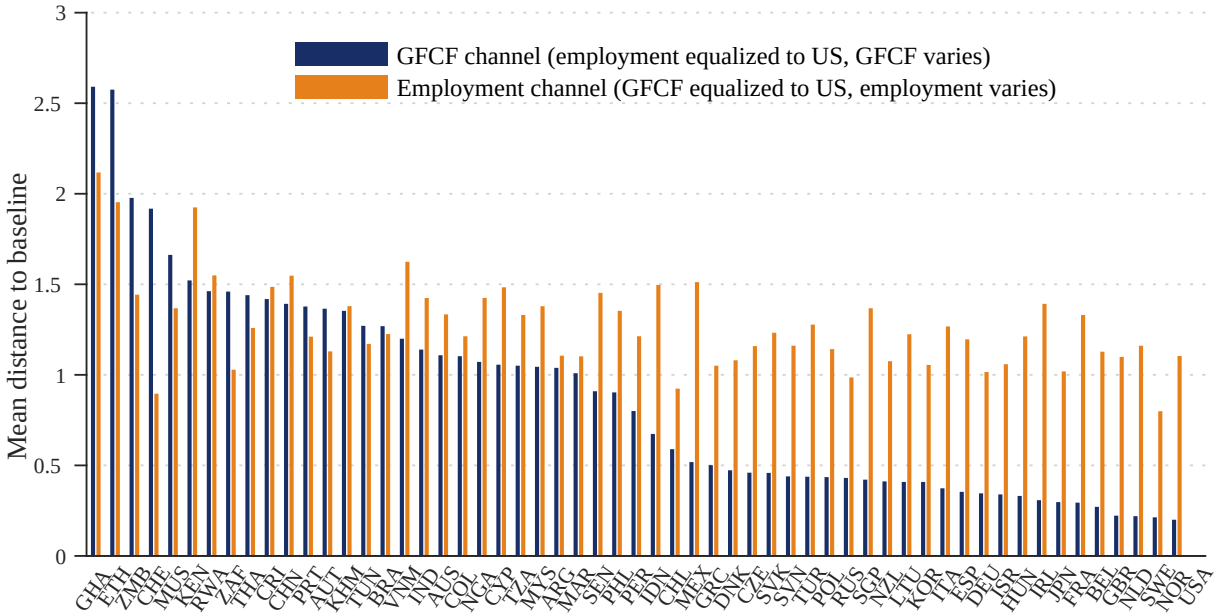
Figure 15: South Korea: Investment Network vs. Input–Output Network, 2014



Notes: Estimates of the investment network and the input–output network in South Korea, 2014.

B.4 Sources of Cross-Country Variation in Investment Networks

Figure 16: Within-country distance between baseline and counterfactual investment networks



Notes: Each bar reports the mean Frobenius distance between the baseline investment network and a counterfactual, averaged over all available years for each country, $\overline{\|\mathbf{\Omega}_t^c - \mathbf{\Omega}_t^{c,\mathrm{CF}}\|_F}$. The navy bars correspond to the GFCF channel: the U.S. occupational employment distribution is imposed on each country while keeping country-own GFCF flows; any deviation from baseline thus reflects GFCF differences. The orange bars correspond to the employment channel: U.S. GFCF levels are imposed while retaining country-own employment composition. Countries are sorted in descending order of their GFCF-channel distance.

Table 15: Cross-country variance in investment networks: contribution of each channel

	Median [p25, p75]
Share of cross-country variance in Ω explained by each channel alone (%)	
GFCF differences (employment equalized to US; only GFCF varies)	126.7 [77.9, 221.7]
Employment differences (GFCF equalized to US; only employment varies)	9.4 [2.9, 32.8]

Notes: Each entry is $\text{var}(\Omega^{\text{CF}}_{ijc})/\text{var}(\Omega^{\text{base}}_{ijc})$, computed element-by-element across countries and summarised over all (i, j) pairs. Cross-section: 58 countries, year 2005. Median, P25 and P75 reported (in parenthesis).

B.5 Data Validation

Table 16: Comparison with vom Lehn and Winberry (2022), US 2012

(a) Outdegrees

Sector	This Paper	VLW
Agriculture	0.00	0.00
Construction	1.65	1.31
Durables	0.30	0.24
Electronics	0.90	0.64
ICT	3.02	3.57
Machinery	1.42	1.94
Nondurables	0.04	0.03
Services	1.16	1.06
Transportation	1.35	1.10
Transportation Services	0.16	0.10

(b) Homophily

Sector	This Paper	VLW
Agriculture	0.00	0.00
Construction	0.04	0.03
Durables	0.04	0.05
Electronics	0.12	0.07
ICT	0.43	0.61
Machinery	0.18	0.26
Nondurables	0.01	0.005
Services	0.11	0.12
Transportation	0.11	0.06
Transportation Services	0.03	0.01

Notes: Comparison between investment network estimates in year 2012, corresponding to the year in which we fix the occupational composition of the labor force. This table reports the row sum of the networks, *outdegrees*, as well as the weight of the diagonal in the network, *homophily*.

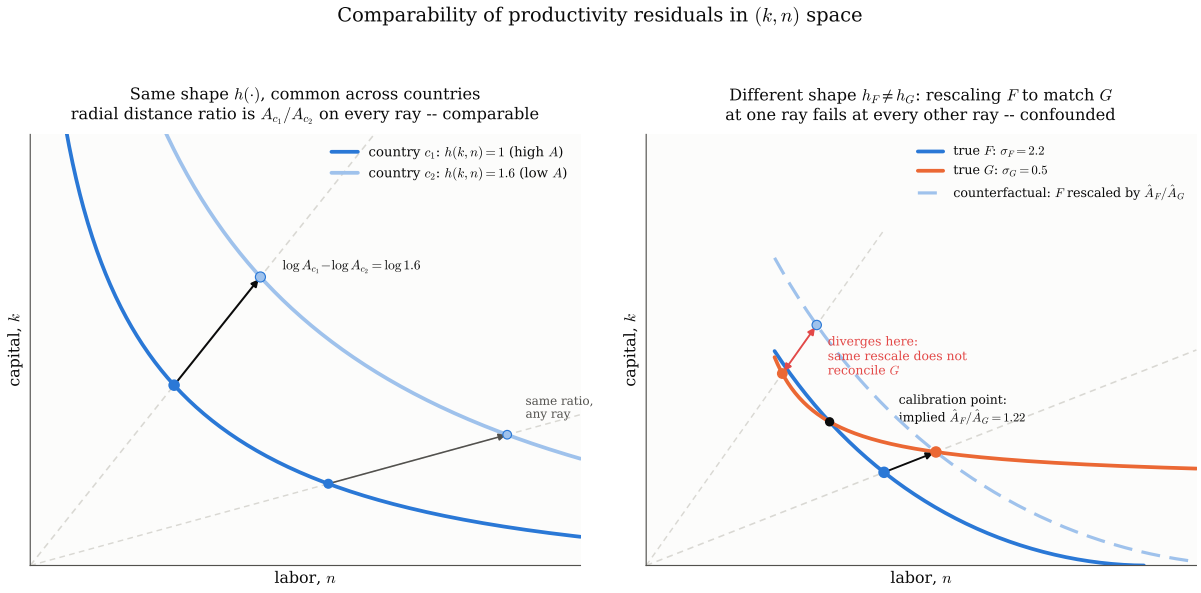
Table 17: Comparison with Gillmore *et al.* (2025), Chile 2015

	Outdegree		Homophily	
Sector	This Paper	Gillmore *et al.* (2025)	This Paper	Gillmore et al. (2025)
Agriculture	0.32	0.07	0.13	0.04
Construction	4.22	5.13	0.68	0.86
Durables	0.44	0.46	0.07	0.07
Electronics	0.06	0.11	0.01	0.05
ICT	2.73	0.15	0.33	0.03
Machinery	0.49	0.28	0.07	0.11
Nondurables	0.34	0.15	0.09	0.04
Services	1.08	4.47	0.06	0.36
Transportation	0.13	0.04	0.01	0.01
TrptServices	0.20	0.11	0.06	0.04

Notes: Comparison between our benchmark (domestic) investment network estimates and Gillmore *et al.* (2025) VAT firm-to-firm transaction estimates. This table reports the row sum of the networks, *outdegrees*, as well as the weight of the diagonal in the network, *homophily*.

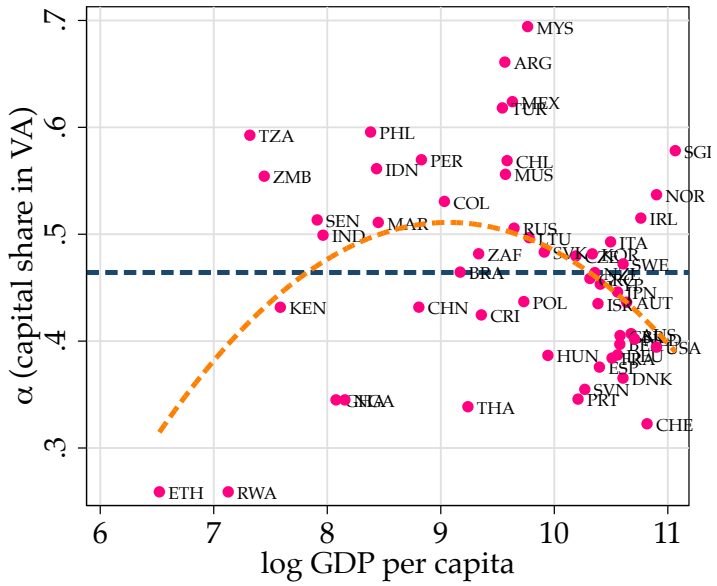
B.6 Income Accounting

Figure 17: Comparability of Productivity Residuals in (k, n) Space



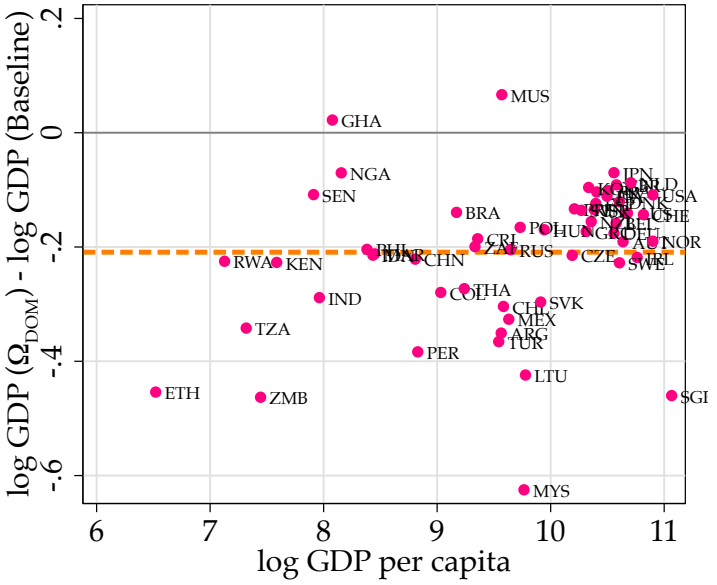
Notes: Isoquants in capital-labor space. *Left panel:* two countries sharing a common technology (same CES shape). The radial distance between their isoquants, measured along any ray from the origin (i.e. holding k/n fixed), equals the Hicks-neutral productivity gap $\log A_{c_1} - \log A_{c_2}$ regardless of which ray is used, so the gap is well defined and comparable across countries. *Right panel:* two technologies with different curvature whose isoquants cross. Calibrating a Hicks-neutral rescaling of F to match G at one input ratio (black arrow) makes the two coincide only at that point; evaluated at any other input ratio (red arrow) the rescaled F isoquant and the true G isoquant diverge, so no single Hicks-neutral gap reconciles the two technologies everywhere.

Figure 18: Capital Share in Value Added Along Development



Notes: Country-level capital share in value added, α, plotted against log GDP per capita. The dashed orange line is a quadratic fit; the horizontal navy line marks the cross-country mean.

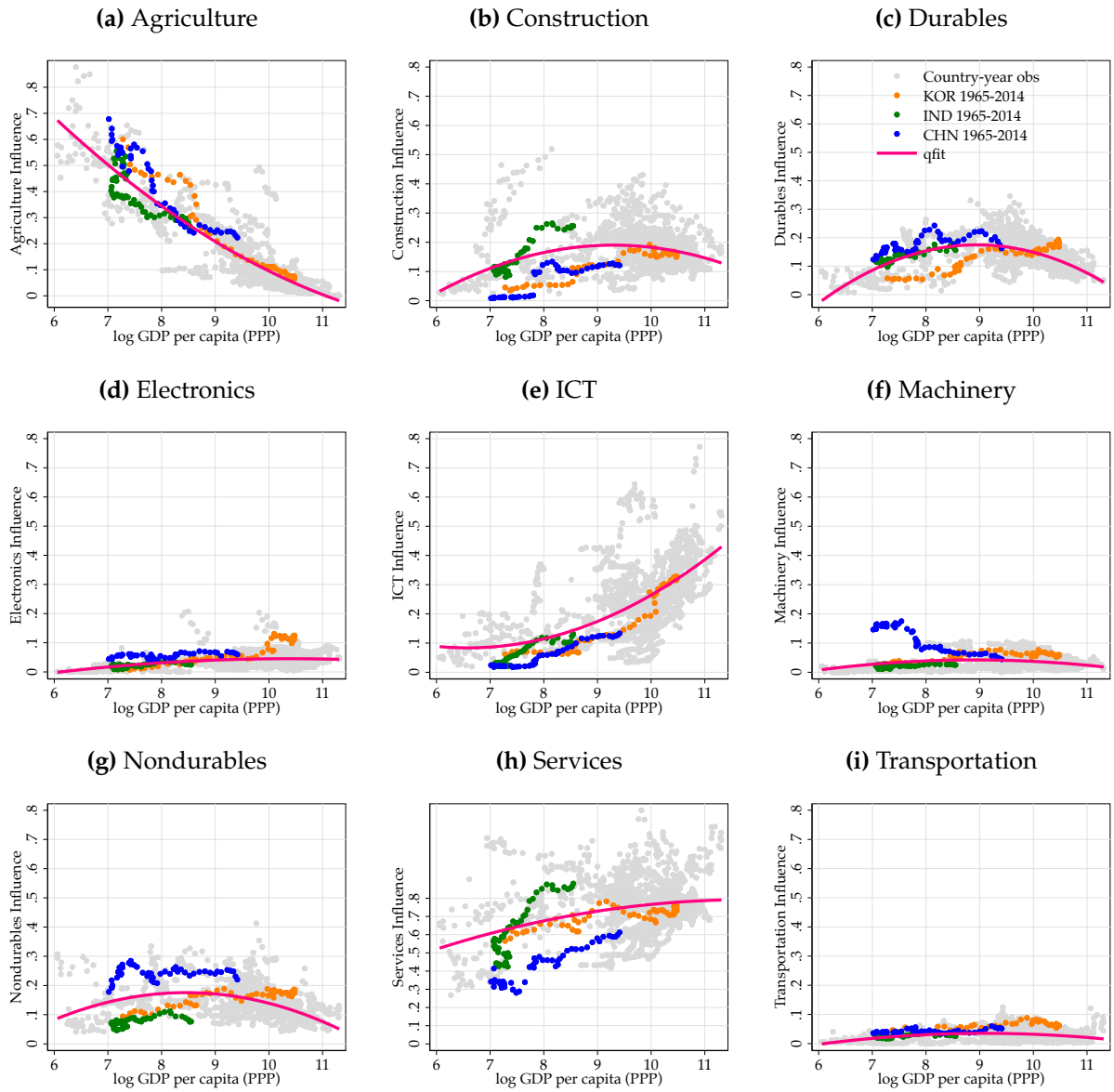
Figure 19: Counterfactual: Domestic Investment Network



Notes: Income per capita computed using domestic investment network for each country, relative to the baseline.

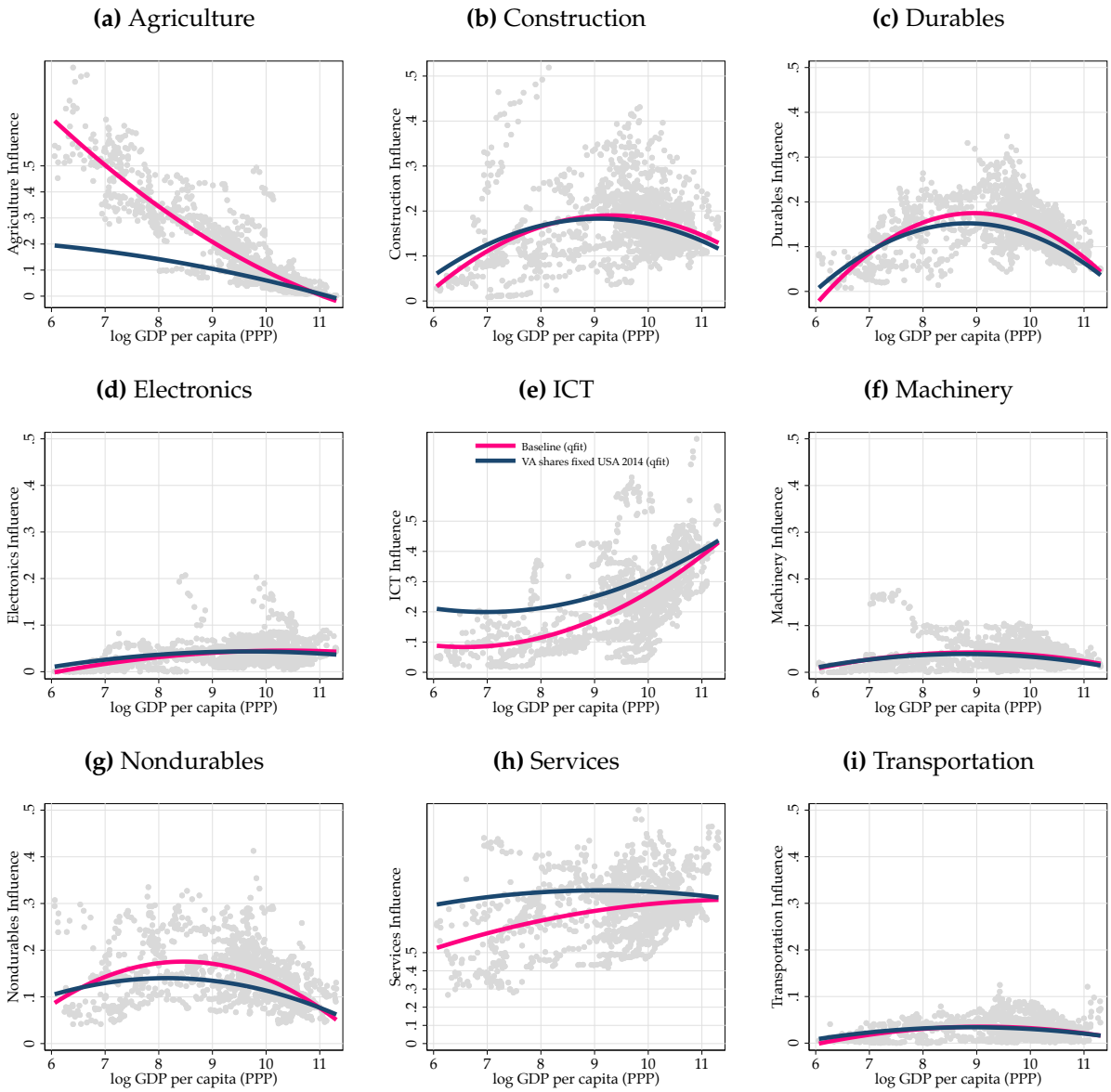
B.7 sectoral Influence Along Development

Figure 20: sectoral Influence Along Development



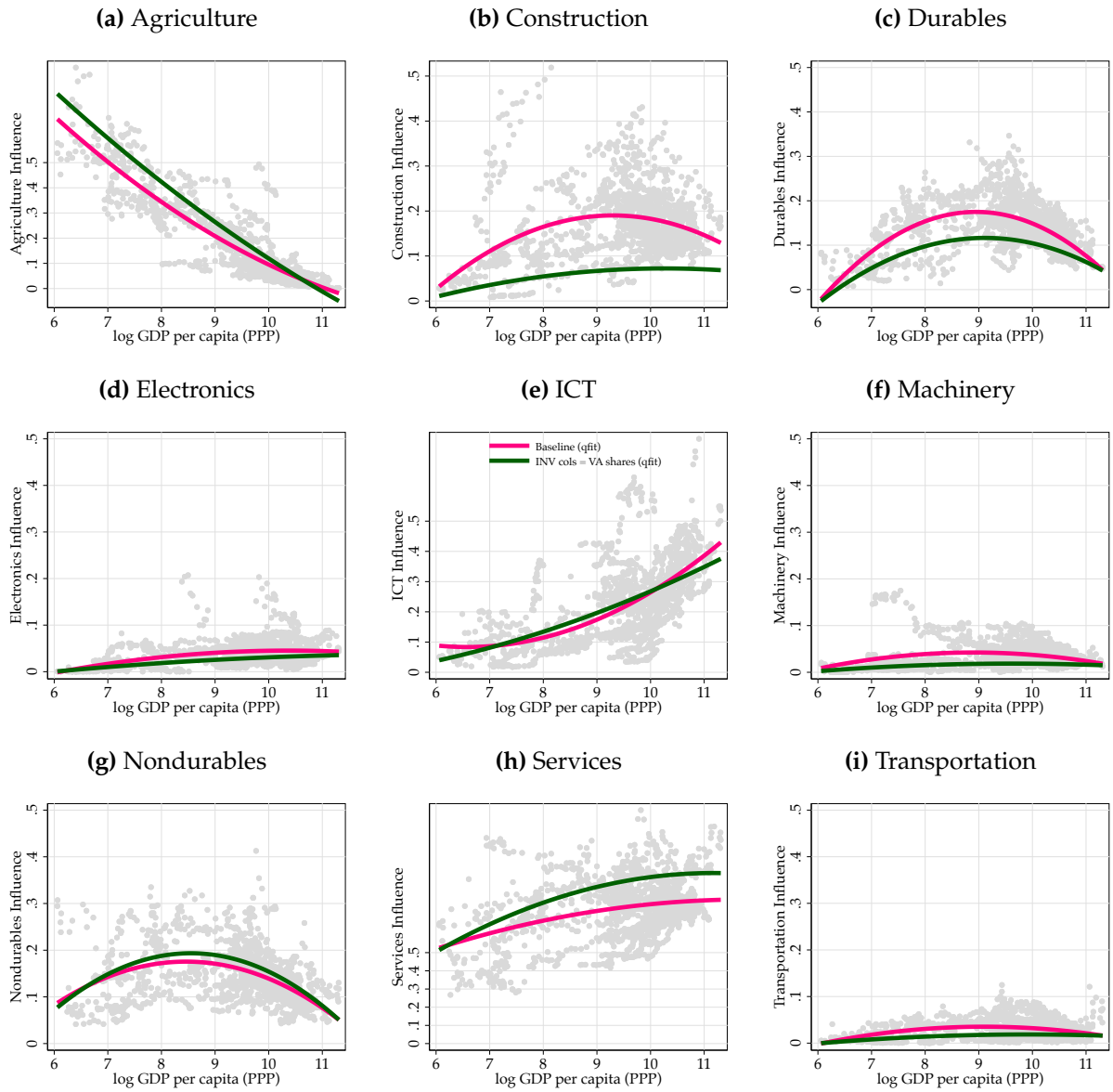
Notes: Data for all country-years in the sample. sectoral influence is defined as $\Phi\eta\Gamma$ in Proposition 4, and measures the change in aggregate value added following a change in sectoral productivity. Dots represent country–year observations; South Korea (1965–2014), India (1965-2014) and China (1965-2014) are highlighted.

Figure 21: Sectoral Influence: Counterfactual with U.S. Value-Added Shares



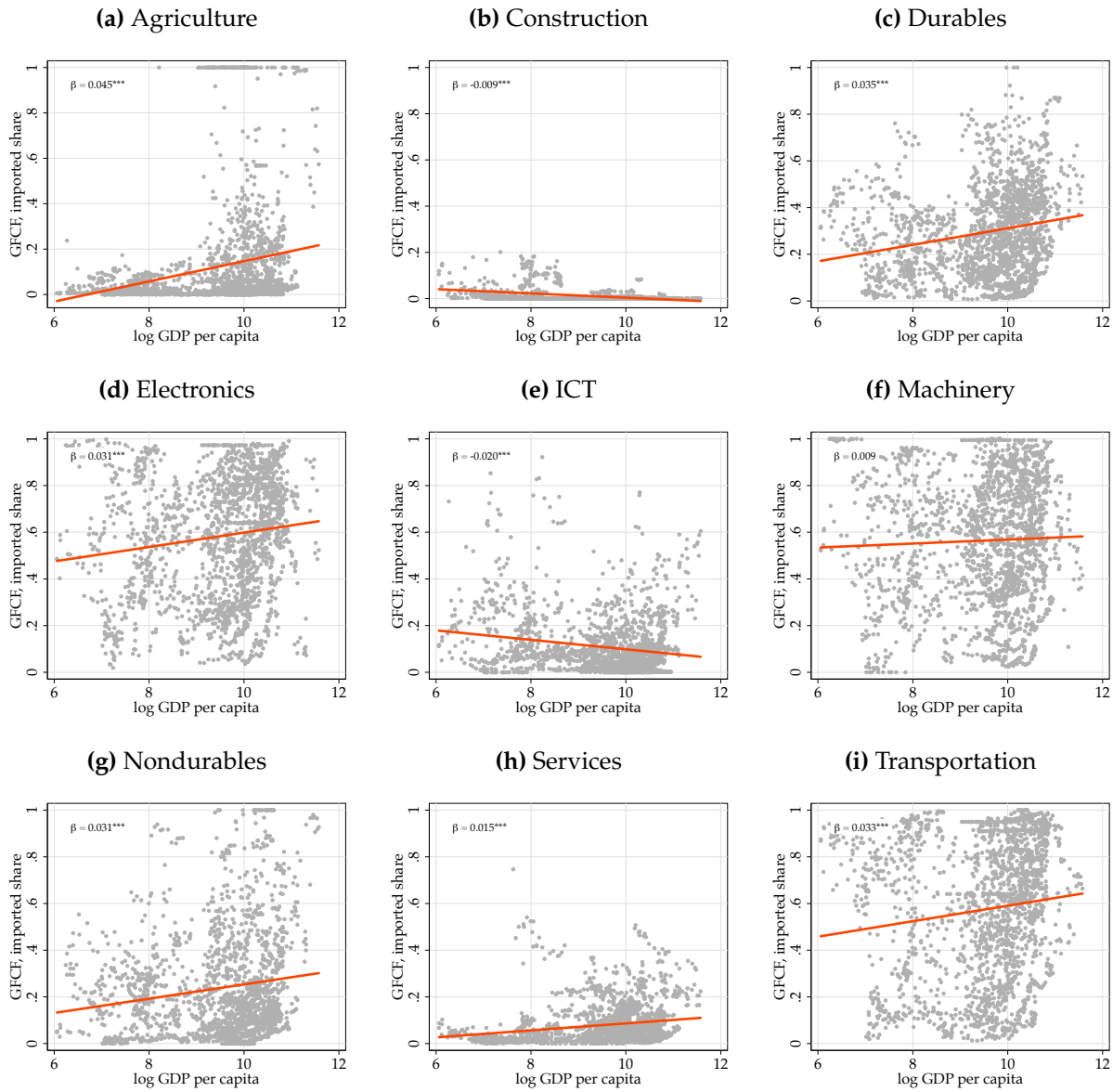
Notes: Sectoral influence is defined as $\Phi\eta\Gamma$ in Proposition 4 and measures the change in aggregate value added following a change in sectoral productivity. Relative to Figure 20, the value-added share vector ζ is held fixed at U.S. values for every country in the sample. Dots represent country–year observations.

Figure 22: Sectoral Influence: Counterfactual with Investment Network = Value-Added Shares



Notes: Sectoral influence is defined as $\Phi\eta\Gamma$ in Proposition 4 and measures the change in aggregate value added following a change in sectoral productivity. Relative to Figure 20, each column of the investment network Ω is set equal to the value-added share vector for every country in the sample. Dots represent country–year observations.

Figure 23: Sectoral Investment, Imported Share



Notes: Data for all country-years in the sample. For each sector, the share of total gross fixed capital formation (GFCF) sourced from abroad. The β coefficient is the slope of the linear fit (in orange), with stars denoting its statistical significance (* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$).

B.8 Transition Dynamics

Tables 18 and 19 report the time-varying primitives used in the transition calibration of Section 7: the CES investment loadings $\tilde{\omega}_{in,t}$ and the input–output network $\mu_{in,t}$, both at the four anchor years 1966, 1990, 2005, and 2014. In the transition, we interpolate $\tilde{\omega}_{in,t}$ log-linearly between anchors (renormalizing each column at every t) and $\mu_{in,t}$ linearly, holding both flat after 2014.

Table 18: Investment Weights, $\tilde{\omega}_{in,t}$ (Korea, 1966/1990/2005/2014)

	1966					1990					2005					2014				
Source i	Agri.	Cons.	ICT	Manuf.	Serv.	Agri.	Cons.	ICT	Manuf.	Serv.	Agri.	Cons.	ICT	Manuf.	Serv.	Agri.	Cons.	ICT	Manuf.	Serv.
Agri.	0.18	0.00	0.00	0.14	0.00	0.06	0.00	0.00	0.05	0.00	0.06	0.00	0.00	0.04	0.00	0.06	0.00	0.00	0.03	0.00
Cons.	0.19	0.11	0.01	0.09	0.56	0.33	0.33	0.12	0.30	0.78	0.35	0.25	0.49	0.43	0.87	0.50	0.30	0.38	0.60	0.71
ICT	0.00	0.00	0.01	0.00	0.00	0.00	0.01	0.06	0.01	0.00	0.06	0.17	0.20	0.08	0.03	0.13	0.35	0.39	0.11	0.13
Manuf.	0.42	0.64	0.92	0.58	0.29	0.37	0.35	0.63	0.27	0.13	0.33	0.31	0.15	0.15	0.05	0.22	0.20	0.09	0.07	0.08
Serv.	0.21	0.25	0.06	0.18	0.15	0.24	0.31	0.19	0.38	0.08	0.20	0.26	0.16	0.30	0.05	0.09	0.15	0.14	0.19	0.08

Notes: Investment weights $\tilde{\omega}_{in,t}$ for Korea at the four anchor years 1966, 1990, 2005, and 2014, recovered to match the investment-network odds ratios and the sectoral composition of aggregate investment at each anchor (common elasticity $\iota = 2.41$). Columns are destination sectors and sum to one; rows are source sectors. Data source: WIOD (see Section 3.2).

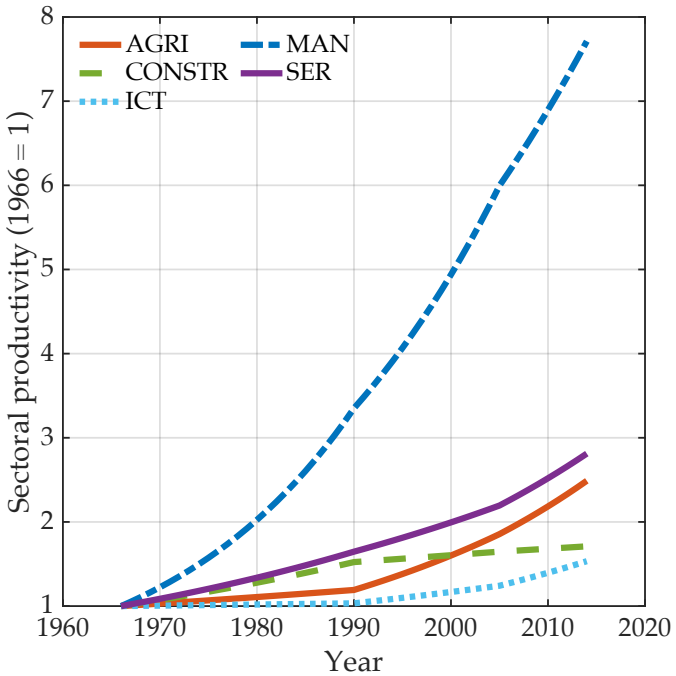
Table 19: Input–Output Network, $\mu_{in,t}$ (Korea, 1966/1990/2005/2014)

	1966					1990					2005					2014				
Source i	Agri.	Cons.	ICT	Manuf.	Serv.	Agri.	Cons.	ICT	Manuf.	Serv.	Agri.	Cons.	ICT	Manuf.	Serv.	Agri.	Cons.	ICT	Manuf.	Serv.
Agri.	0.50	0.01	0.00	0.40	0.01	0.23	0.00	0.00	0.10	0.00	0.16	0.01	0.00	0.04	0.01	0.16	0.01	0.00	0.03	0.01
Cons.	0.00	0.00	0.00	0.00	0.03	0.00	0.00	0.00	0.00	0.04	0.00	0.00	0.00	0.00	0.02	0.00	0.00	0.00	0.00	0.02
ICT	0.00	0.03	0.07	0.01	0.07	0.01	0.08	0.16	0.04	0.13	0.02	0.12	0.54	0.07	0.15	0.01	0.12	0.38	0.03	0.17
Manuf.	0.37	0.58	0.63	0.48	0.23	0.54	0.62	0.58	0.67	0.25	0.62	0.69	0.20	0.75	0.28	0.66	0.68	0.27	0.75	0.33
Serv.	0.12	0.37	0.30	0.11	0.66	0.22	0.29	0.26	0.19	0.57	0.20	0.18	0.26	0.14	0.54	0.17	0.18	0.35	0.18	0.48

Notes: 5-sector input–output network for Korea at the four anchor years 1966, 1990, 2005, and 2014; data source: WIOD (see Section 3.2). Columns are destination sectors and sum to one.

Figure 24 plots the calibrated sectoral productivity paths.

Figure 24: Sectoral Productivity Paths (South Korea, 1966–2014)



Notes: Calibrated sectoral productivity $z_{j,t}$, with each sector indexed to one in 1966. Manufacturing grows fastest, rising roughly eightfold over the period, while the other sectors rise between one-and-a-half- and threefold; these differences in slope drive the reallocation of value added and the evolution of the investment network. We interpolate productivities log-linearly between the anchor years 1966, 1990, 2005, and 2014, and hold them constant thereafter.

Figure 25 compares the model's sectoral value-added shares with the data.

Figure 25: Value Added Shares: Model vs Data (South Korea, 1965–2014)

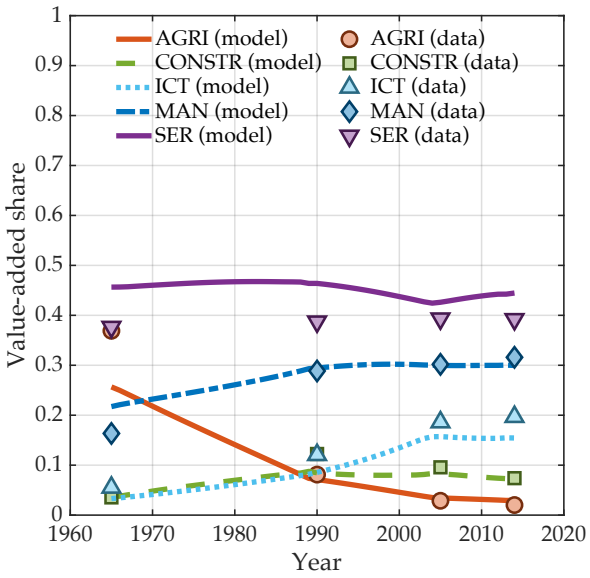
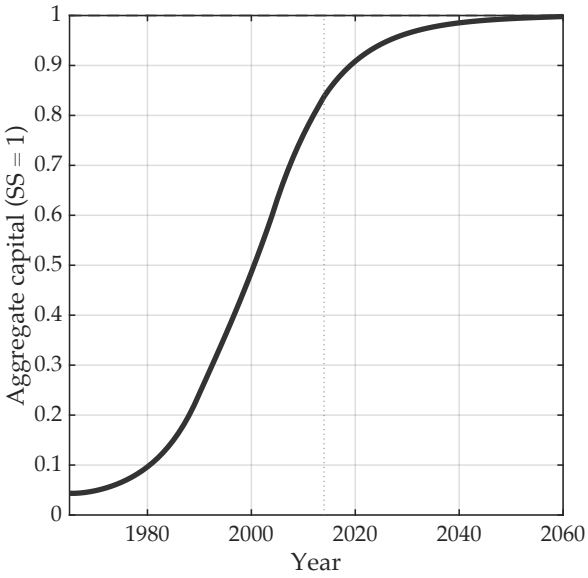


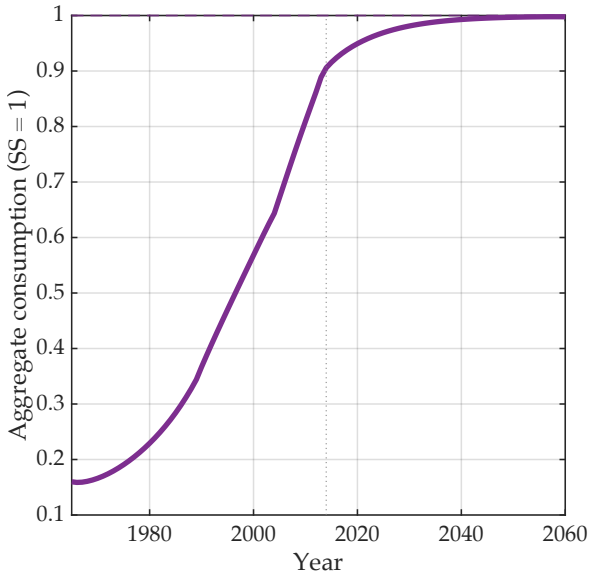
Figure 26 reports the aggregate capital and consumption paths, both relative to the steady state.

Figure 26: Aggregate Capital and Consumption Relative to Steady State (South Korea)

(a) Aggregate Capital

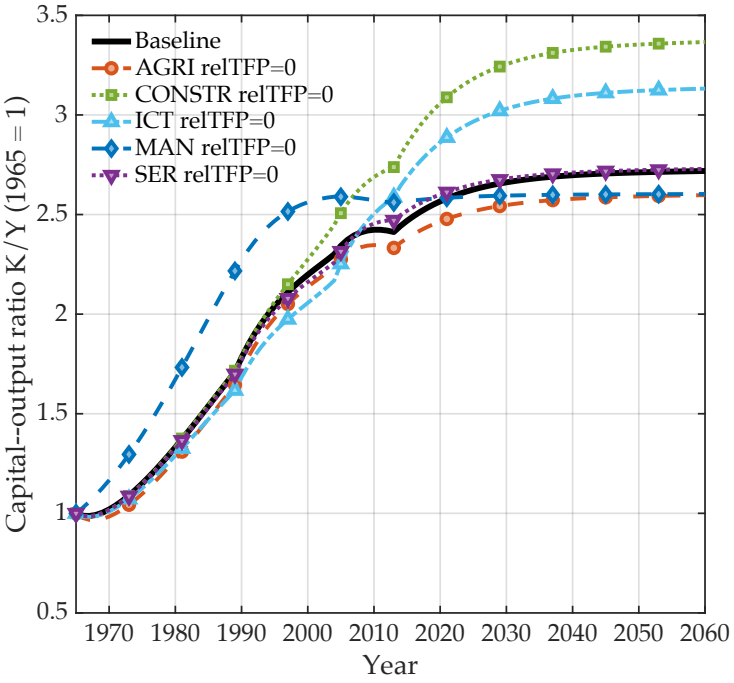


(b) Aggregate Consumption



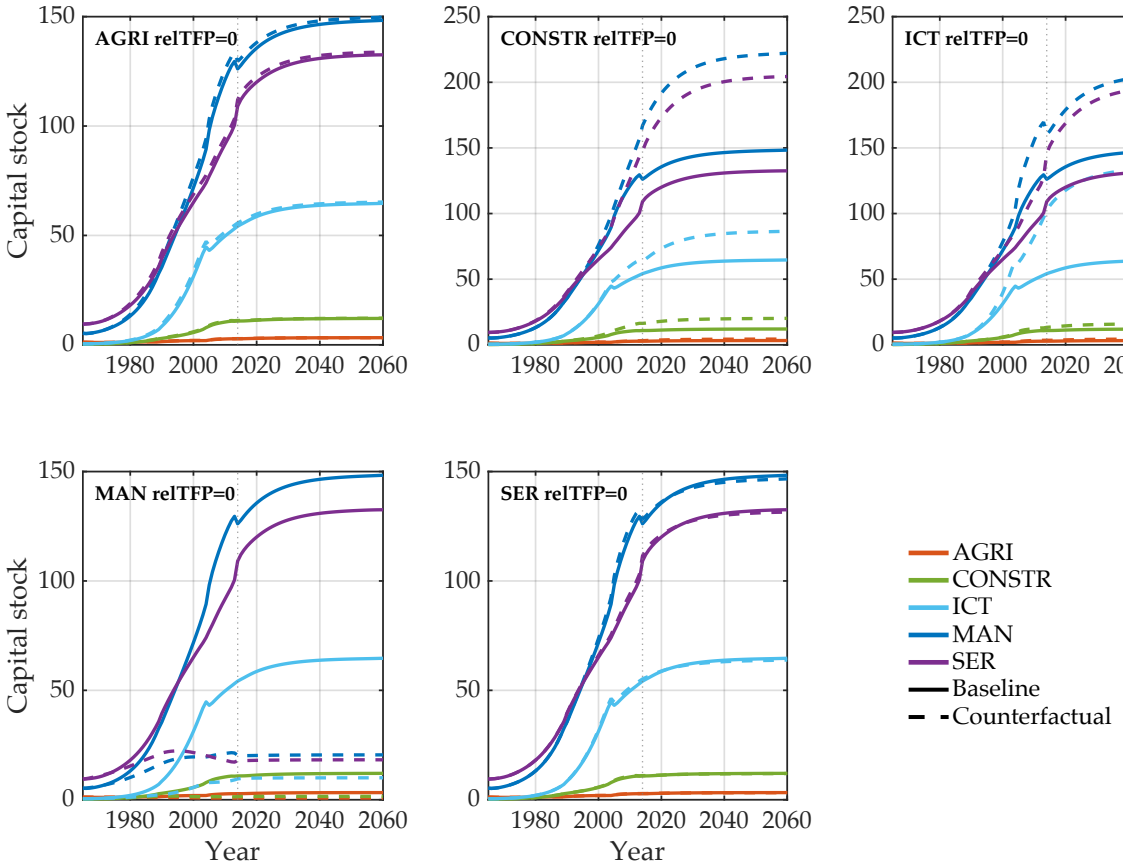
Counterfactual: No Sectoral Relative Productivity Growth

Figure 27: Capital–Output Ratio: Baseline and No-Relative-Growth Counterfactuals



Notes: Aggregate capital–output ratio K/Y (chained Törnqvist capital over chained Törnqvist output, indexed to one in 1965), for the baseline and for each counterfactual that freezes one sector's manufacturing-relative TFP at its 1966 level. Paths run to 2060, by which point each economy has reached its steady state.

Figure 28: Sectoral Capital along the Transition: Baseline vs. No-Relative-Growth Counterfactuals



Notes: Sectoral capital levels along the transition (to 2060). Each panel freezes one sector's manufacturing-relative TFP at its 1966 level; solid lines are the baseline, dashed lines the counterfactual.

Table 20: Output and the Capital–Output Ratio across No-Relative-Growth Counterfactuals

Scenario	$\Delta \ln Y$	$\Delta \ln(K/Y)$	$\dfrac{\Delta \ln Y}{\Delta \ln(K/Y)}$
Baseline	2.04	0.88	2.31
Agriculture	2.10	0.85	2.47
Construction	2.17	1.01	2.15
ICT	2.28	0.95	2.39
Manufacturing	0.23	0.91	0.25
Services	2.04	0.91	2.25

Notes: Detrended log changes, 1965–2014, with output and capital both aggregated as chained Törnqvist indices. Each row freezes one sector's manufacturing-relative TFP at its 1966 level, holding all other primitives at baseline. $\Delta \ln Y$ is the change in aggregate output, $\Delta \ln(K/Y)$ the change in the capital–output ratio.